

Mathematical Logic (UCLA Math 114L)

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August 14, 2026

Abstract

These are the notes I produced while teaching Mathematical Logic (Math 114L) at University of California, Los Angeles in Spring 2026. Xiang Li was in charge of the discussions.

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1 Vocabulary and Notation

- 0 is a natural number, i.e.

$$\mathbb{N} = \{0, 1, 2, \dots, n, n+1, \dots\}.$$

- “ $x \leq y$ ” is read: “ x is less than y ” or “ x is less or equal than y ”.
- “ $x < y$ ” is read: “ x is strictly less than y ”.
- Similarly for “ $x > y$ ” and “ $x \geq y$ ”.
- “ x is positive” means “ $x \geq 0$ ” and “ x is strictly positive” means “ $x > 0$ ”; similarly for “negative” and “strictly negative”.
- “ x is a subset of y ” means “ $x \subseteq y$ ” and “ x is a strict subset of y ” means “ $x \subset y$ ”; similarly for “superset” and “strict superset”.
- For all $n \in \mathbb{N}$, we will denote by $[n]$ the set of all natural numbers strictly lesser than n :

$$[n] := \{0, 1, \dots, n-1\}.$$

- For sets E and F , we denote by $\mathcal{F}(E, F)$ the set of all functions from E to F .
- The set of all n -tuples of elements of E , denoted by E^n , is formally defined to be the set $\mathcal{F}([n], E)$. Notice that for $n = 0$, we have $E^0 = \{\emptyset\}$.
- The set $E^{<\mathbb{N}}$ denotes the set of all finite tuples of elements E , sometimes also called “the set of all finite sequences of elements of E ”, and is formally defined to be

$$\bigcup_{n \in \mathbb{N}} E^n.$$

- The set of all (infinite) sequences of elements of E is denoted by $E^{\mathbb{N}}$ and is defined to be $\mathcal{F}(\mathbb{N}, E)$. Notice that for all $n \in \mathbb{N}$, there is a canonical function

$$E^{\mathbb{N}} \rightarrow E^n : a \mapsto a \upharpoonright [n].$$

- If a is a sequence (finite or infinite), then it is formally a function, so for every i in its domain, the notation $a(i)$ makes sense. When talking about sequences, we prefer to write a_i instead of $a(i)$.
- A family is just a function $f : I \rightarrow E$. We call I its index set and we say that the values of f are in E . In the case of a family, we also use the notation f_i to mean $f(i)$. In fact, a sequence is just a family whose index set is $[n]$ or $\mathbb{N}$.
- When we think of a function $f : I \rightarrow E$ as a family, we often prefer to write $(f_i : i \in I)$ instead of simply f .

- Thus, the set of all families with index set I and values in E is just the set $\mathcal{F}(I, E)$. Its general element is of the form $a = (a_i : i \in I)$. Sometimes in order to make clear that a is a family, we use the notation $\vec{a}$. We consider this to have the same formal meaning as a and do not write $(\vec{a})(i)$ or $(\vec{a})_i$, but simply $a(i)$ and a_i .

- An n -ary operation on E is a function

$$f : E^n \rightarrow E.$$

In this case, the natural number n is called “the arity of f ”.

- A 0-ary operation is just a function

$$f : \{\emptyset\} \rightarrow E,$$

and it can be identified with the element $f(\emptyset)$ of E .

- A 1-ary operation, pronounced “unary operation”, is just a function

$$f : E \rightarrow E.$$

- A 2-ary operation, pronounced “binary operation”, is a function

$$f : E^2 \rightarrow E.$$

For $x, y \in E$, we prefer to write $f(x, y)$ instead of $f((x, y))$, and even further, we prefer to write xfy instead of $f(x, y)$.

- In general, if $f : E^n \rightarrow E$ and $s_0, \dots, s_{n-1} \in E$, we prefer to write

$$f(s_0, \dots, s_{n-1})$$

over $f((s_0, \dots, s_{n-1}))$ (whenever there is no risk of confusion).

- An n -ary relation on a set E is any subset of E^n . If R is a binary relation on E , we write “ xRy ” to mean “ $(x, y) \in R$ ”.
- If σ is a relation or an operation on E , then we denote by $\text{ar}(\sigma)$ its arity.

2 Basic Mathematical Reasoning

2.1 Propositions and Predicates

Terminology 2.1.1. A **proposition** is a mathematical sentence that is either true or false. □

Example 2.1.2. The sentence: “10 is a prime number.” is a proposition (which is false). □

Exercise 2.1.3 (Liar’s paradox). *Prove that the sentence: “This sentence is false.” is **not** a proposition.* □

Terminology 2.1.4. A **predicate** or a **property** is a mathematical sentence which depends on variables and which becomes a proposition we substitute those variables with concrete values. $\square$

Notation 2.1.5. We may use letters P, Q, R , etc. to denote general propositions. We also write $P(x)$ to denote predicate that depends on x , we write $Q(y, z)$ to denote a predicate that depends on y and z , etc. $\square$

Example 2.1.6. The sentence $P(x)$: “ x is an even natural number.” is a predicate. The sentence $P(10)$: “10 is an even natural number.” is a proposition (which is true). $\square$

If P and Q are propositions, then so are:

- “ P and Q ”;
- “ P or Q ”;
- “not P ”;
- “if P , then Q ”, written also “ $P \implies Q$ ”;
- “ P if and only if Q ”, written also “ $P \iff Q$ ”.

We assume that the student is familiar with how the truth value of each of these complex propositions depends on the truth value of P and Q .

We also have two distinguished propositions: the **tautology** and the **contradiction**. The tautology is “the pure truth”, while the contradiction is “the pure falsity”.

If $P(x)$ is a predicate, then the following sentences are propositions:

- “for all x , $P(x)$ holds”, written also “ $\forall x, P(x)$ ”
- “there is x such that $P(x)$ holds”, written also “ $\exists x, P(x)$ ”;
- “there is a unique x such that $P(x)$ holds”, written also “ $\exists! x, P(x)$ ”.

If E is a set, then we also have propositions:

- “for all x in E , $P(x)$ holds”, written also “ $\forall x \in E, P(x)$ ”
- “there is x in E such that $P(x)$ holds”, written also “ $\exists x \in E, P(x)$ ”;
- “there is a unique $x \in E$ such that $P(x)$ holds”, written also “ $\exists! x \in E, P(x)$ ”.

The above notions are called **quantifiers**. The quantifier $\forall$ is called the **universal quantifier**, $\exists$ is called the **existential quantifier**, and $\exists!$ is called the **uniqueness quantifier**. Let us note that the quantifiers $\forall$ and $\exists$ are considered basic, while the other ones are derived. Here is how to express the derived ones in terms of the basic ones:

- $\exists! x, P(x)$ holds if and only if both $\exists x, P(x)$ and

$$\forall x, \forall y, (P(x) \text{ and } P(y) \implies x = y)$$

hold;

- $\forall x \in E, P(x)$ holds if and only if

$$\forall x, (x \in E \implies P(x));$$

- $\exists x \in E, P(x)$ holds if and only if

$$\exists x, (x \in E \text{ and } P(x));$$

- $\exists! x \in E, P(x)$ holds if and only if both $\exists x \in E, P(x)$ and

$$\forall x \in E, \forall y \in E, (P(x) \text{ and } P(y) \implies x = y)$$

hold.

2.2 Rules of Inference

Logical steps that we are allowed to make inside a mathematical proof are called **rules of inference**. Pedagogically, we can divide these rules into *simple* and *complex* ones (not a standard terminology). The complex ones will be reviewed in the subsections that follow; here, we will give a quick overview of the simple ones.

The simple rules are simple in the sense that they correspond to just one logical step. We customarily write them as

$$\frac{P_0 \quad P_1 \quad \dots \quad P_{n-1}}{Q},$$

where $P_0, P_1, \dots, P_{n-1}, Q$ are propositions/predicates. The intended reading behind this kind of notation is the following: “If we have already proven $P_0, P_1, \dots$, and P_n , then we may conclude that Q holds as well.” The following are some basic simple rules of inference:

$$\frac{P \quad Q}{P \text{ and } Q};$$

$$\frac{P \text{ and } Q}{P};$$

$$\frac{P \text{ and } Q}{Q};$$

$$\frac{P}{P \text{ or } Q};$$

$$\frac{Q}{P \text{ or } Q};$$

$$\frac{\text{not } P \quad P}{\text{contradiction}};$$

$$\frac{\text{not not } P}{P};$$

$$\begin{array}{c}
\frac{P}{\text{not not } P}; \\
\\
\frac{\text{contradiction}}{\text{any } P \text{ whatsoever}}; \\
\\
\frac{}{\text{tautology}}; \\
\\
\frac{P \Rightarrow Q \quad P}{Q}; \\
\\
\frac{P \Leftrightarrow Q \quad P}{Q}; \\
\\
\frac{P \Leftrightarrow Q \quad Q}{P}; \\
\\
\frac{P \Rightarrow Q \quad Q \Rightarrow P}{P \Leftrightarrow Q}; \\
\\
\frac{R(c)}{\exists x, R(x)}; \\
\\
\frac{\forall x, R(x)}{R(c)}; \\
\\
\frac{}{c = c}; \\
\\
\frac{c = d \quad R(c)}{R(d)}.
\end{array}$$

In these rules, we denoted by P and Q arbitrary propositions, by $R(x)$ an arbitrary predicate, and by c and d arbitrary objects.

Let us remark that these listed rules are not chosen randomly. To these basic simple rules, one should add five basic complex rules (proving implication, proof by contradiction, proof by cases, using the existential quantifier, and proving the universal quantifier), and two basic principles (the principle of assumption¹ and the principle of lemma²) in order to get **the complete logical system** behind all of mathematical reasoning. One of the goals of this course is to understand why this is true.

Exercise 2.2.1. *One might doubt that we have written down everything we need. For example, can we really derive, from what has been said, all standard properties of the equality? The answer is yes, and this is the content of this exercise.*

Let a , b , and c be arbitrary objects. Prove the following propositions.

¹At every point in a proof, we can use any of the assumptions that we are making.

²Once we prove a proposition, we can use it in any future proof without the need to reprove it.

a. It holds that $a = a$.

b. If $a = b$, then $b = a$.

c. If $a = b$ and $b = c$, then $a = c$. □

Below we review the complex rules of inference. We call them complex because they require one or more subproofs in order to reach the conclusion. For example, to prove $P \implies Q$, we first have to enter a subproof where we assume P and prove Q , and then exit this subproof, **discharge** the assumption P , and conclude $P \implies Q$ in the main proof. Discharging the additional assumption is characteristic of complex proof. The point is that, in the above scenario, we have proven $P \implies Q$ to hold without P as an underlying assumption.

2.3 Proving Implication

The basic way to prove $P \implies Q$ is as follows. First, we enter a sub-proof where we assume additionally that P holds and where we prove that Q holds as well. This where we exit the sub-proof and conclude $P \implies Q$ in the main proof. We observe the following form.

Proof. Let us assume P and prove Q . [We provide a proof of Q .] Thus, Q is true.

Therefore, $P \implies Q$. □

Definition 2.3.1. A natural number n is a **perfect square** if $\exists k \in \mathbb{N} \quad n = k^2$. □

Example 2.3.2. Let a and b be natural numbers. Prove that if a and b are both perfect squares, then ab is also a perfect square.

Proof. Let us assume that a and b are perfect squares and let us prove that ab is also a perfect square. Since a is a perfect square, we can find $x \in \mathbb{N}$ such that $a = x^2$. Similarly, we can find $y \in \mathbb{N}$ such that $b = y^2$.

Let $z = xy$. We have that $z \in \mathbb{N}$ and

$$ab = x^2 y^2 = (xy)^2 = z^2$$

Thus, ab is a perfect square.

Therefore, we may conclude that the implication: “If a and b are perfect squares, then ab is a perfect square.” is true. □

Remark 2.3.3. In practice, we would normally omit the last sentence of the previous proof, because it is very much obvious from what is said before it. □

2.4 Proving Equivalence

The basic way to prove $P \iff Q$ is to **reason by double implication**, i.e. to prove both $P \implies Q$ (“the direct implication”) and $Q \implies P$ (“the reverse implication”). This type of argument is written as follows.

Proof. Let us prove $P \iff Q$ by reasoning by double implication.

Direct implication. [We provide a proof of $P \implies Q$.]

Reverse implication. [We provide a proof of $Q \implies P$.]

Conclusion. Therefore, $P \iff Q$ is true. $\square$

Remark 2.4.1. In general, the implication “ $Q \implies P$ ” is referred to as **the reverse implication** of the implication “ $P \implies Q$ ”. Often when we prove $P \implies Q$, we may ask whether the reverse implication is true; to answer this question, we proceed to either prove or refute $Q \implies P$. $\square$

Example 2.4.2. Let n be a natural number. Prove that n is even if and only if n^2 is even.

Proof. We reason by double implication.

Direct implication. Let us assume that n is even and let us prove that n^2 is also even. Since n is even, there is $k \in \mathbb{N}$ such that $n = 2k$. Let $l = 2k^2$. Then l is a natural number and

$$n^2 = 4k^2 = 2l,$$

whence n^2 is even.

Reverse implication. Let us assume that n^2 is even and let us prove that n is also even.

Assume towards contradiction that n is odd. Then for some $k \in \mathbb{N}$, we have $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Let $l = 2k^2 + 2k$. Then $l \in \mathbb{N}$ and $n^2 = 2l + 1$, whence n^2 is odd – contradiction.

Thus, n is even, as required. $\square$

2.5 Proof by Contradiction

To prove that P holds, you may **reason by contradiction**. In such a proof, we first enter a sub-proof where we assume that P is *false*, then we derive a contradiction. This is where we exit the sub-proof and conclude (in the main proof) that P is true. We should use the following form when reasoning by contradiction.

Proof. Let us assume towards contradiction that P is false. [We provide a proof of contradiction.]

Thus, we derived contradiction.

Therefore, we conclude that P is true. $\square$

Example 2.5.1. Prove that $\sqrt{2}$ is an irrational number.

Proof. Let us assume towards contradiction that $\sqrt{2}$ is rational. Then there are $a \in \mathbb{Z}$ and $b \in \mathbb{N} - \{0\}$ such that

$$\sqrt{2} = \frac{a}{b}.$$

Up to reducing the fraction, we may assume that a and b are coprime.

After squaring and multiplying by b^2 , we get that

$$a^2 = 2b^2.$$

It follows that a^2 is even, so a is even. In other words, for some $x \in \mathbb{Z}$, we have $a = 2x$. After substituting and dividing by 2, we get

$$2x^2 = b^2.$$

Now, b^2 is even, whence b is even. We have proven that both a and b are even, which means they are **not** coprime. This is a contradiction.

Therefore, $\sqrt{2}$ is irrational. □

2.6 Proof by Cases

If we know “ P or Q ” and if we want to prove R , then we may **reason by cases**. We proceed as follows:

- we start a sub-proof called “Case 1” where we assume additionally that P holds and we prove that R holds as well (this is where we finish the sub-proof);
- we start another sub-proof called “Case 2”, where we assume additionally that Q holds and we prove that R holds as well (this is where we finish this sub-proof);
- we conclude (in the main proof) that R holds.

We should observe the following form when doing a proof by cases.

Proof. Since “ P or Q ” holds, we will reason by cases.

Case 1. Assume P . [We provide a proof of R .] Thus, R is true.

Case 2. Assume Q . [We provide a proof of R .] Thus, R is true.

Conclusion. The proof by Cases 1 and 2 yields that R is true. □

Example 2.6.1. Prove that there are strictly positive **irrational** numbers a and b such that a^b is **rational**.

Proof. Since $(\sqrt{2})^{\sqrt{2}}$ is either rational or irrational, we will reason by cases.

Case 1. Assume that $(\sqrt{2})^{\sqrt{2}}$ is rational. Let $a = b = \sqrt{2}$. Then a and b are irrational numbers such that a^b is rational.

Case 2. Assume that $(\sqrt{2})^{\sqrt{2}}$ is irrational. Let $a = (\sqrt{2})^{\sqrt{2}}$ and $b = \sqrt{2}$. Then a and b are irrational numbers such that

$$a^b = \left((\sqrt{2})^{\sqrt{2}} \right)^{\sqrt{2}} = (\sqrt{2})^{\sqrt{2} \cdot \sqrt{2}} = (\sqrt{2})^2 = 2$$

is a rational number.

Conclusion. The proof by Cases 1 and 2 shows that there are irrational numbers a and b such that a^b is rational. □

Remark 2.6.2. If we know “ P or Q or R ”, then we may reason by **three** cases: Case 1 – P is true; Case 2 – Q is true; Case 3 – R is true. In fact, we can have any number of cases (and also sub-cases), but we have to make sure that they cover all possibilities, i.e. that their disjunction is true. □

2.7 Using Existential Quantifier

If we have proven “ $\exists x, P(x)$ ” and we want to use it to prove some proposition R , then we may proceed as follows. First, we enter a sub-proof where we use a new name c and assume additionally that $P(c)$ holds. (We think of this as *picking a witness* for the existential statement.) Inside the sub-proof, we provide a reasoning that leads to the conclusion that R holds. It is important here that R does not mention c , i.e. that c is really a new name from the point of view of the main proof. When we have achieved this, we may exit the sub-proof, discharge the assumption that $P(c)$ holds, and conclude in the main proof that R holds. We propose the following form of writing.

Proof. Let c be such that $P(c)$ holds. [We provide a proof of R ; make sure R does not mention c .]
Therefore, R holds. [The hypothesis “ $P(c)$ ” is discharged.] $\square$

Example 2.7.1. Let a and b be reals. Define a function f from $\mathbb{R}$ to $\mathbb{R}$ by:

$$\forall x \in \mathbb{R}, f(x) = ax^2 + b.$$

Assume that $\exists x \in \mathbb{R}, f(x) = 0$. Then $ab < 0$.

Proof. Let $c \in \mathbb{R}$ be such that $f(c) = 0$. Then

$$ac^2 + b = f(c) = 0.$$

This means that the quadratic equation

$$aX^2 + 0 \cdot X + b = 0$$

has a real solution. Thus, its discriminant

$$0^2 - 4ab = -4ab$$

is positive. It follows that $ab < 0$.

Therefore, $ab < 0$. $\square$

Remark 2.7.2. In practice, we would normally omit the last line of the above proof. When it is omitted, it is clear that we intend an implicit application of the rule for using the existential quantifier. $\square$

2.8 Proving Universal Quantifier

To prove “ $\forall x, P(x)$ ”, we start by entering a sub-proof where a new name, for an arbitrary object, is provided. Say this name is c . Then we go on to provide a proof of $P(c)$, while assuming nothing about c . When this is done, we may exit the sub-proof, discharge the new name, and conclude that $\forall x, P(x)$. We propose the following form.

Proof. Let c be given. [c has to be a new name.] We want to prove $P(c)$. [We provide a proof of $P(c)$.]

Therefore, $\forall x, P(x)$. [We discharge the name c .] $\square$

Example 2.8.1. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by:

$$\forall x \in \mathbb{R}, f(x) = x^2 + x + 1.$$

Then

$$\forall x \in \mathbb{R}, f(x) > 0.$$

Proof. Let c be given. We want to prove that

$$c \in \mathbb{R} \implies f(c) > 0.$$

Let us assume $c \in \mathbb{R}$ and prove $f(c) > 0$. Since $c \in \mathbb{R}$,

$$f(c) = c^2 + c + 1 = \left(c + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4} > 0.$$

Therefore:

$$c \in \mathbb{R} \implies f(c) > 0.$$

Therefore:

$$\forall x \in \mathbb{R}, f(x) > 0.$$

□

Remark 2.8.2. Again, very often we would omit the last line. As remarked in the context of implication, we would normally also omit the line before the last. □

Remark 2.8.3. Also, if the variable x does not appear elsewhere without a quantifier (i.e. “free”), then we may treat it as a new name and use it instead of the name c . In other words, in practice, we very often start a proof of the above type by saying: “Let x be given.” □

Remark 2.8.4. In the above proof, we start one sub-proof when we introduce the new name c , and then another one when we assume $c \in \mathbb{R}$. This is normally joined together into a single sub-proof. We would start by saying: “Let $c \in \mathbb{R}$ be given.” This sub-proof would end with the line: “Thus, $f(c) > 0$.” □

A grownup version of the proof. Let $x \in \mathbb{R}$ be given. Then

$$f(x) = x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4} > 0.$$

□

2.9 Formatting tip: Analysis-Synthesis

When we need to determine the set of all objects x for which a property $P(x)$ holds, we may proceed by breaking our solution into the “analysis” and “synthesis” parts.

Analysis. We start by assuming that x is such that $P(x)$ holds. Then we proceed to determine properties of x that would allow us to narrow down the set of possible candidates.

Synthesis. For each of the candidates obtained in the analysis, we verify if they indeed satisfy $P(x)$.

Conclusion. The set of all x such that $P(x)$ holds is exactly the set of all candidates obtained in the analysis which survived the verification process given in the synthesis.

Example 2.9.1. Determine the set of all reals x strictly greater than $\frac{1}{2}$ that satisfy $x^3 = x$.

Solution. We proceed by analysis-synthesis.

Analysis. Suppose x is such that

$$x \in \mathbb{R} \quad \wedge \quad x > \frac{1}{2} \quad \wedge \quad x^3 = x.$$

Since $x > \frac{1}{2}$, we conclude that $x \neq 0$. Since $x^3 = x$ and $x \neq 0$, we conclude that $x^2 = 1$. It follows that

$$(x - 1)(x + 1) = 0,$$

so $x = 1$ or $x = -1$.

Synthesis. We have to verify which of the candidates $x \in \{1, -1\}$ satisfies:

$$x \in \mathbb{R} \quad \wedge \quad x > \frac{1}{2} \quad \wedge \quad x^3 = x.$$

If $x = -1$, we have that $x > \frac{1}{2}$ is false, so $x = -1$ is not a solution.

If $x = 1$, an easy computation shows that x is a solution.

Conclusion. We conclude that the set of all real numbers x strictly greater than $\frac{1}{2}$ such that $x^3 = x$ is equal to $\{1\}$. □

Proving Uniqueness Quantifier

Recall that “ $\exists!x, P(x)$ ” is equivalent to:

$$\exists x, P(x) \quad \wedge \quad \forall x, \forall y, (P(x) \wedge P(y) \implies x = y).$$

A basic proof of “ $\exists!x, P(x)$ ” thus consists of two parts (presented in any order): the “existence” part, where “ $\exists x, P(x)$ ” is proved, and the “uniqueness” part, where

$$\forall x, \forall y, (P(x) \wedge P(y) \implies x = y)$$

is proved. We propose the following form.

Proof.

Existence. [We provide a proof of $\exists x, P(x)$.]

Uniqueness. [We provide a proof of

$$\forall x, \forall y, (P(x) \wedge P(y) \implies x = y).]$$

□

Definition 2.9.2. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$. We say that f is **linear** iff there is a real α such that

$$\forall x \in \mathbb{R}, f(x) = \alpha x.$$

□

Example 2.9.3. Let a and b be reals. Assume $a \neq 0$. Then there is a unique linear function f from $\mathbb{R}$ to $\mathbb{R}$ such that $f(a) = b$.

Solution.

Uniqueness. Let f and g be two linear functions from $\mathbb{R}$ to $\mathbb{R}$ and let us prove that $f = g$. Since f and g have the same domain, we only need to prove that

$$\forall x \in \mathbb{R}, f(x) = g(x).$$

Let $\alpha \in \mathbb{R}$ be such that

$$\forall x \in \mathbb{R}, f(x) = \alpha x$$

and let $\beta \in \mathbb{R}$ be such that

$$\forall x \in \mathbb{R}, g(x) = \beta x.$$

Since $f(a) = b$, we conclude $\alpha a = b$; since $g(a) = b$, we conclude, $\beta a = b$. Thus, $\alpha a = \beta a$. Now, since $a \neq 0$, this yields $\alpha = \beta$.

Let $x \in \mathbb{R}$ be given. We have

$$f(x) = \alpha x = \beta x = g(x).$$

Therefore, $\forall x \in \mathbb{R}, f(x) = g(x)$, which finishes the proof of the uniqueness.

Existence. We have to find a linear function f from $\mathbb{R}$ to $\mathbb{R}$ such that $f(a) = b$. Since we do not know what the answer is, we proceed by analysis-synthesis.³

Analysis. Suppose f is a linear function from $\mathbb{R}$ to $\mathbb{R}$ such that $f(a) = b$. Since f is linear, there is $\alpha \in \mathbb{R}$ such that

$$\forall x \in \mathbb{R} \quad f(x) = \alpha x.$$

It follows that $\alpha a = b$. Since $a \neq 0$, we conclude $\alpha = b/a$.

Synthesis. Since $a \neq 0$, we may let $\alpha = b/a$. Let f be the function from $\mathbb{R}$ to $\mathbb{R}$ defined by:

$$\forall x \in \mathbb{R}, \quad f(x) = \alpha x.$$

We have that f is a linear function. It remains to prove that $f(a) = b$. For this, we simply compute:

$$f(a) = \alpha a = (b/a)a = b.$$

Conclusion. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\forall x \in \mathbb{R}, \quad f(x) = \alpha x$$

is a linear function with the property $f(a) = b$. □

Remark 2.9.4. The previous example shows how the analysis-synthesis style can be leveraged to write down a proof of existence when we intend to exhibit a witness, but do not yet know what the witness is. □

³The analysis part contains the search for f , while the synthesis part contains the proof of existence using the result of the search.

2.10 Definitions

Definitions are used to introduce new words into our language. They **cannot be used to prove new facts** – they can only be used to determine the meaning of phrases which otherwise would not have any meaning. There are two kinds of things that can be defined in mathematics: *properties* and *expressions*.

If we are considering a composite property, then we may wish to introduce a new phrase whose meaning is *asserted by definition* to be the same as the composite property. If the property depends on $x_0, x_1, \dots, x_{n-1}$, then the new phrase should display this dependence. Let $P(x_0, x_1, \dots, x_{n-1})$ be the property we are interested in. Then to introduce a new phrase with the same meaning, we write:

Definition. We say $\text{phrase}(x_0, x_1, \dots, x_{n-1})$ iff $P(x_0, x_1, \dots, x_{n-1})$. □

The phrase “iff” here means, as usual, “if and only if”, but its nature is different: we are not asserting the truth of a proposition, but are assigning the meaning to “ $\text{phrase}(x_0, x_1, \dots, x_{n-1})$ ”.

Example 2.10.1. Consider the property $P(E)$ that a set E may or may not have:

$$\exists n \in \mathbb{N}, \exists f \in \mathcal{F}(E, [n]), f \text{ is bijective.}$$

We may wish to introduce a phrase involving the variable E whose meaning is equivalent to the above mathematical sentence. To do so, we give a definition: “We say that a set E is **finite** iff there are $n \in \mathbb{N}$ and a bijection f from E to $[n]$.”

The other type of object that can be defined are expressions. An expression $e(x_0, \dots, x_{n-1})$ takes as input objects $x_0, \dots, x_{n-1}$ and returns another object. When defining an expression $e(x_0, \dots, x_{n-1})$, we start from a property $P(x_0, \dots, x_{n-1}, y)$ such that

$$\forall x_0, \dots, \forall x_{n-1}, \exists y, P(x_0, \dots, x_{n-1}, y),$$

and then, for given $x_0, \dots, x_{n-1}$, we define $e(x_0, \dots, x_{n-1})$ to be the unique y such that $P(x_0, \dots, x_{n-1}, y)$ holds. It is indispensable to make sure that the uniqueness condition holds; with it, we are simply naming an object which has been completely identified; without it, we are in the realm of *choosing* an object of certain kind, which is logically of completely different nature. To define an expression, we write as follows.

Definition. Let $x_0, \dots, x_{n-1}$ be given objects. Then we define $e(x_0, \dots, x_{n-1})$ to be the unique y such that $P(x_0, \dots, x_{n-1}, y)$ holds. □

Example 2.10.2. After having verified that for all sets A and B , there is a unique set C with the property

$$\forall x, (x \in C \iff x \in A \wedge x \in B),$$

we may introduce the expression “ $A \cap B$ ” as follows: “For given sets A and B , we define $A \cap B$ to be the unique set C satisfying:

$$\forall x, (x \in C \iff x \in A \wedge x \in B).”$$

□

We also very often compose existing expressions to define a new one. If $t(x_0, \dots, x_{n-1})$ is a composite expression for which we want to introduce a new name, say $e(x_0, \dots, x_{n-1})$, then this is a special case of the above scenario where we consider the predicate

$$P(x_0, \dots, x_{n-1}, y): "y = t(x_0, \dots, x_{n-1})".$$

Indeed, it is then trivially true that

$$\forall x_0, \dots, \forall x_{n-1}, \exists! y, P(x_0, \dots, x_{n-1}, y).$$

In this case, in stead of writing: “We define $e(x_0, \dots, x_{n-1})$ to be the unique y such that $y = t(x_0, \dots, x_{n-1})$.”, we use the following abbreviated form.

Definition. For given $x_0, \dots, x_{n-1}$, we define

$$e(x_0, \dots, x_{n-1}) := t(x_0, \dots, x_{n-1}).$$

The spacial cases of defining a property and an expression is defining a 0-ary property and a 0-ary expression: a 0-ary property is just a proposition (which is true or false), while 0-ary expression is just a constant (a name for a single distinguished object).

Example 2.10.3. We may define RH to be the proposition: “For all complex number z with $0 < \operatorname{Re} z < 1$, if $\zeta(z) = 0$, then $\operatorname{Re} z = \frac{1}{2}$.” This proposition is called **Riemann’s Hypothesis**, and it is not known whether it is true or false. Many people believe that it is true, so they may be interested in working out some consequences of it. In this case, it is practical to have the 0-ary predicate RH so one can efficiently refer to Riemann’s Hypothesis. $\square$

Example 2.10.4. An example of a 0-ary function, i.e. constant, is $\emptyset$. This constant is defined by saying: “ $\emptyset$ is the unique x such that $\forall y, y \notin x$.” $\square$

Example 2.10.5. One often encounters definitions of functions of the following form: “We define $f : \mathbb{R} \rightarrow \mathbb{R}$ to be the function defined by

$$\forall x \in \mathbb{R}, f(x) = \sqrt{1 + x^2}.”$$

Alternatively, we may use the following formalism as well: “We define the following function:

$$f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \sqrt{1 + x^2}.”$$

Both of these forms are variations of the standard form of defining a **constant**: “We define f to be the unique object with the following properties:

- f is a function;
- the domain of f is $\mathbb{R}$ and its codomain is contained in $\mathbb{R}$;
- for all $x \in \mathbb{R}$, we have $f(x) = \sqrt{1 + x^2}$.”

Of course, such a definition has to be preceded by a proof that there is, indeed, a unique such object (which, in this case, is obvious once we know basic Analysis). $\square$

Example 2.10.6. Very often, we define a sequence by *recursion*. For example, the **Fibonacci sequence** ($f_n : n < \omega$) is defined by:

- $f_0 = 0$ and $f_1 = 1$;
- for all $n \in \mathbb{N}$, we have $f_{n+2} = f_{n+1} + f_n$.

This definition is, again, just simple variation for the more formal version, which completely fits the general form of defining a **constant**: “The Fibonacci sequence is the unique object x with the following properties:

- x is a sequence;
- the values of x are natural numbers;
- $x_0 = 0$;
- $x_1 = 1$;
- for all $n \in \mathbb{N}$, we have $x_{n+2} = x_{n+1} + x_n$.”

Again, it is important to verify, before asserting this definition, that there is, indeed, a unique such x . However, in this case, **proving this is not obvious**. Luckily, there is a general theorem that can be used in many similar situations (see Example 3.1.1). $\square$

2.11 Exercises

Exercise 2.11.1. Let n be a natural number. Prove that $\frac{n(n+1)}{2}$ is a natural number. $\square$

Exercise 2.11.2. Let x and y be real numbers. Prove that

$$\min(x, y) = \frac{1}{2}(x + y - |x - y|)$$

and

$$\max(x, y) = \frac{1}{2}(x + y + |x - y|).$$

$\square$

Exercise 2.11.3. Let x be a real number. Prove that if x is irrational and $x > 0$, then $\sqrt{x}$ is irrational. $\square$

Exercise 2.11.4. Prove that the equation

$$x^3 + x + 1 = 0$$

has no rational solutions. $\square$

Exercise 2.11.5. Let n be a natural number. Prove that n is even if and only if $7n + 4$ is even. $\square$

Exercise 2.11.6. Prove or disprove: “For all rational numbers a and b , the number a^b is rational.” $\square$

Exercise 2.11.7. Prove that

$$\forall x \in \mathbb{R} \quad (x \leq 0 \iff \forall \varepsilon > 0 \quad x < \varepsilon).$$

□

Exercise 2.11.8. Prove that there is no greatest real number.

□

Exercise 2.11.9. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by:

$$\forall x \in \mathbb{R}, f(x) = x^2 + x + 1.$$

Prove that

$$\exists x \in \mathbb{R}, f(x) > 2.$$

Exercise 2.11.10. Let a and b be reals. Define the function f from $\mathbb{R}$ to $\mathbb{R}$ by:

$$\forall x \in \mathbb{R}, f(x) = ax^2 + b.$$

Assume that $\forall x \in \mathbb{R}, f(x) = 0$. Prove that $a = b = 0$.

□

Exercise 2.11.11. Let $P(x), Q(x), R(x), S(x)$ be predicates. Assume all of the following:

- a. $\forall x, (P(x) \vee Q(x))$
- b. $\forall x, (\neg Q(x) \vee S(x))$
- c. $\forall x, (R(x) \Rightarrow \neg S(x))$
- d. $\exists x, \neg P(x)$.

Prove **rigorously** that $\exists x, \neg R(x)$.

□

Exercise 2.11.12. Provide **rigorous proofs** for the following equivalences.

- a. $\forall x, (P(x) \wedge Q(x)) \iff (\forall x, P(x) \wedge \forall x, Q(x))$
- b. $\exists x, (P(x) \vee Q(x)) \iff (\exists x, P(x) \vee \exists x, Q(x))$
- c. $\forall x, \forall y, R(x, y) \iff \forall y, \forall x, R(x, y)$
- d. $\exists x, \exists y, R(x, y) \iff \exists y, \exists x, R(x, y)$
- e. $\neg \forall x, P(x) \iff \exists x, \neg P(x)$
- f. $\neg \exists x, P(x) \iff \forall x, \neg P(x)$

□

Exercise 2.11.13. Provide **rigorous proofs** for the following implications.

- a. $\forall x, P(x) \vee \forall x, Q(x) \implies \forall x, (P(x) \vee Q(x))$
- b. $\exists x, (P(x) \wedge Q(x)) \implies \exists x, P(x) \wedge \exists x, Q(x)$
- c. $\exists x, \forall y, R(x, y) \implies \forall y, \exists x, R(x, y)$

In each of the cases, give an example of $P(x), Q(x), R(x, y)$ such that the reverse implication fails. (Prove rigorously that it fails.)

□

Exercise 2.11.14. Let $P(x)$ be an arbitrary predicate. Prove **rigorously** that:

$$\exists x, \forall y, (P(x) \implies P(y)).$$

□

3 Basic Mathematics

3.1 Recursion

Example 3.1.1. The standard example of ordinary recursion is the **Fibonacci sequence**, which is defined as the unique sequence $(f_n : n \in \mathbb{N})$ of natural numbers such that:

- $f_0 = 0$;
- $f_1 = 1$;
- for all $n \in \mathbb{N}$, $f_{n+2} = f_{n+1} + f_n$.

Why is this an acceptable definition? In other words, why do these condition describe a unique sequence of natural numbers? (See also Example 2.10.6.) $\square$

Theorem 3.1.2 (Recursion). *Suppose:*

1. E is a set;
2. $H : E^{<\mathbb{N}} \rightarrow E$.

Then there is a unique sequence $(a_n : n \in \mathbb{N}) \in E^{\mathbb{N}}$ such that for all $n \in \mathbb{N}$,

$$a_n = H((a_i : i < n)).$$

Proof. Uniqueness. Let a and b be two such sequences. Assume towards contradiction that $a \neq b$. Then there is $n \in \mathbb{N}$ such that $a_n \neq b_n$. Pick the least such n . Then

$$(a_i : i < n) = (b_i : i < n),$$

whence

$$a_n = H((a_i : i < n)) = H((b_i : i < n)) = b_n;$$

this is a contradiction.

Existence. For every $n \in \mathbb{N}$, we call an n -tuple $(e_i : i < n) \in E^n$ *good* iff for all $k < n$,

$$e_k = H((e_i : i < k)).$$

We denote $P(n)$: “There is a unique good n -tuple.” and prove by induction that for all $n \in \mathbb{N}$, $P(n)$ holds.

Induction basis. The property $P(0)$ is true because the empty tuple is indeed the unique good 0-tuple.

Induction step. Let $n \in \mathbb{N}$ be such that $P(n)$ holds and let us prove $P(n+1)$.

Uniqueness. Let $(e_i : i < n+1)$ and $(f_i : i < n+1)$ be two good $(n+1)$ -tuples. Then the n -tuples $(e_i : i < n)$ and $(f_i : i < n)$ are also good, whence

$$(e_i : i < n) = (f_i : i < n)$$

(by $P(n)$). Finally,

$$e_n = H((e_i : i < n)) = H((f_i : i < n)) = f_n,$$

whence

$$(e_i : i < n + 1) = (f_i : i < n + 1).$$

Existence. By $P(n)$, there is a good n -tuple $(e_i : i < n)$. Let $e_n := H((e_i : i < n))$. Then the $(n + 1)$ -tuples $(e_i : i < n + 1)$ is also good.

Conclusion. The induction yields that for all $n \in \mathbb{N}$, $P(n)$ holds. For a given $n \in \mathbb{N}$, let $(e_i^n : i < n)$ be the unique good n -tuple. Now, for all natural numbers $m \leq n$, we have that

$$(e_i^n : i < m)$$

is a good m -tuple, whence

$$(e_i^n : i < m) = (e_i^m : i < m).$$

Letting $a := (e_n^n : n \in \mathbb{N}) \in E^{\mathbb{N}}$, we see that for all $n \in \mathbb{N}$,

$$(a_i : i < n) = (e_i^n : i < n).$$

We now conclude that a is as required: for all $n \in \mathbb{N}$,

$$a_n = e_n^{n+1} = H((e_i^{n+1} : i < n)).$$

□

Example 3.1.3. Let us show how to use the recursion theorem to prove that the Fibonacci sequence is well-defined. Let $E := \mathbb{N}$ and let

$$H : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{N}$$

be defined, for a given tuple $(e_i : i < n) \in \mathbb{N}^n$, as follows:

- if $n = 0$, then $H((e_i : i < n)) := 0$;
- if $n = 1$, then $H((e_i : i < n)) := 1$;
- if $n \geq 2$, then $H((e_i : i < n)) := e_{n-1} + e_{n-2}$.

Now, for all sequences $a = (a_n : n \in \mathbb{N}) \in \mathbb{N}^{\mathbb{N}}$, we have that a satisfies the conditions for the Fibonacci sequence if and only if for all $n \in \mathbb{N}$,

$$a_n = H((a_i : i < n)).$$

On the other hand, the recursion theorem yields that there is exactly one such sequence. □

Exercise 3.1.4.

a. Prove that there is a unique function $f : \mathbb{N} \rightarrow \text{Nat}$ with the following properties:

- $f(0) = \emptyset$;
- for all $n \in \mathbb{N}$, $f(n + 1) = f(n) + “|”$.

b. Prove that f identifies monoids $(\mathbb{N}, +, 0)$ and $(\text{Nat}, +, \emptyset)$ in the following precise sense:⁴

- f is a bijection from $\mathbb{N}$ to Nat ;
- for all $m, n \in \mathbb{N}$, $f(m + n) = f(m) + f(n)$;
- $f(0) = \emptyset$.

Notation 3.1.5. Let A and B be two sets. We denote by $\mathcal{F}(A, B)$ the set of all function from A to B . □

Exercise 3.1.6. Prove that there is a unique sequence

$$(\sigma_n : n \in \mathbb{N})$$

such that:

- for all $n \in \mathbb{N}$, we have $\sigma_n : \mathbb{R}^n \rightarrow \mathbb{R}$;
- $\sigma_0(\emptyset) = 0$;
- for all $n \in \mathbb{N}$ and all $(a_i : i < n + 1) \in \mathbb{R}^{n+1}$, we have

$$\sigma_{n+1}((a_i : i < n + 1)) = \sigma_n((a_i : i < n)) + a_n.$$

□

Notation 3.1.7. Given the sequence $(\sigma_n : n \in \mathbb{N})$ of the previous exercise, for all

$$(a_i : i < n) \in \mathbb{R}^{<\mathbb{N}},$$

we define

$$\sum_{i < n} a_i := \sigma_n((a_i : i < n)).$$

□

Exercise 3.1.8. Give a rigorous definition for

$$\prod_{i < n} a_i,$$

where $(a_i : i < n) \in \mathbb{R}^{<\mathbb{N}}$. □

3.2 Orders

Definition 3.2.1. Let E be a set and let R be a binary relation on E . We say that R is an **order relation** on E iff the following conditions are satisfied:

a. R is reflexive, i.e.

$$\forall x \in E, \quad xRx;$$

⁴We also say that f is an **isomorphism of monoids**.

b. R is anti-symmetric, i.e.

$$\forall x, y \in E, (xRy \wedge yRx \implies x = y);$$

c. R is transitive, i.e.

$$\forall x, y, z \in E, (xRy \wedge yRz \implies xRz).$$

□

Example 3.2.2. The relation $\leq$ is an order relation on $\mathbb{N}$; it is considered *the standard order relation* on $\mathbb{N}$. It satisfies:

$$\forall m, n \in \mathbb{N}, (m \leq n \iff \exists k \in \mathbb{N}, n = m + k).$$

Remark 3.2.3. $\leq$ is not the only natural order relation on $\mathbb{N}$. For example, we may consider its multiplicative analog. □

Exercise 3.2.4. Prove that $|$ is an order relation on $\mathbb{N}$, where $|$ is defined by:

$$\forall m, n \in \mathbb{N}, (m|n \iff \exists k \in \mathbb{N}, n = mk).$$

Exercise 3.2.5. Let $|$ be the binary relation on $\mathbb{Z}$ such that:

$$\forall m, n \in \mathbb{N}, (m|n \iff \exists k \in \mathbb{Z}, n = mk).$$

Is $|$ an order relation on $\mathbb{Z}$? □

Exercise 3.2.6. Let E be a set and let R be the binary relation on $\mathcal{P}(E)$ such that

$$\forall X, Y \in \mathcal{P}(E), (XRY \iff X \subseteq Y).$$

Prove that R is an order relation on $\mathcal{P}(E)$. □

Notation 3.2.7. We denote the relation R from the previous exercise also by $\subseteq$ and we say: “ $\subseteq$ is an order relation on $\mathcal{P}(E)$ ”. □

Exercise 3.2.8. Let E be a set and let R be an order relation on it. Let $F \subseteq E$ and let $R \upharpoonright F$ be the relation on F such that

$$\forall x, y \in F, (x(R \upharpoonright F)y \iff xRy).$$

Prove that $R \upharpoonright F$ is an order relation on F . □

Notation 3.2.9. Very often we write R for $R \upharpoonright F$ and say: “ R is an order relation on F ”. □

Definition 3.2.10. Let R be an order relation on E . Then we define R^{-1} to be the binary relation on E such that:

$$\forall x, y \in E, (xR^{-1}y \iff yRx).$$

□

Example 3.2.11.

- On $\mathbb{N}$, we have $\leq^{-1} = \geq$.

- On $\mathcal{P}(E)$, we have $\subseteq^{-1} = \supseteq$. □

Exercise 3.2.12. Let R be an order relation on E . Prove that R^{-1} is also an order relation on E . □

Definition 3.2.13. Let E be a set and let R be an order relation on it. We say that R is **total** iff

$$\forall x, y \in E, (xRy \vee yRx).$$
□

Example 3.2.14. $\leq$ is a total order relation on $\mathbb{N}$. □

Exercise 3.2.15. Is $|$ a total order relation on $\mathbb{N}$? □

Exercise 3.2.16. Let E be a set. Characterize exactly when $\subseteq$ is a total order relation on $\mathcal{P}(E)$. □

Exercise 3.2.17. Let E be a set and let R be a total order relation on it. Prove that R^{-1} is also total. □

Exercise 3.2.18. Let E and F be sets and let R and S be total order relations on E and F , respectively. Denote by T the binary relation on $E \times F$ such that⁵

$$\forall (x_0, y_0), (x_1, y_1) \in E \times F, ((x_0, y_0)T(x_1, y_1) \iff x_0Rx_1 \wedge y_0Sy_1).$$

Prove that T is a total order relation on $E \times F$. □

Notation 3.2.19. The **strict-order relation** associated to an order relation R on E is the binary relation R° on E such that:

$$\forall x, y \in E, (xR^\circ y \iff xRy \wedge x \neq y).$$
□

Example 3.2.20.

- On $\mathbb{N}$, we have $\leq^\circ = <$.
- Let E be an arbitrary set. Then on $\mathcal{P}(E)$, we have $\subseteq^\circ = \subset$. □

Terminology 3.2.21. An **order** is a pair (E, R) , where E is a set and R is an order relation on E . A **total order** is an order (E, R) such that the order relation R is total. □

Definition 3.2.22. Let (E, R) be an order.

- A **minimal element** of (E, R) is any element $x \in E$ such that

$$\neg \exists y \in E, yR^\circ x.$$

- A **maximal element** of (E, R) is any element x such that

$$\neg \exists y \in E, xR^\circ y.$$

□

⁵This is the so-called **lexicographic order relation** on $E \times F$.

Example 3.2.23. 0 is the unique minimal element of $(\mathbb{N}, \leq)$ and there are no maximal elements. Furthermore, for all non-empty sets $X \subseteq \mathbb{N}$, we have that:

- $(X, \leq)$ has a unique minimal element;
- $(X, \leq)$ has a maximal element if and only if X is finite. □

Exercise 3.2.24. Identify all minimal and maximal elements of $(\mathbb{N}, |)$. □

Exercise 3.2.25. Identify all minimal and maximal elements of $(\mathbb{N} - \{0, 1\}, |)$. □

Exercise 3.2.26. Let E be a set. Identify all minimal and maximal elements of $(\mathcal{P}(E), \subseteq)$. □

Exercise 3.2.27. Let E be a non-empty set. Identify all minimal and maximal elements of $(\mathcal{P}(E) - \{\emptyset, E\}, \subseteq)$. □

Exercise 3.2.28. Let (E, R) be a total order. Prove that there is at most one minimal and at most one maximal element. □

Exercise 3.2.29. Let (E, R) be an order and let $x \in E$. Prove:

- a. x is a minimal element of (E, R) if and only if it is maximal of (E, R^{-1}) ;
- b. x is a maximal element of (E, R) if and only if it is minimal of (E, R^{-1}) . □

Definition 3.2.30. Let (E, R) be an order and let $x \in E$. Then:

- x is a **minimum** of (E, R) iff $\forall y \in E, \quad xRy$;
- x is a **maximum** of (E, R) iff $\forall y \in E, \quad yRx$.

Exercise 3.2.31. Let (E, R) be an order and let $a \in E$. Prove:

- a is a minimum of (E, R) if and only if it is a maximum of (E, R^{-1}) ;
- a is a maximum of (E, R) if and only if it is a minimum of (E, R^{-1}) . □

Exercise 3.2.32. Let (E, R) be an order. Prove that (E, R) has at most one minimum and at most one maximum. □

Terminology 3.2.33. Suppose a is a minimum of (E, R) . Then we say that a is the minimum of (E, R) . Similarly for the maximum. □

Exercise 3.2.34. Let (E, R) be an order and let a be its minimum. Prove that a is the unique minimal element of (E, R) . □

Exercise 3.2.35. Let (E, R) be an order. Suppose $a \in E$ is a unique minimal element of (E, R) . Does it follow that a is the minimum of (E, R) . □

Exercise 3.2.36. Let (E, R) be a *total* order and let $a \in E$. Prove that a is a minimal element of (E, R) if and only if a is the minimum of (E, R) . $\square$

Terminology 3.2.37. The minimum of (E, R) is also called its *least element*; its maximum is called the *greatest element*. $\square$

Definition 3.2.38. Let (E, R) be an order.

- We say that (E, R) is *well-founded* iff for all non-empty subsets A of E , the order (A, R) has a minimal element.
- We say that (E, R) is *ill-founded* iff it is not well-founded.
- We say that (E, R) is a *well-order* iff it is total and well-founded. $\square$

Exercise 3.2.39. Prove that an order (E, R) is a well-order if and only if for every non-empty subset A of E , the order (A, R) has a *minimum*. $\square$

Example 3.2.40. The order $(\mathbb{N}, \leq)$ is a well-order. For all $n \in \mathbb{N}$, the order $([n], \leq)$ is also well-order. On the other hand, the order $(\mathbb{N}, \geq)$ is not a well-order, because it is ill-founded. $\square$

Exercise 3.2.41.

- Prove that the order $(\mathbb{N}, |)$ is well-founded.
- Let E be a finite set. Prove that the order $(\mathcal{P}(E), \subseteq)$ is well-founded.
- Prove that the order $(\mathcal{P}(\mathbb{N}), \subseteq)$ is ill-founded. $\square$

Exercise 3.2.42. Prove that every finite order is well-founded. $\square$

Exercise 3.2.43. Let (X, R) be a well-founded order and $A \subseteq X$. Prove that (A, R) is a well-founded order. $\square$

Exercise 3.2.44. Prove that an order (E, R) is ill-founded if and only if there is a sequence $\vec{a} \in E^{\mathbb{N}}$ such that

$$\forall i \in \mathbb{N}, a_{i+1} R^\circ a_i.$$

$\square$

Comment. If the relation R is denoted by $\leq$, then we are asserting in the second part that there is a sequence

$$a_0 > a_1 > a_2 > \dots.$$

$\square$

Exercise 3.2.45 (Xiang Li; induction for well-founded orders). Let (E, R) be a well-founded order and let $P(x)$ be a property. Assume that for all $x \in E$, the following holds:

$$(\forall y \in E)(y R^\circ x \implies P(y)) \implies P(x).$$

Prove:

$$\forall x \in E, P(x).$$

$\square$

Exercise 3.2.46 (Xiang Li; recursion for well-founded orders). *In this exercise, we are using the following notation:*

- for sets A and B , the set $\mathcal{F}\mathcal{P}(A, B)$ consists of all partial functions from A to B , i.e. functions f such that $\text{dom}(f) \subseteq A$ and $\text{ran}(f) \subseteq B$;
- for an order (E, R) and an element a of E , we denote by $R \downarrow a$ the set $\{b \in E : bR^\circ a\}$.

Now, suppose:

1. (E, R) is a well-founded order;
2. S is an arbitrary set;
3. $H : E \times (\mathcal{F}\mathcal{P}(E, S)) \rightarrow S$.

Prove that there is a unique function $F : E \rightarrow S$ such that

$$\forall a \in E, F(a) = H(a, F \upharpoonright (R \downarrow a)).$$

□

Axiom of Choice. For every set X , there exists a well-order on X .

□

Remark 3.2.47. The above formulation of the Axiom of Choice is known as the **well-ordering principle**. Usually, one formulates the axiom of choice via the existence of **choice functions**. The two formulations are, in fact, equivalent (which is a theorem due to Zermelo). □

Definition 3.2.48. A **choice function** for a set X is a function

$$f : \mathcal{P}(X) - \{\emptyset\} \rightarrow X$$

such that for every $A \in \mathcal{P}(X) - \{\emptyset\}$, it holds that $f(A) \in A$.

□

Exercise 3.2.49. Prove that every set has a choice function. □

Exercise 3.2.50. A **directed graph** is a pair (V, E) , where:

- V is a set;
- E is a binary relation on V ;
- E is **irreflexive**, i.e.

$$\forall v \in V, \neg(vEv).$$

Let $G = (V, E)$ be a directed graph.

- A **path** in G is a sequence $p \in V^{<\mathbb{N}}$ such that $\text{len}(p) \leq 1$ and

$$\forall i < \text{len}(p), (p_i, p_{i+1}) \in E.$$

- A **cycle** in G is a sequence $p \in V^{<\mathbb{N}}$ such that $\text{len}(p) \leq 2$,

$$\forall i < \text{len}(p), (p_i, p_{i+1}) \in E,$$

and the first and the last element of the sequence p are equal.

- The binary relation E^* on V is defined by: for all $u, v \in V$, it holds uE^*v iff there is a path in G whose first element is u and whose last element is v .
- The graph G is **acyclic** iff it does not contain a cycle.

a. Let $G = (V, E)$ be a directed graph. Prove that E^* is an order relation on V if and only if G is acyclic.

b. Let (V, R) be a finite order. Prove that the set

$$\{E \subseteq R : (V, E) \text{ is a directed graph and } E^* = R\}$$

has a minimum, when ordered by $\subseteq$.

Comments. One way this fact can be useful is when we have to give examples of finite ordered sets. Namely, to do so, one only needs to draw an acyclic directed graph. Furthermore, the exercise is saying that this is really the only way to give a finite order. $\square$

3.3 Equivalence Relations

Definition 3.3.1. An **equivalence relation** on a set X is a binary relation E on X which is:

- **reflexive:**

$$\forall x \in X, xEx;$$

- **symmetric:**

$$\forall (x, y) \in X^2, (xEy \implies yEx);$$

- **transitive:**

$$\forall (x, y, z) \in X^3, (xEy \wedge yEz \implies xEz).$$

$\square$

Example 3.3.2. Let X be a set. Consider the following binary relation E on X :

$$\forall (x, y) \in X^2, (xEy \iff x = y).$$

Then E is an equivalence relation on X . We denote it by $=_X$, or simply $=$, if there is no risk of confusion. $\square$

Exercise 3.3.3. Let X be a set and let R be a binary relation on X . Prove that the following are equivalent:

- R is both an order relation and equivalence relation;

- R is equal to $=_X$. □

Definition 3.3.4. Let $n \in \mathbb{N} - \{0\}$. The **congruence modulo n** is the binary relation R on $\mathbb{Z}$ defined as follows:

$$\forall (x, y) \in \mathbb{Z}^2, (xRy \iff n|(x - y)).$$
□

Notation 3.3.5. We write $x \equiv y \pmod{n}$ or $x \equiv_n y$ for xRy . □

Exercise 3.3.6. Let $n \in \mathbb{N} - \{0\}$. Prove that $\equiv_n$ is an equivalence relation on $\mathbb{Z}$. □

Definition 3.3.7. Let X be a set and let E be an equivalence relation on it. Then $[x]_E$ denotes the equivalence class of $x \in X$, and is defined to be:

$$[x]_E := \{y \in X : yEx\}.$$
□

Notation 3.3.8. When E is clear from context, we write $[x]$ instead of $[x]_E$. □

Terminology 3.3.9. A set C is an **equivalence class** of E iff there is $x \in X$ such that $C = [x]_E$. In that case, x is called a **representative** of C . □

Remark 3.3.10. For all $x, y \in X$:

$$\begin{aligned} x &\in [x]_E; \\ xEy &\iff x \in [x]_E. \end{aligned}$$
□

Example 3.3.11. For all $x \in X$, we have

$$[x]_{=_X} = \{x\}.$$
□

Exercise 3.3.12. Let $n > 0$. How many equivalence classes does $\equiv_n$ have? What are they? □

Exercise 3.3.13. Let X be a set and let E be an equivalence relation on it. Prove that for all $x, y \in X$:

$$xEy \iff [x] = [y].$$
□

Definition 3.3.14. Let X be a set and let E be an equivalence relation on it. We denote by X/E the set of all equivalence classes:

$$X/E := \{[x]_E : x \in X\}.$$

This is called the **quotient** of X by E . □

Definition 3.3.15. Let X be a set and $\mathcal{U} \subseteq \mathcal{P}(X)$. We say that $\mathcal{U}$ is a partition of X iff:

- a. every $A \in \mathcal{U}$ is non-empty;
- b. for all $A, B \in \mathcal{U}$, we have $A \cap B = \emptyset$;
- c. $\bigcup \mathcal{U} = X$. □

Exercise 3.3.16. For any partition $\mathcal{U}$ of a set X , we denote by $\equiv_{\mathcal{U}}$ the binary relation on X given by

$$\forall x, y \in X, (x \equiv_{\mathcal{U}} y \iff (\exists U \in \mathcal{U})(x, y \in U)).$$

Let us fix a set X , an equivalence relation on E it, and a partition $\mathcal{U}$ of X . Prove:

- a. X/E is a partition of X and $\equiv_{X/E} = E$;
- b. $\equiv_{\mathcal{U}}$ is an equivalence relation on X and $X/\equiv_{\mathcal{U}} = \mathcal{U}$. □

Notation 3.3.17. Let X be a set and let E be an equivalence relation on it. We denote by π_E the canonical projection:

$$\pi_E : X \rightarrow X/E : x \mapsto [x]_E.$$

□

Exercise 3.3.18. For a function $f : X \rightarrow Y$, we denote by $\equiv_f$ the binary relation on X given by:

$$\forall x, y \in X, (x \equiv_f y \iff f(x) = f(y)).$$

Let us fix a set X , an equivalence relation E on it, and a function $f : X \rightarrow Y$. Prove:

- a. $\equiv_{\pi_E} = E$;
- b. $\equiv_f$ is an equivalence relation on X ;
- c. the following are equivalent:
 - i. there is a function $f_* : X/E \rightarrow Y$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow \pi_E & \nearrow f_* & \\ X/E & & \end{array}$$

commutes;

- ii. for all $E \subseteq \equiv_f$;
- d. if $E \subseteq \equiv_f$, then there is a **unique** function $f_* : X/E \rightarrow Y$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow \pi_E & \nearrow f_* & \\ X/E & & \end{array}$$

commutes;

e. if $E \subseteq \equiv_f$, then:

- i. f_* is injective if and only if $E = \equiv_f$
- ii. f_* is surjective if and only if f is surjective. □

4 Syntax

4.1 Strings

Notation 4.1.1. A “character” is a basic notion in course. We will denote by Char the set of all characters (that we will use in this course). When there is a risk of confusion, we will delimit a character by the apostrophes (‘ ’).

Example 4.1.2. Consider the following sentences:

- a. x is a natural number.
- b. ‘ x ’ is a natural number.

The first sentence may be true or false depending on the value that the variable x takes. However, the second sentence is false: ‘ x ’ is not a natural number, but a character.

Definition 4.1.3. Let n be a natural number. Then a **string** is of length n is a function

$$s : [n] \rightarrow \text{Char}.$$

Notation 4.1.4.

- We denote by Str the set of all strings.
- When there is a risk of confusion, we delimit individual strings by the quotation marks (“ ”).
- The length of a string s is denoted by $\text{len}(s)$.

Remark 4.1.5. Notice that $\text{Str} = \text{Char}^{<\mathbb{N}}$. □

Remark 4.1.6. Note that the domain of a string s is exactly $[\text{len}(s)]$. □

Example 4.1.7. Str is the set of all strings, while “ Str ” is a string. For example: “ Str ”(1) = t and $\text{len}(\text{“Str”}) = 3$.

Example 4.1.8. The character ‘ p_0 ’ will be one of the building blocks of the formal language that we will soon describe. From it, we can build the string “ p_0 ” of length 1, which will be one of the (well formed) sentences of that language. Notice that, formally, ‘ p_0 ’ $\neq$ “ p_0 ”. However, we will often write p_0 and let the reader decide, based on context, which exact meaning we intend.

Definition 4.1.9. Let s be a string of length m and let t be a string of length n . Then **their concatenation** is the string $s + t$ of length $m + n$ such that: for all $i \in [m + n]$,

$$(s + t)(i) := \begin{cases} s(i), & i < m \\ t(i - m), & i \geq m. \end{cases}$$

Notation 4.1.10. We also use the notation st for the concatenation of the string s and t .

Example 4.1.11. We will soon discuss propositional formulas. Given a propositional formula ϕ , we can construct another propositional formula: $(\neg\phi)$. In this last notation, we are being a bit sloppy. What we really mean is:

$$“(\neg” + \phi + “),”$$

where $+$ denotes the concatenation.

Notation 4.1.12. Let s be a string and let $i \leq j \leq \text{len}(s)$. We denote by $s[i : j]$ the string

$$[j - i] \rightarrow \text{Char} : k \mapsto s(i + k).$$

□

Definition 4.1.13. Let s and t be strings. We say that s is a **substring** of t iff there are $i \leq j \leq \text{len}(t)$ such that $s = t[i : j]$. □

4.2 Grammars

Definition 4.2.1. Let $n \in \mathbb{N}$. An n -ary **grammar rule** is a function

$$R : \text{Str}^n \rightarrow \text{Str}.$$

Notation 4.2.2. An n -ary grammar rule R is often denoted by

$$\frac{s_0 \ s_1 \ \cdots \ s_{n-1}}{R(s_0, s_1, \dots, s_{n-1})},$$

especially when the value $R(s_0, s_1, \dots, s_{n-1})$ can be described as an expression depending on $s_0, \dots, s_{n-1}$ (which is most of the cases).

Example 4.2.3. We will soon encounter the rule:

$$\frac{\phi \ \psi}{(\phi \wedge \psi)}.$$

The intended meaning behind it is: “If ϕ and ψ are well formed propositional formulas, then $(\phi \wedge \psi)$ is also a well formed propositional formula.”

Remark 4.2.4. The arity of a grammar rule can be 0, in which case it is written

$$\frac{}{t}.$$

Here, t is a fixed string, so the rule can be identified with t . The intended meaning is: “The string t is a well-formed sentence of the language under consideration.”

Definition 4.2.5. A **grammar** is a set of grammar rules.

Definition 4.2.6. Let G be a grammar and let X be a set of strings. We say that X **respects** G iff for every $n \in \mathbb{N}$, for all $R \in G$ of arity n and for all $s_0, \dots, s_{n-1} \in X$, we have that

$$R(s_0, \dots, s_{n-1}) \in X.$$

□

Example 4.2.7. The set Str respects every grammar.

Fact 4.2.8. Let G be a grammar and let $\mathcal{X}_G$ be the set of all sets of strings X that respect G . Then $(\mathcal{X}_G, \subseteq)$ has a minimum.

Proof. Let $X_G := \bigcap \mathcal{X}_G$. We will show that X_G is the minimum of the set $\mathcal{X}_G$. More precisely, we will prove that:

- a. $X_G \in \mathcal{X}_G$;
- b. for all $X \in \mathcal{X}_G$, we have that $X_G \subseteq X$.

Part a. Let $n \in \mathbb{N}$, $R \in G$, and $s_0, \dots, s_{n-1} \in X_G$. We want to see that $R(s_0, \dots, s_{n-1}) \in X_G$. Let $X \in \mathcal{X}_G$ be given, We want to see that $R(s_0, \dots, s_{n-1}) \in X$.

Since $s_0, \dots, s_{n-1} \in X_G$, we have that $s_0, \dots, s_{n-1} \in X$. Since X respects G , we conclude $R(s_0, \dots, s_{n-1}) \in X$.

Part b. This is obvious: since $X \in \mathcal{X}_G$, we have:

$$X \supseteq \bigcap \mathcal{X}_G = X_G.$$

□

Definition 4.2.9. Let G be a grammar. The **language generated** by G is the least set of strings that respects G .

Exercise 4.2.10 (Xiang Li). Let G be a grammar.

a. Prove that there is a unique sequence $(Y_n : n < \omega)$ such that:

- $Y_0 = \emptyset$;
- $Y_{n+1} = Y_n \cup \{R(\vec{s}) : R \in G, \vec{s} \in Y_n^{\text{ar}(R)}\}$.

b. Prove that the set

$$\bigcup_{n < \omega} Y_n$$

is equal to the language generated by G .

□

4.3 Propositional Logic

Terminology 4.3.1. A **set of propositional letters** is any infinite set of characters that does not contain the characters: ' $\wedge$ ', ' $\vee$ ', ' $\neg$ ', ' $\implies$ ', ' $\iff$ ', ' $\perp$ ', '(', ')'. $\square$

Definition 4.3.2. Let $\mathbb{P}$ be a set of propositional letters. The **vocabulary of propositional logic** with propositional letters $\mathbb{P}$ is the set of the following characters:

- the **propositional letters** of $\mathbb{P}$;
- the **logical connectives**: ' $\wedge$ ', ' $\vee$ ', ' $\neg$ ', ' $\implies$ ', ' $\iff$ ', ' $\perp$ ';
- the **auxiliary characters**: '(', ')'.

Terminology 4.3.3.

- $\wedge$ is called **conjunction** and read *and*.
- $\vee$ is called **disjunction** and read *or*.
- $\neg$ is called **negation** and read *not*.
- $\implies$ is called **implication** and read *if..., then...*
- $\iff$ is called **equivalence** and read *if and only if*.
- $\perp$ is called **contradiction** and read *bottom*.

Definition 4.3.4. Let $\mathbb{P}$ be a set of propositional letters. The **grammar of propositional logic** (over $\mathbb{P}$) is the grammar consisting of the following rules:

- for all $p \in \mathbb{P}$: $\frac{}{p}$;
- $\frac{}{\perp}$;
- $\frac{\phi}{(\neg\phi)}$;
- $\frac{\phi \quad \psi}{(\phi\wedge\psi)}$;
- $\frac{\phi \quad \psi}{(\phi\vee\psi)}$;
- $\frac{\phi \quad \psi}{(\phi\implies\psi)}$;
- $\frac{\phi \quad \psi}{(\phi\iff\psi)}$.

Definition 4.3.5. Let $\mathbb{P}$ be a set of propositional letters. The set of **propositional formulas** over $\mathbb{P}$, denoted $\text{Prop}_{\mathbb{P}}$, is the language generated by the grammar for propositional logic.

Example 4.3.6. Obviously: for all $p \in \mathbb{P}$, p is a propositional formula. Also, $\perp$ is a propositional formula. These propositional formulas are called **atomic**.

Example 4.3.7. Every propositional formula uses only characters from the vocabulary of propositional logic. Indeed, this follows from the fact that the set of all strings which are made up of characters from the vocabulary of propositional logic (with no other restrictions imposed) respects the grammar of propositional logic.

Notation 4.3.8. Unless said otherwise, we will assume that

$$\mathbb{P} = \{p_i : i \in \mathbb{N}\},$$

where p_i ($i \in \mathbb{N}$) are fixed characters playing the role of propositional letters. In this case, we write **Prop** instead of longer $\text{Prop}_{\mathbb{P}}$. $\square$

Example 4.3.9. “ $(p_7 \implies p_0)$ ” $\in \text{Prop}$

Proof. Let X be an arbitrary set of strings that respects the grammar of propositional logic. We want to see that $(p_7 \implies p_0) \in X$. Since X respects the rule

$$\frac{}{p_7},$$

we get $p_7 \in X$. Since it also respects the rule

$$\frac{}{p_0},$$

we get $p_0 \in X$. Finally, since it respects the rule

$$\frac{\phi \quad \psi}{(\phi \implies \psi)},$$

we conclude $(p_7 \implies p_0) \in X$. $\square$

Example 4.3.10. “ $p_7 \implies p_0$ ” $\notin \text{Prop}$

Proof. Assume towards contradiction that $p_7 \implies p_0 \in \text{Prop}$. Let $X := \text{Prop} - \{p_7 \implies p_0\}$ and let us prove that X respects the grammar of propositional logic.

Let $i \in \mathbb{N}$ and let us prove that $p_i \in X$. We already observed that $p_i \in \text{Prop}$. Since $\text{len}(p_i) = 1$ and

$$\text{len}(p_7 \implies p_0) = 3,$$

these two strings are not equal, so $p_i \in X$.

We similarly prove that $\perp \in X$.

Now, let $\phi \in X$ and let us prove that $(\neg\phi) \in X$. Since $X \subseteq \text{Prop}$, we get $\phi \in \text{Prop}$, whence $(\neg\phi) \in \text{Prop}$. On the other hand, the characters at the position 0 of “ $(\neg\phi)$ ” and “ $p_7 \implies p_0$ ” are ‘(’ and ‘ p_7 ’, respectively. This shows that these two strings are different, so $(\neg\phi) \in X$.

Finally, let $\phi, \psi \in X$ and let

$$\square \in \{\wedge, \vee, \implies, \iff\}.$$

We want to show that $(\phi\square\psi) \in X$. Since $X \subseteq \text{Prop}$, we get $\phi, \psi \in \text{Prop}$, whence $(\phi\square\psi) \in \text{Prop}$. On the other hand, the strings $(\phi\square\psi)$ and $p_7 \implies p_0$ begin with ‘(’ and ‘ p_7 ’, respectively, whence $(\phi\square\psi) \in X$.

This shows that X respects the grammar of propositional logic, whence $\text{Prop} \subseteq X$. This is obviously a contradiction. $\square$

Exercise 4.3.11. *Prove:*

- a. “ $((\perp \vee p_3) \wedge (\neg p_2))$ ” $\in$ Prop;
- b. “ $((\implies \wedge)$ ” $\notin$ Prop;
- c. $\emptyset \notin$ Prop.

4.4 Formation Sequences

Notation 4.4.1. Let R be a grammar rule. We denote by $\text{ar}(R)$ its arity, i.e.

$$R : \text{Str}^{\text{ar}(R)} \rightarrow \text{Str}.$$

Definition 4.4.2. Let G be a grammar. Then a **formation sequence** according to G is a sequence

$$(s_0, s_1, \dots, s_{n-1})$$

of strings (where $n \in \mathbb{N}$) such that: for all $k < n$, there are $R \in G$ and $i_0, \dots, i_{\text{ar}(R)-1} < k$ such that

$$s_k = R(s_{i_0}, \dots, s_{i_{\text{ar}(R)-1}}).$$

□

Example 4.4.3. Here are two formation sequences in the context of propositional formulas:

- a. $(\perp, p_2, p_3, (\perp \vee p_2), (\neg(\perp \vee p_2)), (\neg p_3))$
- b. $(p_3, (\neg p_3))$.

We think of both sequence as “ways to form $(\neg p_3)$ ”, but the first one contains “unnecessary parts”.

Remark 4.4.4. Every initial segment of every formation sequence is also a formation sequence. Also, the concatenation of finitely many formation sequences is a formation sequence. □

Terminology 4.4.5. We say that a string s **has a formation sequence** in a grammar G iff s appears in some formation sequence according to G . □

Remark 4.4.6. A string s has a formation sequence if and only if it is the last string in some formation sequence. □

Fact 4.4.7. *Let G be a grammar and let s be a string. Then s belongs to the language generated by G if and only if s has a formation sequence according to G .*

Proof. Let X be the language generated by G and let Y be the set of all strings that have a formation sequence. We reason by double implication.

Direct implication. We are showing $X \subseteq Y$, for which it suffices to prove that Y respects G . Let $R \in G$ and let $s_0, \dots, s_{\text{ar}(R)-1} \in Y$. We want to show that $s := R(s_0, \dots, s_{\text{ar}(R)-1})$ belongs to Y . For every $i < \text{ar}(R)$, since $s_i \in Y$, we can pick a formation sequence

$$(s_0^i, \dots, s_{m_i}^i)$$

such that $s_{m_i}^i = s_i$. Then the sequence

$$(s_0^0, \dots, s_{m_0}^0, s_0^1, \dots, s_{m_1}^1, \dots, s_0^{\text{ar}(R)-1}, \dots, s_{m_{\text{ar}(R)-1}}^{\text{ar}(R)-1}, s)$$

is a formation sequence ending with s , whence $s \in Y$.

Reverse implication. We will prove by induction that for all $n \in \mathbb{N}$ the property $P(n)$: “Every string s that appears in a formation sequence of length n belongs to X .” holds.

Let $n \in \mathbb{N}$ and let us assume that for all $k < n$, $P(k)$ holds. We want to prove that $P(n)$ holds. Let s be a string that appears in a formation sequence of length n and let us prove that $s \in X$.

Pick one such formation sequence $(s_i : i < n)$; so there is $k < n$ such that $s_k = s$. Now, there are $R \in G$ and $i_0, \dots, i_{\text{ar}(R)-1} < k$ such that

$$s = s_k = R(s_0, \dots, s_{\text{ar}(R)-1}).$$

Let $j < \text{ar}(R)$. We have that s_{i_j} appears in a formation sequence of length $i_j + 1$, namely: $(s_0, \dots, s_{i_j})$. Since $i_j + 1 < n$, we get that $P(i_j + 1)$ is true, whence $s_{i_j} \in X$.

Therefore, for all $s_0, \dots, s_{\text{ar}(R)-1} \in X$. Since X respects G , we conclude that

$$s = R(s_0, \dots, s_{\text{ar}(R)-1}) \in X.$$

□

4.5 Structural Induction

Theorem 4.5.1 (Structural induction). *Let G be a grammar and let $A(s)$ be a property depending on the variable s . Suppose that for all $R \in G$ and all $s_0, \dots, s_{\text{ar}(R)-1} \in \text{Str}$, if for all $i < \text{ar}(R)$, $A(s_i)$ holds, then $A(R(s_0, \dots, s_{\text{ar}(R)-1}))$ holds. Then for every string s in the language generated by G , the property $A(s)$ holds.*

Proof. Let X_G be the language generated by G and let

$$X := \{s \in X_G : A(s)\}.$$

We will prove that X respects G .

Let $R \in G$, let $s_0, \dots, s_{\text{ar}(R)-1} \in X$, and let $t := R(s_0, \dots, s_{\text{ar}(R)-1})$. We want to see that $t \in X$. We have that for all $i < \text{ar}(R)$,

$$s_i \in X_G \text{ and } A(s_i).$$

Since X_G respects G , we get $t \in X_G$. Since for all $i < \text{ar}(R)$, the property $A(s_i)$ holds, we get that $A(t)$ holds. Therefore, $t \in X$.

This shows that X respects G . Since $X \subseteq X_G$ and X_G is the least set of strings that respects G , it follows that $X = X_G$. □

Corollary 4.5.2 (Induction on complexity of a formula). *Let $\mathbb{P}$ be a set of propositional letters. Let $A(\phi)$ be a property depending on the variable ϕ . Suppose:*

1. *for all $p \in \mathbb{P}$, $A(p)$;*

2. $A(\perp)$;
3. for all $\phi \in \text{Prop}_{\mathbb{P}}$, if $A(\phi)$ holds, then $A(\neg\phi)$ holds;
4. for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ and all $\square \in \{\wedge, \vee, \implies, \iff\}$, if $A(\phi)$ and $A(\psi)$ hold, then $A((\phi\square\psi))$ holds.

Then for all $\phi \in \text{Prop}_{\mathbb{P}}$, the property $A(\phi)$ holds. □

Example 4.5.3. Let $\mathbb{P}$ be a set of propositional letters. Every propositional formula over $\mathbb{P}$ has an even number of parentheses.

Proof. We prove this by induction on complexity of a formula. More precise, let $A(\phi)$: “ ϕ has an even number of parentheses” and let us prove, using Corollary 4.5.2, that for all $\phi \in \text{Prop}_{\mathbb{P}}$, $A(\phi)$ holds.

1. Let $p \in \mathbb{P}$. Then “ p ” has 0 parentheses, so $A(p)$ holds.
2. The formula “ $\perp$ ” also has 0 parentheses, so $A(\perp)$ holds.
3. Let $\phi \in \text{Prop}_{\mathbb{P}}$ be such that $A(\phi)$ holds. Then ϕ has an even number of parentheses, say n . It follows that $(\neg\phi)$ has $n + 2$ parentheses, which is also an even number. Thus: $A((\neg\phi))$.
4. Let $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ be such that $A(\phi)$ and $A(\psi)$ and let $\square \in \{\wedge, \vee, \implies, \iff\}$. Let m and n be the numbers of parentheses of ϕ and ψ , respectively. Then m and n are even. Now, $(\phi\square\psi)$ has $m + n + 2$ parentheses, which is also an even number. Therefore: $A((\phi\square\psi))$ holds.

It follows by induction on complexity of a formula that $A(\phi)$ holds for all $\phi \in \text{Prop}_{\mathbb{P}}$. □

Definition 4.5.4. A **monoid** is an algebraic structure $(M, *, e)$, where:

- $*$ is a binary operation on M ;
- $e \in M$ (i.e. a 0-ary operation);
- for all $x, y, z \in M$:

$$(x * y) * z = x * (y * z);$$

- for all $x \in M$:

$$x * e = e * x = x.$$

□

Example 4.5.5. The algebraic structures $(\mathbb{N}, +, 0)$ and $(\mathbb{N}, \cdot, 1)$ are monoids. □

Exercise 4.5.6. Let $*$ be the binary operation on $\mathbb{N}$ given by:

$$\forall x, y \in \mathbb{N}, x * y = x^y.$$

Prove that $(\mathbb{N}, *, 1)$ is **not** a monoid. □

Exercise 4.5.7. Recall that $+$ denotes the operation of concatenation of strings. Prove that

$$(\text{Str}, +, \emptyset)$$

is a monoid (where $\emptyset$ denotes the empty string, which is the same thing as the empty set). □

Definition 4.5.8. A monoid $(M, *, e)$ is **commutative** iff for all $x, y \in M$:

$$x * y = y * x.$$

□

Example 4.5.9. Monoids $(\mathbb{N}, +, 0)$ and $(\mathbb{N}, \cdot, 1)$ are commutative. □

Example 4.5.10. The monoid $(\text{Str}, +, \emptyset)$ is not commutative because “ $\wedge$ ”+“(” = “ $\wedge$ ”(“ $\neq$ “(“ $\wedge$ ” = “(”+“ $\wedge$ ”. □

Exercise 4.5.11. Let C be a set of characters, i.e. $C \subseteq \text{Char}$, and let us denote by L the language generated by the following grammar rules:

- $\frac{}{\emptyset}$;
- for every $c \in S$, the rule $\frac{s}{sc}$.

Answer the following questions.

- a. Prove that $\emptyset \in L$.
- b. Prove that for all $s, t \in L$, we have $s + t \in L$.

This means that we may consider the operation

$$L \times L \rightarrow L : (s, t) \mapsto s + t,$$

which we will also denote by $+$.

- c. Prove that $(L, +, \emptyset)$ is a monoid.
- d. Is this monoid commutative? Give an exact characterization of when it is.

Definition 4.5.12. Continuing the scenario of the previous exercise, consider the case where $C = \{‘|’\}$. We call the grammar thus obtained the **counting grammar** and denote by Nat the language generated by it. □

Exercise 4.5.13. Prove that $(\text{Nat}, +, \emptyset)$ is a commutative monoid. □

4.6 Unique Readability

Definition 4.6.1. Let G be a grammar and let us denote by L the language generated by it. We say that G has **unique readability** iff for all $t \in L$, there is a unique pair

$$(R, s) \in G \times L^{<\mathbb{N}}$$

such that $\text{len}(s) = \text{ar}(R)$ and $t = R(s_0, \dots, s_{\text{ar}(R)-1})$. □

Exercise 4.6.2. Let C be a set of characters. We denote by G the grammar consisting of the following rules:

- $\frac{}{\emptyset}$;
- for every $c \in C$, the rule $\frac{s}{sc}$;

and by H the grammar consisting of the following rules:

- $\frac{}{\emptyset}$;
- for every $c \in C$, the rule $\frac{}{c}$;
- $\frac{s \quad t}{st}$.

Prove:

- a. G has unique readability;
- b. H does not have unique readability;
- c. the languages generated by G and H coincide.

Remark 4.6.3. The previous exercise shows that the unique readability is really a property of the grammar, and not of the language generated by it.

Remark 4.6.4. The next goal is to prove that the propositional logic has unique readability.

Exercise 4.6.5. Let $\mathbb{P}$ be a set of propositional letters. Prove that no propositional formula over $\mathbb{P}$ begins with $\neg$. □

Exercise 4.6.6. Let $\mathbb{P}$ be a set of propositional letters. Prove that every propositional formula over $\mathbb{P}$ has the same number of open and closed parentheses. □

Exercise 4.6.7. Let $\mathbb{P}$ be a set of propositional letters. Prove that a strict initial segment of a propositional formula over $\mathbb{P}$ is not a propositional formula.

Exercise 4.6.8. Let $\mathbb{P}$ be a set of propositional letters. Prove that the grammar of propositional logic over $\mathbb{P}$ has unique readability.

Hint. This comes down to showing that for a given propositional formula ϕ , exactly one of the following is true:

- a. there is a unique $p \in \mathbb{P}$ such that $\phi = "p"$;
- b. $\phi = "\perp"$;
- c. there is a unique $\psi \in \text{Prop}_{\mathbb{P}}$ such that $\phi = (\neg\psi)$;
- d. there is a unique triple

$$(\psi, \theta, \square) \in \text{Prop}_{\mathbb{P}} \times \text{Prop}_{\mathbb{P}} \times \{\wedge, \vee, \implies, \iff\}$$

such that $\phi = (\psi \square \theta)$.

□

4.7 Formation Sets

Exercise 4.7.1. Let G be a grammar with unique readability and let L be the language generated by it. For every $s \in L$, denote by n_s the least $n \in \mathbb{N}$ for which there is a formation sequence of length n that contains s . Is it always true that for all $s \in L$, there is a unique formation sequence of length n_s that contains s ? □

Definition 4.7.2. Let G be a grammar. A **formation set** (with respect to G) is a set S of strings such that

$$\forall t \in S, \exists R \in G, \exists (s_i : i < \text{ar}(R)) \in S^{\text{ar}(R)}, t = R(s_0, \dots, s_{\text{ar}(R)-1}).$$

□

Exercise 4.7.3. Let G be a grammar. Prove:

- a. for every formation sequence $(s_i : i < n)$, the set

$$\{s_i : i < n\}$$

is a formation set;

- b. for every **finite** formation set S , there is a formation sequence $(s_i : i < n)$ such that

$$S = \{s_i : i < n\}.$$

□

Exercise 4.7.4. Let G be a grammar and let S be a formation set according to it. Prove that for all $s \in S$, there is a **finite** formation set S' such that $s \in S' \subseteq S$. □

Exercise 4.7.5. Let G be a grammar and let L be the language generated by it. For every $s \in L$, we denote by $\mathcal{F}_s$ the set of all formation sets that contain s . Consider the following propositions.

- a. G has unique readability.
- b. For every $s \in L$, the order $(\mathcal{F}_s, \subseteq)$ has a minimum.

c. Every $s \in L$, the order $(\mathcal{F}_s, \subseteq)$ has a unique minimal element.

Prove that $a \Rightarrow b$ and $b \Leftrightarrow c$. Show that, in general, the implication $b \Rightarrow a$ does **not** hold. $\square$

Exercise 4.7.6 (Xiang Li). Let G be a grammar and let L be the language generated by G . For $s \in L$, we denote by $\mathcal{F}_s$ the set of all formation sets that contain s and by $\mathcal{M}_s$ the set of minimal elements of $(\mathcal{F}_s, \subseteq)$.

a. Prove: $\forall s \in L, \mathcal{M}_s \neq \emptyset$.

b. Prove that for all $s \in L$ and all $F \in \mathcal{M}_s$, the set F is finite.

c. Prove that for all $s \in L$, we have

$$\bigcap \mathcal{M}_s = \bigcup \mathcal{M}_s \iff |\mathcal{M}_s| = 1.$$

d. Prove that if G has unique readability, then

$$\forall s \in L, \bigcap \mathcal{M}_s = \bigcup \mathcal{M}_s.$$

Denote by R the following binary relation on L :

$$\forall s, t \in L, (sRt \iff s \in \bigcup \mathcal{M}_t).$$

e. Prove that if G has unique readability, then (L, R) is a well-founded order.

f. Give an example of G which does **not** have unique readability, but for which (L, R) is still a well-founded order.

g. Give an example of G for which (L, R) is **not** an order.

h. Give an example of G for which (L, R) is an **ill-founded** order. $\square$

Comment. One might call the strings occurring in $\bigcap \mathcal{M}_s$ “necessary for formation” of s because they appear in all minimal formation sets containing s . Similarly, one could think of the strings in $\bigcup \mathcal{M}_s$ as “possible in formation” of s because they appear in some minimal formation set containing s . $\square$

4.8 Structural Recursion

Definition 4.8.1. Let G be a grammar and let s be a string from the language generated by G . Then the **formation rank** of s (with respect to G), denote by $\rho_G(s)$, is the least natural number n such that there is a formation sequence (according to G) of length $n + 1$ ending with s . $\square$

Exercise 4.8.2. Let G be a grammar with unique readability, let L be the language generated by G , and let $t \in L$. Prove:

a. $\rho_G(t) = 0$ if and only if there is a rule of the form $\frac{}{t}$ in G ;

b. if $t = R(s)$ for some $R \in G$ such that $\text{ar}(R) = 1$ and some $s \in L$, then $\rho_G(t) = \rho_G(s) + 1$;

c. if $t = R(s_0, \dots, s_{\text{ar}(R)-1})$ for some $R \in G$ and some $s_0, \dots, s_{\text{ar}(R)-1} \in L$, then

$$\max\{\rho_G(s_0), \dots, \rho_G(s_{\text{ar}(R)-1})\} < \rho_G(t) \leq \sum_{i < \text{ar}(R)} \rho_G(s_i) + \text{ar}(R).$$

Exercise 4.8.3. Let G be a grammar and let L be the language generated by G . For every $n \in \mathbb{N}$, let $P_n := \{s \in L : \rho_G(s) = n\}$. Prove:

a. for all distinct $m, n \in \mathbb{N}$, we have $P_m \cap P_n = \emptyset$;

b. $L = \bigcup_{n \in \mathbb{N}} P_n$. □

Exercise 4.8.4. Suppose:

1. G is a grammar;
2. E is a set;
3. $H : G \times \text{Str}^{<\mathbb{N}} \times E^{<\mathbb{N}} \rightarrow E$;
4. L is the language generated by G .

Then there is at most one function

$$F : L \rightarrow E$$

such that for all $R \in G$ and all $s_0, \dots, s_{\text{ar}(R)-1} \in L$:

$$F(R(s_0, \dots, s_{\text{ar}(R)-1})) = H(R, (s_i : i < n), (F(s_i) : i < n)).$$

Hint. Let F_0 and F_1 be two such functions and let $A(s)$: “ $F_0(s) = F_1(s)$ ”. Prove by structural induction that for all $s \in L$, $A(s)$ holds. □

Theorem 4.8.5 (Structural recursion). Suppose:

1. G is a grammar with unique readability;
2. E is a set;
3. $H : G \times \text{Str}^{<\mathbb{N}} \times E^{<\mathbb{N}} \rightarrow E$;
4. L is the language generated by G .

Then there is a unique function

$$F : L \rightarrow E$$

such that for all $R \in G$ and all $s_0, \dots, s_{\text{ar}(R)-1} \in L$:

$$F(R(s_0, \dots, s_{\text{ar}(R)-1})) = H(R, (s_i : i < n), (F(s_i) : i < n)).$$

Proof.

Uniqueness. This is Exercise 4.8.4.

Existence. For a given $n \in \mathbb{N}$, we denote

$$P_n := \{s \in L : \rho_G(s) = n\}.$$

By (ordinary) recursion on $n \in \mathbb{N}$, we will define a function

$$F_n : P_n \rightarrow E,$$

and then set

$$F : L \rightarrow E : s \mapsto F_{\rho_G(s)}(s).$$

Let $n \in \mathbb{N}$ be such that the sequence $(F_i : i < n)$ has been defined and let $t \in P_n$. We want to define $F_n(t)$. By the unique readability, there is a unique pair

$$(R, s) \in G \times L^{<\mathbb{N}}$$

such that $\text{len}(s) = \text{ar}(R)$ and $t = R(s_0, \dots, s_{\text{ar}(R)-1})$. We define

$$F(t) := H(R, (s_i : i < n), (F_{\rho_G(s_0)}(s_0), \dots, F_{\rho_G(s_{\text{ar}(R)-1})}(s_{\text{ar}(R)-1}))).$$

Notice that for all $i < \text{ar}(R)$, we have that

$$\rho_G(s_i) < \rho_G(t) = n,$$

so the value $F_{\rho_G(s_i)}(s_i) \in E$ is well-defined at this point in the recursion.

This finishes the recursive definition of the sequence $(F_n : n \in \mathbb{N})$. Now define:

$$F : L \rightarrow E : s \mapsto F_{\rho(s)}(s).$$

Let us prove that F is as required.

Let $R \in G$ and $s_0, \dots, s_{\text{ar}(R)-1} \in L$. Letting $n := \rho_G(R(s_0, \dots, s_{\text{ar}(R)-1}))$, we have $R(s_0, \dots, s_{\text{ar}(R)-1}) \in P_n$ and:

$$\begin{aligned} F(R(s_0, \dots, s_{\text{ar}(R)-1})) &= F_n(R(s_0, \dots, s_{\text{ar}(R)-1})) = H(R, (s_i : i < \text{ar}(R)), (F_{\rho_G(s_0)}(s_0), \dots, F_{\rho_G(s_{\text{ar}(R)-1})}(s_{\text{ar}(R)-1}))) = \\ &= H(R, (s_i : i < \text{ar}(R)), (F(s_0), \dots, F(s_{\text{ar}(R)-1}))). \end{aligned}$$

□

Exercise 4.8.6. In the previous proof, explain exactly how the (ordinary) recursion theorem is used to define the sequence $(F_n : n \in \mathbb{N})$.

Example 4.8.7.

a. Prove there is a unique binary operation

$$* : \text{Nat} \times \text{Nat} \rightarrow \text{Nat}$$

such that:

- for all $s \in \text{Nat}$, we have $s * \emptyset = \emptyset$;
- for all $s, t \in \text{Nat}$, we have

$$s * (t + " | ") = (s * t) + s.$$

b. Prove:

- i. for all $s, t \in \text{Nat}$, $s * t = t * s$;
- ii. for all $s, t, u \in \text{Nat}$, $(s * t) * u = s * (t * u)$;
- iii. for all $s, t, u \in \text{Nat}$, $s * (t + u) = (s * t) + (s * u)$.

c. Prove there is a function $f : \text{Nat} \rightarrow \mathbb{N}$ which identifies the monoids $(\text{Nat}, \cdot, " | ")$ and $(\mathbb{N}, \cdot, 1)$, in the following precise sense:

- f is a bijection from Nat to $\mathbb{N}$;
- for all $s, t \in \text{Nat}$, we have $f(s * t) = f(s) \cdot f(t)$;
- $f(" | ") = 1$.

□

Exercise 4.8.8. Let C be a non-empty set of characters and let G be the grammar consisting of the following rules:

- $\frac{}{\emptyset}$;
- for every $c \in S$, the rule $\frac{}{c}$;
- $\frac{s \quad t}{st}$.

We denote by L the language generated by G . Let $E := G \times \text{Str}^{<\mathbb{N}}$ and let

$$H : G \times \text{Str}^{<\mathbb{N}} \times E^{<\mathbb{N}} \rightarrow E$$

be defined by:

$$H(R, \vec{s}, \vec{t}) := (R, \vec{s}).$$

Prove that there is no function $F : L \rightarrow E$ such that for all $R \in G$ and all $s_0, \dots, s_{\text{ar}(R)-1} \in L$,

$$F(R(s_0, \dots, s_{\text{ar}(R)-1})) = H(R, (s_i : i < \text{ar}(R)), (F(s_i) : i < \text{ar}(R))).$$

Remark 4.8.9. The previous exercise shows that the structural recursion can fail if there is no unique readability. The next exercise shows that the structural recursion, in fact, implies unique readability.

Exercise 4.8.10. Let G be a grammar and let L be the language generated by it. Suppose that for all sets E and all

$$H : G \times E^{<\mathbb{N}} \rightarrow E,$$

there is $F : L \rightarrow E$ such that for all $R \in G$ and all $(s_i)_i \in L^{\text{ar}(R)}$,

$$F(R(s_0, \dots, s_{\text{ar}(R)-1})) = H(R, (F(s_i) : i < \text{ar}(R))).$$

Prove that G has unique readability.

Corollary 4.8.11 (Recursion on complexity of a formula). *Suppose:*

1. $\mathbb{P}$ is a set of propositional letters;
2. E is an arbitrary set;
3. for all $p \in \mathbb{P}$, we are given $e_p \in E$;
4. $e_{\perp} \in E$;
5. $H_{\neg} : \text{Prop}_{\mathbb{P}} \times E \rightarrow E$;
6. for all $\square \in \{\wedge, \vee, \implies, \iff\}$, $H_{\square} : \text{Prop}_{\mathbb{P}} \times \text{Prop}_{\mathbb{P}} \times E \times E \rightarrow E$.

Then there is a unique function $F : \text{Prop}_{\mathbb{P}} \rightarrow E$ such that:

- a. for all $p \in \mathbb{P}$, it holds that $F(p) = e_p$;
- b. $F(\perp) = e_{\perp}$;
- c. for all $\phi \in \text{Prop}_{\mathbb{P}}$, $F(\neg\phi) = H_{\neg}(\phi, F(\phi))$;
- d. for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ and all $\square \in \{\wedge, \vee, \implies, \iff\}$,

$$F((\phi \square \psi)) = H_{\square}(\phi, \psi, F(\phi), F(\psi)).$$

□

Definition 4.8.12. Let $\mathbb{P}$ be a set of propositional letters. The function $\text{Sub} : \text{Prop}_{\mathbb{P}} \rightarrow \mathcal{P}(\text{Prop}_{\mathbb{P}})$ is defined by recursion on complexity of a formula as follows:

- a. $\text{Sub}(\perp) := \{\perp\}$;
 - b. for all $p \in \mathbb{P}$, $\text{Sub}(p) := \{p\}$;
 - c. for all $\phi \in \text{Prop}_{\mathbb{P}}$,
- $$\text{Sub}(\neg\phi) := \text{Sub}(\phi) \cup \{\neg\phi\};$$
- d. for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ and all $\square \in \{\wedge, \vee, \implies, \iff\}$,

$$\text{Sub}((\phi \square \psi)) := \text{Sub}(\phi) \cup \text{Sub}(\psi) \cup \{(\phi \square \psi)\}.$$

□

Exercise 4.8.13. Explain how exactly the recursion on complexity of a formula is being used to define the function Sub . □

Terminology 4.8.14. Let $\mathbb{P}$ be a set of propositional letters. For $\phi \in \text{Prop}_{\mathbb{P}}$, the set $\text{Sub}(\phi)$ is the **set of its subformulas**; a formula $\psi \in \text{Prop}_{\mathbb{P}}$ is a **subformula** of ϕ iff $\psi \in \text{Sub}(\phi)$. □

Exercise 4.8.15. Compute the set of subformulas of the propositional formula

$$((\neg(p_7 \vee (\neg(p_1 \implies p_{131})))) \wedge (p_0 \iff (\neg(\neg\perp)))).$$

□

Exercise 4.8.16. Let ϕ and ψ be propositional formulas over $\mathbb{P}$. Prove that ϕ is a subformula of ψ if and only if ϕ is a substring of ψ . □

Exercise 4.8.17. Since the grammar for propositional logic over $\mathbb{P}$ has unique readability, it follows that for every formula ϕ , there exists the least formation set F_ϕ that contains ϕ (see Exercise 4.7.5). Prove that for all $\phi \in \text{Prop}_{\mathbb{P}}$, we have

$$F_\phi = \text{Sub}(\phi).$$

□

Exercise 4.8.18. Let ϕ be a propositional formula over $\mathbb{P}$ and let

$$n := |\{i < \text{len}(\phi) : \phi_i \in \{\neg, \wedge, \vee, \implies, \iff\}\}|.$$

Prove that $|\text{Sub}(\phi)| \leq 2n + 1$. □

Exercise 4.8.19. Let $\preceq$ denote the following relation on Str :

$$s \preceq t \iff (s \text{ is a substring of } t) \vee s = t.$$

a. Prove that $\preceq$ is a well-founded order relation on Str .

b. Let G be a grammar and L the language generated by it. Then $(L, \preceq)$ is also a well-founded order (Exercise 3.2.43). This means that we can use Exercise 3.2.46 to do recursive definitions. However, Exercise 4.8.10 tells us that the possibility of recursion implies unique readability. Would this not mean that every grammar has unique readability? Explain. □

Exercise 4.8.20. Let $\mathbb{P}$ be a set of propositional letters.

a. Prove that there is a unique function

$$r : \text{Prop}_{\mathbb{P}} \rightarrow \mathbb{N}$$

such that:

- for all $p \in \mathbb{P}$, $r(p) = 0$;
- $r(\perp) = 0$;
- for all $\phi \in \text{Prop}_{\mathbb{P}}$, it holds that $r(\neg\phi) = 1 + r(\phi)$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ and all $\square \in \{\wedge, \vee, \implies, \iff\}$, it holds that

$$r((\phi \square \psi)) = 1 + \max\{r(\phi), r(\psi)\}.$$

b. Prove that for all $\phi \in \text{Prop}_{\mathbb{P}}$,

$$r(\phi) \leq |\{i < \text{len}(\phi) : \phi_i \in \{\neg, \wedge, \vee, \implies, \iff\}\}|.$$

c. Show by examples that both $<$ and $=$ can occur.

d. Let $\psi \in \text{Prop}_{\mathbb{P}}$ and $\phi \in \text{Sub}(\psi) - \{\psi\}$. Prove that $r(\phi) < r(\psi)$.

e. Let $A(\phi)$ be a property. Suppose that for all $\phi \in \text{Prop}_{\mathbb{P}}$, it holds that

$$(\forall \psi \in \text{Prop}_{\mathbb{P}})(r(\psi) < r(\phi) \implies A(\psi)) \implies A(\phi).$$

Prove that $\forall \phi \in \text{Prop}_{\mathbb{P}}, A(\phi)$.

Comments. The value $r(\phi)$ corresponds to the largest “depth of connectives” in the formula ϕ . $\square$

5 Propositional Logic

5.1 Omitting Parentheses

We abbreviate our writing of formulas by omitting some parentheses, according to the following rules.

- We omit the outermost parentheses.
- We the parentheses associated to the negation.
- We assume that $\neg$ binds more strongly than $\wedge$ and $\vee$, which in turn bind more strongly than $\implies$ and $\iff$, and we omit the unnecessary parentheses accordingly. $\square$

Some examples follow.

Abbreviation	Full formula
$\neg\phi \vee \phi$	$((\neg\phi) \vee \phi)$
$\neg(\neg\neg\neg\phi \wedge \perp)$	$(\neg((\neg(\neg(\neg(\phi)))) \wedge \perp))$
$\phi \vee \psi \implies \phi$	$((\phi \vee \psi) \implies \phi)$
$\phi \implies \phi \vee (\psi \implies \chi)$	$(\phi \implies (\phi \vee (\psi \implies \chi)))$

Remark 5.1.1. These abbreviations, seen as simple strings, are not themselves propositional formulas. $\square$

5.2 Semantics for Propositional Logic

Definition 5.2.1. Let $\mathbb{P}$ be a set of propositional letters. A **valuation** for $\mathbb{P}$ is just a function

$$\mu : \mathbb{P} \rightarrow \{0, 1\}.$$

$\square$

Exercise 5.2.2. Let μ be a valuation for $\mathbb{P}$. Prove that there is a unique function

$$f : \text{Prop}_{\mathbb{P}} \rightarrow \{0, 1\} : \phi \mapsto \|\phi\|_{\mu}$$

with the properties:

- $f(\perp) = 0$;
- for every $p \in \mathbb{P}$, $f(p) := \mu(p)$;
- for every $\phi \in \text{Prop}_{\mathbb{P}}$,

$$f((\neg\phi)) := 1 - f(\phi);$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$f((\phi \wedge \psi)) := \min\{f(\phi), f(\psi)\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$f((\phi \vee \psi)) := \max\{f(\phi), f(\psi)\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$f((\phi \implies \psi)) := \max\{1 - f(\phi), f(\psi)\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$f((\phi \iff \psi)) := |f(\phi) - f(\psi)|.$$

□

Remark 5.2.3. We can now define, for a given valuation μ , the function

$$\text{Prop}_{\mathbb{P}} \rightarrow \{0, 1\} : \phi \mapsto \|\phi\|_{\mu}$$

to be the unique function $f : \text{Prop}_{\mathbb{P}} \rightarrow \{0, 1\}$ with the properties listed above. However, it is customary to write such a definition in the following “more explicit” form. □

Definition 5.2.4. Let μ be a valuation for $\mathbb{P}$. The function

$$\text{Prop}_{\mathbb{P}} \rightarrow \{0, 1\} : \phi \mapsto \|\phi\|_{\mu}$$

is defined by recursion on complexity of a formula as follows:

- $\|\perp\|_{\mu} := 0$;
- for every $p \in \mathbb{P}$, $\|p\|_{\mu} := \mu(p)$;
- for every $\phi \in \text{Prop}_{\mathbb{P}}$,

$$\|(\neg\phi)\|_{\mu} := 1 - \|\phi\|_{\mu};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$\|(\phi \wedge \psi)\|_{\mu} := \min\{\|\phi\|_{\mu}, \|\psi\|_{\mu}\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$\|(\phi \vee \psi)\|_{\mu} := \max\{\|\phi\|_{\mu}, \|\psi\|_{\mu}\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$\|(\phi \implies \psi)\|_{\mu} := \max\{1 - \|\phi\|_{\mu}, \|\psi\|_{\mu}\};$$

- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$\|(\phi \iff \psi)\|_{\mu} := 1 - |\|\phi\|_{\mu} - \|\psi\|_{\mu}|.$$

□

Example 5.2.5. Let ϕ and ψ be propositional formulas over $\mathbb{P}$ and let μ be a valuation for $\mathbb{P}$. Then $\|(\phi \implies \psi)\|_{\mu}$ is completely determined by the values $\|\phi\|_{\mu}$ and $\|\psi\|_{\mu}$ in a way that can be represented by the following **truth table**:

$\ \phi\ _{\mu}$	$\ \psi\ _{\mu}$	$\ (\phi \implies \psi)\ _{\mu}$
1	1	1
1	0	0
0	1	1
0	0	1

□

Exercise 5.2.6. Prove that for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$:

- $\|\phi \wedge \psi\|_{\mu} = \|\phi\|_{\mu} \cdot \|\psi\|_{\mu}$;
- $\|\phi \vee \psi\|_{\mu} = \|\phi\|_{\mu} + \|\psi\|_{\mu} - \|\phi\|_{\mu} \cdot \|\psi\|_{\mu}$;
- $\|\phi \implies \psi\|_{\mu} = 1$ if and only if $\|\phi\|_{\mu} \leq \|\psi\|_{\mu}$;
- $\|\phi \iff \psi\|_{\mu} = 1$ if and only if $\|\phi\|_{\mu} = \|\psi\|_{\mu}$.

□

Definition 5.2.7. A propositional letter $p \in \mathbb{P}$ is said to **occur** in a propositional formula ϕ iff there is $k < \text{len}(\phi)$ such that $\phi(k) = p$.

□

Exercise 5.2.8. Prove that a propositional letter occurs in a propositional formula if and only if it is its subformula.

□

Exercise 5.2.9. Let ϕ be a propositional formula and let μ and ν be two valuations. Suppose μ and ν agree on the set of propositional letters that occur in ϕ . Prove that $\|\phi\|_{\mu} = \|\phi\|_{\nu}$.

□

Example 5.2.10. Consider the propositional formula

$$(p_2 \implies \neg p_1) \vee p_7,$$

which we will denote by ϕ . The set of propositional letters occurring in ϕ is the set

$$\{p_2, p_1, p_7\}.$$

This means that, for a given valuation μ , the value $\|\phi\|_{\mu}$ is entirely determined by

$$\mu \upharpoonright \{p_2, p_1, p_7\}.$$

There are exactly 8 possible values for

$$\mu \upharpoonright \{p_2, p_1, p_7\}.$$

For each of those values, we can compute exactly $\|\phi\|_{\mu}$. The result is represented by the following **truth table**:

$\mu(p_1)$	$\mu(p_2)$	$\mu(p_7)$	$\ \neg p_1\ _\mu$	$\ p_2 \implies \neg p_1\ _\mu$	$\ (p_2 \implies \neg p_1) \vee p_7\ _\mu$
1	1	1	0	0	1
1	1	0	0	0	0
1	0	1	0	1	1
1	0	0	0	1	1
0	1	1	1	1	1
0	1	0	1	1	1
0	0	1	1	1	1
0	0	0	1	1	1

In the table, we included all subformulas of ϕ to make it clear how to compute $\|\phi\|_\mu$ “step-by-step”. $\square$

Exercise 5.2.11. Let $\mathbb{P}$ be a set of propositional letters and let V be the set of all valuations for $\mathbb{P}$, i.e. $V = \mathcal{F}(\mathbb{P}, [2])$. For a propositional formula ϕ , we denote by $\hat{\phi}$ the function

$$V \rightarrow [2] : \mu \mapsto \|\phi\|_\mu.$$

In this way, we obtain the following function:

$$F : \text{Prop}_{\mathbb{P}} \rightarrow \mathcal{F}(V, [2]) : \phi \mapsto \hat{\phi}.$$

Prove that the following are equivalent for every $f : V \rightarrow [2]$:

- there is $\phi \in \text{Prop}_{\mathbb{P}}$ such that $f = \hat{\phi}$;
- there is a finite set $X \subseteq \mathbb{P}$ such that for all valuation μ and ν :

$$\mu \upharpoonright X = \nu \upharpoonright X \implies f(\mu) = f(\nu).$$

$\square$

5.3 Models and Theories

Notation 5.3.1. Let ϕ be a propositional formula over $\mathbb{P}$ and let μ be a valuation for $\mathbb{P}$. We say that μ is a **model of ϕ** , and write $\mu \models \phi$, iff $\|\phi\|_\mu = 1$. $\square$

Definition 5.3.2. We say that a propositional formula ϕ is a **tautology**, written also $\models \phi$, iff every valuation μ is a model for ϕ . $\square$

Exercise 5.3.3. Check which of the following formulas are tautologies.

- $(\neg p_0 \vee p_1) \iff (p_1 \implies p_0)$
- $p_0 \implies ((p_1 \implies p_2) \implies ((p_0 \implies p_1) \implies (p_0 \implies p_2)))$
- $(p_0 \implies \neg p_0) \iff \neg p_0$
- $\neg(p_0 \implies \neg p_0)$

$$e. (p_0 \implies (p_1 \implies p_2)) \iff (p_0 \wedge p_1 \implies p_2)$$

$$f. p_0 \vee \neg p_0$$

$$g. \perp \iff (p_0 \wedge \neg p_0)$$

$$h. \perp \implies p_0$$

□

Definition 5.3.4. Let $p \in \mathbb{P}$ and $\theta \in \text{Prop}_{\mathbb{P}}$. The **substitution** map

$$\text{Prop}_{\mathbb{P}} \rightarrow \text{Prop}_{\mathbb{P}} : \phi \mapsto \phi[p := \theta]$$

is defined by recursion on complexity of the formulas as follows:

- $p[p := \theta] = \theta$;
- for all $q \in \mathbb{P} - \{p\}$, $q[p := \theta] = q$;
- $\perp[p := \theta] = \perp$;
- for all $\phi \in \text{Prop}_{\mathbb{P}}$, $(\neg\phi)[p := \theta] = (\neg \phi[p := \theta])$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$ and all $\square \in \{\wedge, \vee, \implies, \iff\}$,

$$(\phi \square \psi)[p := \theta] = (\phi[p := \theta] \square \psi[p := \theta]).$$

□

Exercise 5.3.5. Let

$$\phi = p_1 \wedge p_0 \implies (p_0 \implies p_3).$$

Compute

$$\phi[p_0 := \neg p_0 \implies p_3].$$

□

Exercise 5.3.6. Let

$$\phi = (p_3 \iff p_0) \vee (p_2 \implies \neg p_0).$$

Compute

$$\phi[p_0 := \neg p_0 \implies p_3].$$

□

Exercise 5.3.7. Let ϕ be a tautology, $p \in \mathbb{P}$ and $\theta \in \text{Prop}_{\mathbb{P}}$. Prove that $\phi[p := \theta]$ is a tautology. □

Example 5.3.8. We can easily verify that the formula

$$p_0 \vee \neg p_0$$

is a tautology. It follows from the previous exercise that for every formula θ , the formula

$$\theta \vee \neg\theta$$

is also a tautology. □

Definition 5.3.9. A **propositional theory** over $\mathbb{P}$ is a set of propositional formulas over $\mathbb{P}$. □

Definition 5.3.10. Let T be a propositional theory and let μ be a valuation. We say that μ is a **model of T** , and write $\mu \models T$, iff μ is a model for every formula in T . □

Exercise 5.3.11. Describe the set of all models of the propositional theory

$$\{p_i \implies p_{i+2} : i \in \mathbb{N}\}.$$

□

Definition 5.3.12. Let T be a propositional theory and let ϕ be a propositional formula. We say that ϕ is a **semantical consequence of T** , and write $T \models \phi$, iff every model of T is also a model of ϕ . □

Exercise 5.3.13. Let ϕ be a propositional formula. Prove: $\emptyset \models \phi$ if and only if $\models \phi$. □

Notation 5.3.14. We write $\phi_0, \phi_1, \dots, \phi_{n-1} \models \psi$ to mean

$$\{\phi_0, \phi_1, \dots, \phi_{n-1}\} \models \psi.$$

□

Exercise 5.3.15. Let ϕ, ψ, θ be propositional formulas. Prove:

a. $\models \phi \implies \phi$

b. $\models \neg\neg\phi \implies \phi$

c. $\models \phi \vee \psi \iff \psi \vee \phi$

d. $\phi, \psi \models \phi \wedge \psi$

e. $\phi, \phi \implies \psi \models \psi$

f. $\phi \implies \psi, \neg\psi \models \neg\phi$

g. $\phi \models \phi$

h. if $\phi \models \psi$ and $\psi \models \theta$, then $\phi \models \theta$;

i. $\models \phi \implies \psi$ if and only if $\phi \models \psi$. □

Exercise 5.3.16 (Xiang Li). Let T be a propositional theory and let ϕ and ψ be propositional formulas. Prove that $T \cup \{\phi\} \models \psi$ if and only if $T \models (\phi \implies \psi)$. □

Definition 5.3.17. A propositional theory T is **satisfiable** iff there is a valuation μ which is a model for T . □

Exercise 5.3.18. Let T be a propositional theory. Prove that the following statements are equivalent:

a. T is satisfiable;

b. $T \not\models \perp$;

c. there is a formula $\phi \in \text{Prop}$ such that $T \not\models \phi$. □

Definition 5.3.19. A theory T is said to be **semantically complete** iff it is satisfiable and for every formula ϕ ,

$$T \models \phi \text{ or } T \models \neg\phi.$$

□

Exercise 5.3.20. Give an example of a theory which is **not** semantically complete. □

Definition 5.3.21. Let μ be a valuation. The **full theory** of μ is the set $\text{Th}(\mu)$ defined as follows:

$$\text{Th}(\mu) := \{\phi \in \text{Prop}_{\mathbb{P}} : \mu \models \phi\}.$$

□

Exercise 5.3.22. Prove that the full theory of every valuation is semantically complete. □

Exercise 5.3.23. Let T be a semantically complete theory. Prove that T has a unique model. □

Exercise 5.3.24. Let T be a theory and let ϕ be a formula. Prove that $T \not\models \phi$ if and only if $T \cup \{\neg\phi\}$ is satisfiable. □

Exercise 5.3.25. Let $\mathcal{S}$ denote the set of all satisfiable propositional theories over $\mathbb{P}$. Prove that the following are equivalent for every theory T :

a. T is a maximal element of $(\mathcal{S}, \subseteq)$;

b. T is satisfiable and

$$\forall \phi \in \text{Prop}, (T \models \phi \iff \phi \in T);$$

c. there is a valuation μ such that $T = \text{Th}(\mu)$. □

Comment. The first part is commonly read: “ T is a maximal satisfiable theory”. The exercise is saying (among other things) that maximal satisfiable theories essentially correspond to valuations. □

5.4 Semantic Equivalence

Definition 5.4.1. We say that propositional formulas ϕ and ψ are **semantically equivalent**, written $\phi \equiv \psi$, iff

$$\models \phi \iff \models \psi.$$

□

Definition 5.4.2. Let T be a propositional theory. We say that propositional formulas ϕ and ψ are **semantically equivalent modulo T** , written $\phi \equiv_T \psi$, iff

$$T \models \phi \iff T \models \psi.$$

□

Remark 5.4.3. The relations $\equiv$ and $\equiv_{\emptyset}$ are the same. □

Exercise 5.4.4. Let T be a propositional theory. Prove that $\equiv_T$ is an equivalence relation on Prop . □

Notation 5.4.5. We write $T, \phi_0, \dots, \phi_{n-1} \models \psi$ to mean

$$T \cup \{\phi_0, \dots, \phi_{n-1}\} \models \psi.$$

□

Exercise 5.4.6. Let T be a propositional theory and $\phi, \psi \in \text{Prop}$. Then $\phi \equiv_T \psi$ if and only if

$$T, \phi \models \psi \text{ and } T, \psi \models \phi.$$

□

Exercise 5.4.7. Let ϕ, ψ, θ be formulas and let $p \in \mathbb{P}$. Prove that if $\phi \equiv \psi$, then $\theta[p := \phi] \equiv \theta[p := \psi]$. □

Example 5.4.8. The formula

$$(p_0 \implies p_1) \iff (\neg p_0 \vee p_1)$$

is a tautology. By Exercise 5.3.7, we have that, for all $\phi, \psi \in \text{Prop}$, the formula

$$(\phi \implies \psi) \iff (\neg \phi \vee \psi)$$

is a tautology, whence

$$(\phi \implies \psi) \equiv (\neg \phi \vee \psi)$$

Now, consider the formula

$$p_0 \wedge \neg p_1.$$

By the previous exercise, we have

$$(\phi \implies \psi) \wedge \neg p_1 \equiv (\neg \phi \vee \psi) \wedge \neg p_1.$$

Applying again the same exercise to $\theta \equiv \theta$, we get

$$(\phi \implies \psi) \wedge \neg \theta \equiv (\neg \phi \vee \psi) \wedge \neg \theta.$$

□

Exercise 5.4.9. Let $\mathbb{P}$ be a set of propositional letters and let V be the set of all valuations, i.e. $V = \mathcal{F}(\mathbb{P}, [2])$. For a propositional formula ϕ , we denote by $\hat{\phi}$ the function

$$V \rightarrow [2] : \mu \mapsto \|\phi\|_{\mu}.$$

In this way, we obtain the following function:

$$F : \text{Prop}_{\mathbb{P}} \rightarrow \mathcal{F}(V, [2]) : \phi \mapsto \hat{\phi}.$$

Prove that for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$,

$$\phi \equiv \psi \iff \hat{\phi} = \hat{\psi}.$$

□

Definition 5.4.10. Propositional theories T_1 and T_2 are said to be **semantically equivalent** iff

$$\forall \phi \in \text{Prop}_{\mathbf{P}}, (T_1 \models \phi \iff T_2 \models \phi).$$

□

Exercise 5.4.11. Let ϕ and ψ be propositional formulas. Prove the formulas ϕ and ψ are semantically equivalent if and only if the theories $\{\phi\}$ and $\{\psi\}$ are semantically equivalent. □

Definition 5.4.12. A theory T is **finitely axiomatizable** iff it is semantically equivalent to a finite theory. □

Exercise 5.4.13. Prove that a theory is finitely axiomatizable if and only if it is semantically equivalent to the theory $\{\phi\}$, for some formula ϕ . □

Exercise 5.4.14. Prove that the propositional theory

$$\{p_i \implies p_{i+2} : i \in \mathbb{N}\}.$$

is **not** finitely axiomatizable. □

Definition 5.4.15. A theory T is **semantically independent** iff it is **not** semantically equivalent any of its proper subsets. □

Exercise 5.4.16 (Xiang Li).

- a. Prove that every **finite** theory contains a subset which is both independent and semantically equivalent to it.
- b. Give an example of an (infinite) theory for which the above conclusion fails.
- c. Prove that every theory is semantically equivalent to some independent theory. □

6 First-order Logic

6.1 First-order Logic

Definition 6.1.1. A **first-order language** is a 4-tuple $\mathcal{L} = (\mathcal{F}_{\mathcal{L}}, \mathcal{C}_{\mathcal{L}}, \mathcal{R}_{\mathcal{L}}, \text{ar}_{\mathcal{L}})$, where:

- $\mathcal{F}_{\mathcal{L}}$, $\mathcal{C}_{\mathcal{L}}$, and $\mathcal{R}_{\mathcal{L}}$ are three disjoint sets of characters, which do not contain $\perp$, $\neg$, $\wedge$, $\vee$, $\implies$, $\iff$, $\forall$, $\exists$;
- $\text{ar}_{\mathcal{L}} : \mathcal{F}_{\mathcal{L}} \cup \mathcal{R}_{\mathcal{L}} \rightarrow \mathbb{N} - \{0\}$. □

Terminology 6.1.2. Elements of $\mathcal{F}_{\mathcal{L}}$ are referred to as **function symbols** or **operation symbols**, elements of $\mathcal{C}_{\mathcal{L}}$ are **constant symbols**, and elements of $\mathcal{R}_{\mathcal{L}}$ are **relation symbols**. For a symbol $\sigma \in \mathcal{F}_{\mathcal{L}} \cup \mathcal{R}_{\mathcal{L}}$, the number $\text{ar}_{\mathcal{L}}(\sigma)$ is its **arity**. □

Example 6.1.3.

- The **language of orders** is the language $(\mathcal{F}, \mathcal{C}, \mathcal{R}, \text{ar})$ given by:
 - $\mathcal{F} = \emptyset$;
 - $\mathcal{C} = \emptyset$;
 - $\mathcal{R} = \{\leq\}$;
 - $\text{ar}(\leq) = 2$.
- The **language of graphs** is the language $(\mathcal{F}, \mathcal{C}, \mathcal{R}, \text{ar})$ given by:
 - $\mathcal{F} = \emptyset$;
 - $\mathcal{C} = \emptyset$;
 - $\mathcal{R} = \{E\}$;
 - $\text{ar}(E) = 2$.
- The **language of monoids** is the language $(\mathcal{F}, \mathcal{C}, \mathcal{R}, \text{ar})$ given by:
 - $\mathcal{F} = \{*\}$;
 - $\mathcal{C} = \{1\}$;
 - $\mathcal{R} = \emptyset$;
 - $\text{ar}(*) = 2$.

Definition 6.1.4. Let $\mathcal{L}$ be a first-order language. A **structure for $\mathcal{L}$** (or, an **$\mathcal{L}$ -structure**) is a pair $M = ([M], \mathcal{J}^M)$, where:

- $[M]$ is a set;
- $\mathcal{J}^M$ is a function with domain $\mathcal{L}$;
- for all $\sigma \in \mathcal{F}_{\mathcal{L}}$, the object $\mathcal{J}^M(\sigma)$ is an operation on $[M]$ of arity $\text{ar}_{\mathcal{L}}(\sigma)$;
- for all $\sigma \in \mathcal{R}_{\mathcal{L}}$, the object $\mathcal{J}^M(\sigma)$ is a relation on $[M]$ of arity $\text{ar}_{\mathcal{L}}(\sigma)$. □

Example 6.1.5. Let $\mathcal{L}$ be the language of orders. An example of an $\mathcal{L}$ -structure is the structure $M = (\mathbb{N}, \mathcal{J}^M)$, where $\mathcal{J}^M(\leq) = |$. Another example is the structure $N = (\mathbb{R}, \mathcal{J}^N)$, where $\mathcal{J}^N(\leq)$ is the relation on $\mathbb{R}$ given by:

$$(\forall x, y \in \mathbb{R})(x \mathcal{J}^N(\leq) y \iff x - y = \pi).$$

Notice that the pair $([M], \mathcal{J}^M(\leq))$ is an order, while $([N], \mathcal{J}^N(\leq))$ is not. □

Notation 6.1.6. We can (and will from now on) understand orders as structures in the language of orders: if (E, R) is an order, then it is naturally understood as the structure $M = (E, \mathcal{J}^M)$, where $\mathcal{J}^M$ is defined so that $\mathcal{J}^M(\leq) = R$. □

Example 6.1.7. Let $\mathcal{L}$ be the language of graphs. An example of an $\mathcal{L}$ -structure is the structure $M = (\mathbb{N}, \mathcal{I}^M)$, where $\mathcal{I}^M(E) = |$. Another example is the structure $N = (\mathbb{R}, \mathcal{I}^N)$, where $\mathcal{I}^N(E)$ is the relation on $\mathbb{R}$ given by:

$$(\forall x, y \in \mathbb{R})(x \mathcal{I}^N(E) y \iff x - y = \pi).$$

Notice that the pair $([N], \mathcal{I}^N(E))$ is a directed graph, while $([M], \mathcal{I}^M(E))$ is not. $\square$

Notation 6.1.8. We can (and will from now on) understand directed graphs as structures in the language of graphs: if (V, E) is a directed graph, then it is naturally understood as the structure $M = (V, \mathcal{I}^M)$, where $\mathcal{I}^M$ is defined so that $\mathcal{I}^M(E) = E$. $\square$

Notation 6.1.9. Let $\mathcal{L}$ be a language and M an $\mathcal{L}$ -structure. Then for $\sigma \in \mathcal{F}_{\mathcal{L}} \cup \mathcal{C}_{\mathcal{L}} \cup \mathcal{R}_{\mathcal{L}}$, we write σ^M to mean $\mathcal{I}^M(\sigma)$. $\square$

Terminology 6.1.10. Let $\mathcal{L}$ be a language and M an $\mathcal{L}$ -structure.

- The set $[M]$ is referred to as the **domain** or the **universe** of the structure M .
- The function $\mathcal{I}^M$ is referred the **interpretation** (of the language, given by the structure).
- For $\sigma \in \mathcal{F}_{\mathcal{L}} \cup \mathcal{C}_{\mathcal{L}} \cup \mathcal{R}_{\mathcal{L}}$, the object σ^M is the **interpretation** of the symbol σ . $\square$

Exercise 6.1.11. Let $\mathcal{L}$ be the language of monoids. Consider the $\mathcal{L}$ -structure M whose universe in $\mathbb{N}$ and whose interpretation is given by:

- for all $x, y \in \mathbb{N}$, $x *^M y := x^2 + y + 1$;
- $1^M := 7$.

Prove that the triple $([M], *^M, 1^M)$ is not a monoid. $\square$

Notation 6.1.12. We can (and will from now on) understand monoids as structures in the language of monoids: if $(E, *, e)$ is a monoid, then it is naturally understood as the structure $M = (E, \mathcal{I}^M)$, where $\mathcal{I}^M$ is defined so that $*^M = *$ and $1^M = e$. $\square$

Notation 6.1.13. We will fix a set $\mathcal{V}$ of **variables** for the first-order logic: $\{v_i : i \in \mathbb{N}\}$. $\square$

Definition 6.1.14. Let $\mathcal{L}$ be a first-order language. The **grammar of $\mathcal{L}$ -terms** consists of the following rules:

- for all $i \in \mathbb{N}$, the rule $\frac{}{v_i}$;
- for all $c \in \mathcal{C}_{\mathcal{L}}$, the rule $\frac{}{c}$;
- for all $f \in \mathcal{C}_{\mathcal{L}}$, the rule

$$\frac{t_0 \quad t_1 \quad \dots \quad t_{n-1}}{f(t_0, t_1, \dots, t_{n-1})},$$

where $n = \text{ar}_{\mathcal{L}}(f)$.

The **set of $\mathcal{L}$ -terms** is the language generated by this grammar, and **$\mathcal{L}$ -terms** are elements of this set. $\square$

Example 6.1.15. An example of a term in the language of monoids is:

$$*(* (1, v_7), v_3);$$

which is to be understood as $(1 * v_7) * v_3$. On the other hand, the only terms in the language of orders are the variables. $\square$

Notation 6.1.16. Let $\mathcal{L}$ be a language, t_0 and t_1 two $\mathcal{L}$ -terms, and f a binary function symbol of $\mathcal{L}$. When there is no risk of confusion, we write $t_0 f t_1$ to be an abbreviation for $f(t_0, t_1)$. $\square$

Exercise 6.1.17. Let $\mathcal{L}$ be a first-order language. Prove that the grammar of $\mathcal{L}$ -terms has unique readability. $\square$

Corollary 6.1.18 (Recursion on complexity of terms). *Suppose:*

1. $\mathcal{L}$ is a language;
2. T is the set of $\mathcal{L}$ -terms;
3. E is a set;
4. $H_V : \mathcal{V} \rightarrow E$;
5. $H_C : \mathcal{C}_{\mathcal{L}} \rightarrow E$;
6. for all $f \in \mathcal{F}_{\mathcal{L}}$, $H_f : T^{\text{ar}_{\mathcal{L}}(f)} \times E^{\text{ar}_{\mathcal{L}}(f)} \rightarrow E$.

Then there is a unique function F from the set of $\mathcal{L}$ -terms to the set E such that:

- for all $i \in \mathbb{N}$, $F(v_i) = H_V(v_i)$;
- for all $c \in \mathcal{C}_{\mathcal{L}}$, $F(c) = H_C(c)$;
- for all $f \in \mathcal{F}_{\mathcal{L}}$, letting $n = \text{ar}_{\mathcal{L}}(f)$, it holds that for all $\mathcal{L}$ -terms $t_0, t_1, \dots, t_{n-1}$,

$$F(f(t_0, \dots, t_{n-1})) = H_f((t_i : i < n), (F(t_i) : i < n)).$$

$\square$

Definition 6.1.19. Let $\mathcal{L}$ be a first-order language and let M be an $\mathcal{L}$ structure. An **M -valuation** is a map $\mu : \mathcal{V} \rightarrow M$. $\square$

Remark 6.1.20. We will often write M to mean both a structure and the underlying set, $[M]$ (if there is no risk of confusion). So, for example, we may speak of a function $\mu : \mathbb{N} \rightarrow M$ when we really mean $\mu : \mathbb{N} \rightarrow [M]$. $\square$

Definition 6.1.21. Suppose:

1. $\mathcal{L}$ is a language;

2. t is an $\mathcal{L}$ -term;
3. M is an $\mathcal{L}$ -structure;
4. μ is an M -valuation.

Then we define the value $t^M[\mu]$ by recursion on complexity of t :

- if $t = v_i$ for some $i \in \mathbb{N}$, then $t^M[\mu] = \mu(v_i)$;
- if $t = c$ for some $c \in \mathcal{C}_{\mathcal{L}}$, then $t^M[\mu] = c^M$;
- if $t = f(t_0, \dots, t_{n-1})$ for some $f \in \mathcal{F}_{\mathcal{L}}$ and some $\mathcal{L}$ -terms $t_0, \dots, t_{n-1}$, then

$$t^M[\mu] = f^M(t_0^M[\mu], \dots, t_{n-1}^M[\mu]).$$

□

Exercise 6.1.22. Let $\mathcal{L}$ be the language of monoids and let M be the $\mathcal{L}$ -structure given by:

- $\lfloor M \rfloor = \mathbb{N}$;
- for all $x, y \in \mathbb{N}$, $x *^M y = x^2 + y + 1$;
- $1^M = 7$.

Let μ be the M -valuation given by:

$$\forall i \in \mathbb{N}, \mu(v_i) = 2i;$$

and let $t = (* (1, v_7), v_3)$. Compute $t^M[\mu]$.

□

Definition 6.1.23. Let $\mathcal{L}$ be a language. The **set of variables** of an $\mathcal{L}$ -term t is the set

$$\{w \in \mathcal{V} : (\exists i < \text{len}(t))(t_i = w)\}.$$

Exercise 6.1.24. Let t be an $\mathcal{L}$ -term, let M be an $\mathcal{L}$ -structure and let μ and ν be M -valuations that agree on the set of variables of t . Prove that $t^M[\mu] = t^M[\nu]$. □

Definition 6.1.25. Let $\mathcal{L}$ be a first-order language. The **atomic $\mathcal{L}$ -formulas** are the following formulas:

- $t_0 = t_1$, where t_0 and t_1 are $\mathcal{L}$ -terms;
- $R(t_0, \dots, t_{n-1})$, where $R \in \mathcal{R}_{\mathcal{L}}$, $n = \text{ar}_{\mathcal{L}}(R)$, and $t_0, \dots, t_{n-1}$ are $\mathcal{L}$ -terms;
- $\perp$.

□

Definition 6.1.26. Suppose:

1. $\mathcal{L}$ is a language;
2. ϕ is an atomic $\mathcal{L}$ -formula;

3. M is an $\mathcal{L}$ -structure;
4. μ is an M -valuation.

We define the value $\|\phi\|_\mu^M$ as follows:

- if ϕ is of the form $t_0 = t_1$, for some terms t_0, t_1 , then

$$\|\phi\|_\mu^M = \begin{cases} 1, & t_0^M[\mu] = t_1^M[\mu] \\ 0, & t_0^M[\mu] \neq t_1^M[\mu]; \end{cases}$$

- if ϕ is of the form $R(t_0, \dots, t_{n-1})$, for some relation symbol R , for $n = \text{ar}_{\mathcal{L}}(R)$, and for some terms $t_0, \dots, t_{n-1}$, then

$$\|\phi\|_\mu^M = \begin{cases} 1, & (t_0^M[\mu], \dots, t_{n-1}^M[\mu]) \in R^M, \\ 0, & (t_0^M[\mu], \dots, t_{n-1}^M[\mu]) \notin R^M; \end{cases}$$

- $\|\perp\|_\mu^M = 0$. □

Definition 6.1.27. Let $\mathcal{L}$ be a first-order language. We define the **grammar of the first-order logic** for $\mathcal{L}$ to consist of the following rules:

- for every atomic $\mathcal{L}$ -formula ϕ , the rule $\frac{}{\phi}$;
- the rule $\frac{\phi}{(\neg\phi)}$;
- for all $\square \in \{\wedge, \vee, \implies, \iff\}$, the rule $\frac{\phi \quad \psi}{(\phi\square\psi)}$;
- for all $Q \in \{\forall, \exists\}$ and all $v_i \in \mathcal{V}$, the rule $\frac{\phi}{((Qv_i)\phi)}$.

The **set of first-order formulas** in the language $\mathcal{L}$, denoted $\text{Form}_{\mathcal{L}}$, is the language generated by the grammar for the first-order logic. A **first-order formula** (in $\mathcal{L}$) is any element of this set. □

Exercise 6.1.28. Let $\mathcal{L}$ be a first-order language. Prove that the grammar of the first-order logic for $\mathcal{L}$ has the unique readability. □

Corollary 6.1.29 (Recursion on complexity of a formula). *Suppose:*

1. $\mathcal{L}$ is a first-order language;
2. E is a set;
3. H_0 is a function from the set of atomic $\mathcal{L}$ -formulas to E ;
4. $H_{\neg} : \text{Form}_{\mathcal{L}} \times E \rightarrow E$;
5. for all $\square \in \{\wedge, \vee, \implies, \iff\}$, $H_{\square} : \text{Form}_{\mathcal{L}}^2 \times E^2 \rightarrow E$;

6. for all $Q \in \{\exists, \forall\}$, $H_Q : \mathcal{V} \times \text{Form}_{\mathcal{L}} \times E \rightarrow E$.

Then there is a unique function $F : \text{Form}_{\mathcal{L}} \rightarrow E$ such that:

- for every atomic $\mathcal{L}$ -formula ϕ , we have $F(\phi) = H_0(\phi)$;
- for every $\mathcal{L}$ -formula ϕ , we have $F(\neg\phi) = H_{\neg}(\phi, F(\phi))$;
- for all $\mathcal{L}$ -formulas ϕ, ψ and all $\square \in \{\wedge, \vee, \implies, \iff\}$, we have $F(\phi \square \psi) = H_{\square}(\phi, \psi, F(\phi), F(\psi))$;
- for all $\mathcal{L}$ -formulas ϕ , all $Q \in \{\exists, \forall\}$, and all variables v_i , we have $F(((Qv_i)\phi)) = H_Q(v_i, \phi, F(\phi))$.

□

Exercise 6.1.30. Prove that the boolean algebra 2 is a complete lattice. Furthermore, for all $(x_i)_{i \in I} \in 2^I$, we have

$$\bigvee_{i \in I} x_i = \max_{i \in I} x_i,$$

$$\bigwedge_{i \in I} x_i = \min_{i \in I} x_i.$$

□

Definition 6.1.31. Suppose:

1. $\mathcal{L}$ is a language;
2. M is an $\mathcal{L}$ -structure;
3. μ is an M valuation;
4. $v_i \in \mathcal{V}$;
5. $a \in M$.

Then we define the M -valuation $\mu[v_i := a]$ as follows:

$$(\mu[v_i := a])(v_j) = \begin{cases} \mu(v_j), & v_j \neq v_i, \\ a, & v_j = v_i. \end{cases}$$

□

Definition 6.1.32. Suppose:

1. $\mathcal{L}$ is a language;
2. M is an $\mathcal{L}$ -structure.

We define the function

$$\text{Form}_{\mathcal{L}} \times \mathcal{F}(\mathcal{V}, M) \rightarrow 2 : (\phi, \mu) \mapsto \|\phi\|_{\mu}^M$$

by recursion on complexity of a formula:

- if ϕ is atomic, then $\|\phi\|_\mu^M$ is already defined;

- if ϕ is of the form $(\neg\psi)$, then

$$\|\phi\|_\mu^M = \neg\|\psi\|_\mu^M;$$

- if ϕ is of the form $(\psi \wedge \theta)$, then

$$\|\phi\|_\mu^M = \|\psi\|_\mu^M \wedge \|\theta\|_\mu^M;$$

- if ϕ is of the form $(\psi \vee \theta)$, then

$$\|\phi\|_\mu^M = \|\psi\|_\mu^M \vee \|\theta\|_\mu^M;$$

- if ϕ is of the form $(\psi \implies \theta)$, then

$$\|\phi\|_\mu^M = \neg\|\psi\|_\mu^M \vee \|\theta\|_\mu^M;$$

- if ϕ is of the form $(\psi \iff \theta)$, then

$$\|\phi\|_\mu^M = (\|\psi\|_\mu^M \wedge \|\theta\|_\mu^M) \vee (\neg\|\psi\|_\mu^M \wedge \neg\|\theta\|_\mu^M);$$

- if ϕ is of the form $((\forall v_i)\psi)$, then

$$\|\phi\|_\mu^M = \bigwedge_{a \in M} \|\psi\|_{\mu[v_i=a]}^M;$$

- if ϕ is of the form $((\exists v_i)\psi)$, then

$$\|\phi\|_\mu^M = \bigvee_{a \in M} \|\psi\|_{\mu[v_i=a]}^M.$$

□

Definition 6.1.33. Let $\mathcal{L}$ be a language. The set of **free variables** of an $\mathcal{L}$ -formula ϕ , denoted $\text{fvar}(\phi)$, is defined by recursion on complexity as follows:

- if ϕ is the formula $\perp$, then $\text{fvar}(\phi) = \emptyset$;
- if ϕ is of the form $t_0 = t_1$ (t_0, t_1 terms), then $\text{fvar}(\phi)$ is the union of the sets of variables of t_0 and t_1 ;
- if ϕ is of the form $R(t_0, \dots, t_{n-1})$ (R a relation symbol, $t_0, \dots, t_{n-1}$ terms), then $\text{fvar}(\phi)$ is the union of the sets of variables of $t_0, t_1, \dots, t_{n-1}$;
- if ϕ is of the form $\neg\psi$ (ψ a formula), then $\text{fvar}(\phi) = \text{fvar}(\psi)$;
- if ϕ is of the form $\psi \square \theta$ (ψ, θ formulas, $\square$ binary connective), then $\text{fvar}(\phi) = \text{fvar}(\psi) \cup \text{fvar}(\theta)$;
- if ϕ is of the form $((Qv_i)\psi)$ (ψ a formula, Q a quantifier, $i \in \mathbb{N}$), then $\text{fvar}(\phi) = \text{fvar}(\psi) - \{v_i\}$. □

Exercise 6.1.34. Suppose:

1. $\mathcal{L}$ is a language;
2. M is an $\mathcal{L}$ -structure;
3. ϕ is an $\mathcal{L}$ -formula;
4. μ and ν are M -valuations;
5. $\mu \upharpoonright \text{fvar}(\phi) = \nu \upharpoonright \text{fvar}(\phi)$.

Prove that $\|\phi\|_\mu^M = \|\phi\|_\nu^M$. □

Definition 6.1.35. Let $\mathcal{L}$ be a language.

- An $\mathcal{L}$ -formula ϕ is a **sentence** iff $\text{fvar}(\phi) = \emptyset$.
- For an $\mathcal{L}$ -sentence ϕ and an $\mathcal{L}$ -structure M , we say that M **satisfies** ϕ , written $M \models \phi$, iff the common value of $\|\phi\|_\mu^M$, for all M -valuations μ , is 1.
- An **$\mathcal{L}$ -theory** is any set of $\mathcal{L}$ -sentences.
- An $\mathcal{L}$ -structure M **satisfies** and $\mathcal{L}$ -theory T , written $M \models T$, iff it satisfies every sentence that belongs to T . □

Notation 6.1.36. We omit the outermost parentheses associated to quantifiers. For example, the abbreviation

$$(\forall v_0)(\exists v_1)\phi$$

stands for $((\forall v_0)((\exists v_1)\phi))$. □

Notation 6.1.37. If R is a binary relation symbol and t_0 and t_1 are terms, then we write $t_0 R t_1$ to be an abbreviation for $R(t_0, t_1)$. □

Example 6.1.38.

- Let $\mathcal{L}$ be the language of graphs. Then an $\mathcal{L}$ -structure is a directed graph if and only if it satisfies the sentence

$$(\forall v_0)\neg(v_0 E v_0).$$

- Let $\mathcal{L}$ be the language of orders. An $\mathcal{L}$ -structure is an order if and only if it satisfies the theory:

$$\left\{ \begin{array}{l} (\forall v_0)(v_0 \leq v_0), \\ (\forall v_0)(\forall v_1)(v_0 \leq v_1 \wedge v_1 \leq v_0 \implies v_0 = v_1), \\ (\forall v_0)(\forall v_1)(\forall v_2)(v_0 \leq v_1 \wedge v_1 \leq v_2 \implies v_0 \leq v_2) \end{array} \right\}.$$

- An order is total if and only if it satisfies the sentence

$$(\forall v_0)(\forall v_1)(v_0 \leq v_1 \vee v_1 \leq v_0).$$

- Let $\mathcal{L}$ be the language of monoids. An $\mathcal{L}$ -structure is a monoid if and only if it satisfies the **theory of monoids**:

$$\left\{ \begin{array}{l} (\forall v_0)(\forall v_1)(\forall v_2)((v_0 * v_1) * v_2 = v_0 * (v_1 * v_2), \\ (\forall v_0)(v_0 * 1 = v_0 \wedge 1 * v_0 = v_0) \end{array} \right\}.$$

- A monoid is commutative if and only if it satisfies the sentence

$$(\forall v_0)(\forall v_1)(v_0 * v_1 = v_1 * v_0).$$

□

Exercise 6.1.39. Let ϕ be an $\mathcal{L}$ -sentence and let M be an $\mathcal{L}$ -structure. Prove that

$$M \not\models \phi \iff M \models \neg \phi.$$

□

Exercise 6.1.40. Prove that a directed graph M is acyclic if and only if it satisfies the theory:

$$\{(\forall v_0)(\forall v_1) \dots (\forall v_{n+1})(v_0 E v_1 \wedge v_1 E v_2 \wedge \dots \wedge v_n E v_{n+1} \implies \neg(v_{n+1} E v_0)) \quad : \quad n \in \mathbb{N}\}.$$

□

Definition 6.1.41. Let $\mathcal{L}$ and $\mathcal{L}'$ be two languages. We say that $\mathcal{L}$ is a **reduction** of $\mathcal{L}'$, or that $\mathcal{L}'$ is an **expansion** of $\mathcal{L}$, and write $\mathcal{L} \subseteq \mathcal{L}'$, iff:

- $\mathcal{C}_{\mathcal{L}} \subseteq \mathcal{C}_{\mathcal{L}'}$;
- $\mathcal{F}_{\mathcal{L}} \subseteq \mathcal{F}_{\mathcal{L}'}$;
- $\mathcal{R}_{\mathcal{L}} \subseteq \mathcal{R}_{\mathcal{L}'}$;
- for every $\sigma \in \mathcal{F}_{\mathcal{L}} \cup \mathcal{R}_{\mathcal{L}}$, we have $\text{ar}_{\mathcal{L}}(\sigma) = \text{ar}_{\mathcal{L}'}(\sigma)$.

□

Definition 6.1.42. Suppose:

1. $\mathcal{L}$ and $\mathcal{L}'$ are two languages;
2. $\mathcal{L}$ is a reduction of $\mathcal{L}'$;
3. M is an $\mathcal{L}$ -structure;
4. M' is an $\mathcal{L}'$ -structure.

We say that M is a **reduction** of M' , or that M' is an **expansion** of M , and write $M = M' \upharpoonright \mathcal{L}$, iff:

- $[M] = [M']$;
- for every symbol σ of $\mathcal{L}$, we have $\sigma^M = \sigma^{M'}$.

□

Definition 6.1.43. The **language of groups** is the language $(\mathcal{F}, \mathcal{C}, \mathcal{R}, \text{ar})$ given by:

- $\mathcal{F} = \{*, i\}$;
- $\mathcal{C} = \{1\}$;
- $\mathcal{R} = \emptyset$;
- $\text{ar}(*) = 2$ and $\text{ar}(i) = 1$.

□

Remark 6.1.44. The language of groups is an expansion of the language of monoids. □

Definition 6.1.45. The **theory of groups** is the following theory in the language of groups:

$$\left\{ \begin{array}{l} (\forall v_0)(\forall v_1)(\forall v_2) (v_0 * v_1) * v_2 = v_0 * (v_1 * v_2), \\ (\forall v_0)(v_0 * 1 = v_0 \wedge 1 * v_0 = v_0), \\ (\forall v_0)(v_0 * i(v_0) = 1 \wedge i(v_0) * v_0 = 1) \end{array} \right\}.$$

□

Definition 6.1.46. A **group** is a structure in the language of groups that satisfies the theory of groups. □

Exercise 6.1.47. Let N be a group and let $\mathcal{L}$ be the language of monoids. Prove that $N \upharpoonright \mathcal{L}$ is a monoid. □

Exercise 6.1.48. Let M be a monoid.

- Prove that if there is at most one expansion of M (to the language of groups) which is a group.
- Prove that there is an expansion of M (to the language of groups) which is a group if and only if

$$M \models (\forall v_0)(\exists v_1)(v_0 * v_1 = 1 \wedge v_1 * v_0 = 1).$$

□

Definition 6.1.49. Let $\mathcal{L}$ be a language, let ϕ be an $\mathcal{L}$ formula, and let M be an $\mathcal{L}$ -structure. Let μ be a **partial** function from $\mathcal{V}$ to M . Then the relation $M \models \phi[\mu]$ holds iff $\text{fvar}(\phi) \subseteq \text{dom}(\mu)$ and the common value of $\|\phi\|_v^M$ is 1, for all M -valuations v extending μ . □

Notation 6.1.50. Let $\mathcal{L}$ be a language, let ϕ be an $\mathcal{L}$ formula, and let M be an $\mathcal{L}$ -structure. Let $w_0, w_1, \dots, w_{n-1}$ be **distinct** variables and let $a_0, a_1, \dots, a_{n-1} \in M$. Then $M \models \phi[w_0 = a_0, w_1 = a_1, \dots, w_{n-1} = a_{n-1}]$ holds iff, letting

$$\mu = \{(w_i, a_i) : i < n\},$$

it holds that $M \models \phi[\mu]$. □

Notation 6.1.51. Let $\mathcal{L}$ be a language, let ϕ be an $\mathcal{L}$ formula, and let M be an $\mathcal{L}$ -structure. Let $a_0, a_1, \dots, a_{n-1} \in M$. Then $M \models \phi[a_0, a_1, \dots, a_{n-1}]$ holds iff $M \models \phi[v_0 = a_0, v_1 = a_1, \dots, v_{n-1} = a_{n-1}]$ holds. □

Exercise 6.1.52. Let M be a monoid. Consider the formula

$$\phi = (\exists v_1)(v_0 * v_1 = 1 \wedge v_1 * v_0 = 1).$$

The set X is defined to be

$$\{a \in M : M \models \phi[a]\}.$$

a. Prove that X is closed in M .

Let N be the unique substructure of M whose universe is X .

b. Prove that there is a unique expansion (to the language of groups) of N which is a group. This unique expansion is denoted $M^\times$. $\square$

Exercise 6.1.53. Let E be an arbitrary set. We denote by $\mathbb{M}_E$ the following structure in the language of monoids:

- $[\mathbb{M}_E] = \mathcal{F}(E, E)$;
- for all $f, g \in [\mathbb{M}_E]$, $f *^{\mathbb{M}_E} g = f \circ g$;
- $1^{\mathbb{M}_E} = \text{Id}_E$.

Exercises:

a. Prove that $\mathbb{M}_E$ is a monoid.

We denote by $\mathbb{S}_E$ the group $\mathbb{M}_E^\times$.

b. Prove that $\mathbb{S}_E$ is the set of all bijections from E to E .

c. Prove that for all $f \in \mathbb{S}_E$, $i^{\mathbb{S}_E}(f) = f^{-1}$. $\square$

6.2 Morphisms of Structures

Definition 6.2.1. Let $\mathcal{L}$ be a language. A **morphism** of $\mathcal{L}$ -structure M and N is a function $\phi : [M] \rightarrow [N]$ satisfying:

- for every function symbol f of $\mathcal{L}$, of arity n , and all $a_0, \dots, a_{n-1} \in M$, it holds that

$$\phi(f^M(a_0, \dots, a_{n-1})) = f^N(\phi(a_0), \dots, \phi(a_{n-1}));$$

- for every constant symbol c of $\mathcal{L}$, it holds that $\phi(c^M) = c^N$;
- for every relation symbol R , of arity n , and all $a_0, \dots, a_{n-1} \in M$, it holds that

$$R^M(a_0, \dots, a_{n-1}) \implies R^N(\phi(a_0), \dots, \phi(a_{n-1})).$$

The morphism $\phi : M \rightarrow N$ is said to be **full** iff for every relation symbol R , of arity n , and all $a_0, \dots, a_{n-1} \in M$, it holds that

$$R^M(a_0, \dots, a_{n-1}) \iff R^N(\phi(a_0), \dots, \phi(a_{n-1})).$$

$\square$

Exercise 6.2.2. Let M be the structure in the language of orders given by:

- $[M] = \mathbb{N}^{<\mathbb{N}}$;
- for all $s, t \in \mathbb{N}^{<\mathbb{N}}$,

$$s \leq^M t \iff \text{dom}(s) \subseteq \text{dom}(t) \wedge t \upharpoonright \text{dom}(s) = s.$$

Let N be the structure in the language of orders given by:

- $[N] = \mathbb{N}$;
- $\leq^N$ is the usual order $\leq$ on $\mathbb{N}$.

Let $\phi : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{N}$ be given by: an element s of $\mathbb{N}^{<\mathbb{N}}$ is mapped to the unique $n \in \mathbb{N}$ such that $\text{dom}(s) = [n]$. Prove that $\phi : M \rightarrow N$ is a morphism, but not a full morphism. $\square$

Terminology 6.2.3. A morphism between orders is also called “**increasing** function”. $\square$

Definition 6.2.4. Let M and N be $\mathcal{L}$ -structures and let ϕ be a morphism between them. We say that ϕ is an **isomorphism** iff there is morphism

$$\psi : N \rightarrow M$$

such that $\psi \circ \phi = \text{Id}_M$ and $\phi \circ \psi = \text{Id}_N$. $\square$

Notation 6.2.5. We write

$$\phi : M \xrightarrow{\sim} N$$

to mean that ϕ is an isomorphism. $\square$

Exercise 6.2.6. Let M and N be $\mathcal{L}$ -structures and let $\phi : [M] \rightarrow [N]$. Prove that ϕ is an isomorphism from M to N if and only if ϕ is a full morphism and a bijection. $\square$

Exercise 6.2.7. Give an example of a language $\mathcal{L}$, two $\mathcal{L}$ -structures M and N , and a morphism $\phi : M \rightarrow N$ which is bijective, but not an isomorphism. $\square$

Exercise 6.2.8. Let M, N, P be $\mathcal{L}$ -structures. Prove:

- $\text{Id}_M : M \rightarrow M$ is an isomorphism;
- if $\phi : M \rightarrow N$ is an isomorphism, then so is $\phi^{-1} : N \rightarrow M$;
- if $\phi : M \rightarrow N$ and if $\psi : N \rightarrow P$ are isomorphisms, then so is

$$\psi \circ \phi : M \rightarrow P.$$

$\square$

Terminology 6.2.9. We say that $\mathcal{L}$ -structures M and N are **isomorphic**, and write $M \simeq N$, iff there is an isomorphism from M to N . $\square$

Exercise 6.2.10. Let M , B , and P be $\mathcal{L}$ -structures. Prove:

- a. $M \simeq M$;
- b. if $M \simeq N$, then $N \simeq M$;
- c. if $M \simeq N$ and $N \simeq P$, then $M \simeq P$. □

Exercise 6.2.11. Let $\mathcal{L}$ be a language and let T be an $\mathcal{L}$ -theory. Let M and N be isomorphic $\mathcal{L}$ -structures. Prove that $M \models T$ if and only if $N \models T$. □

Definition 6.2.12. Let M be an $\mathcal{L}$ -structure. An **automorphism** of M is any isomorphism from M to M . □

Exercise 6.2.13. Let M be an $\mathcal{L}$ -structure. Let A be the set of all automorphisms of M . Prove that A is closed in $\mathbb{S}_M$. We denote by $\text{Aut}(M)$ the unique substructure of $\mathbb{S}_M$ whose universe is A . □

Exercise 6.2.14. Let M be a monoid. For $a \in M$, we denote

$$m_a : M \rightarrow M : x \mapsto a * x.$$

Prove that the map

$$M \rightarrow \mathbb{M}_M : a \mapsto m_a$$

is an injective morphism of monoids. □

Exercise 6.2.15. Let M be a group. For $a \in M$, we denote

$$m_a : M \rightarrow M : x \mapsto a * x.$$

- a. Prove that for all $a \in M$, the map m_a is an automorphism of M .
- b. Prove that the map

$$M \rightarrow \text{Aut}(M) : a \mapsto m_a$$

is an injective morphism of groups. □

6.3 Substructures

Remark 6.3.1. Every morphism of monoids is full; every morphism of groups also. This because the fullness refers to relation symbols, of which there are none in the case of the language of monoids/groups. □

Exercise 6.3.2. Let M , N and P be $\mathcal{L}$ -structures. Prove:

- a. $\text{Id}_{[M]} : M \rightarrow M$ is a full morphism;
- b. if $\phi : M \rightarrow N$ is a bijective full morphism, then $\phi^{-1} : N \rightarrow M$ is also a full morphism;
- c. if $\phi : M \rightarrow N$ and $\psi : N \rightarrow P$ are morphisms, then so is

$$\psi \circ \phi : M \rightarrow P;$$

furthermore, if ϕ and ψ are full, then so is $\psi \circ \phi$. □

Definition 6.3.3. Let M and N be $\mathcal{L}$ -structures. Then M is a **substructure** of N iff:

- $[M] \subseteq [N]$;
- the map $i_{M,N} : M \rightarrow N : x \mapsto x$ is a morphism.

The structure M is a **full substructure** of N iff $i_{M,N}$ is a full morphism. □

Exercise 6.3.4. Let M and M' be full substructures of N such that $[M] = [M']$. Prove that $M = M'$. □

Exercise 6.3.5.

- Show by example that a substructure of an order need not be an order.
- Prove that full substructure of an order is itself an order. □

Terminology 6.3.6. A full substructure of an order is called its **suborder**. □

Exercise 6.3.7. Let L be an order. Prove that there is a **total** order L' with the same universe as L such that L is a substructure of L' . Is L a full substructure of L' ? □

Exercise 6.3.8. Let G_0 and G_1 be two directed graphs. Prove that G_0 is a substructure of G_1 if and only if $[G_0] \subseteq [G_1]$ and $E^{G_0} \subseteq E^{G_1}$. □

Exercise 6.3.9. Prove that every substructure of a directed graph is itself a directed graph. □

Terminology 6.3.10. A substructure of a directed graph is called its **subgraph**. □

Exercise 6.3.11. Prove that a substructure of a monoid is itself a monoid. □

Terminology 6.3.12. A substructure of a monoid is called its **submonoid**. □

Exercise 6.3.13. Prove that a substructure of a group is itself a group. □

Terminology 6.3.14. A substructure of a group is called its **subgroup**. □

Remark 6.3.15. It is important here that we have the symbol i in the language of groups. We could think of groups as monoids which satisfy

$$(\forall v_0)(\exists v_1)(v_0 * v_1 = 1 \wedge v_1 * v_0 = 1),$$

but then it would not be true that a substructure of a group is necessarily a group (why?). □

Definition 6.3.16. Let M be an $\mathcal{L}$ -structure and let X be a subset of $[M]$. We say that X is **closed** in M iff:

- for all function symbols f , of arity n , we have that for all $\vec{a} \in X^n$,

$$f^M(a_0, \dots, a_{n-1}) \in X;$$

- for every constant symbol c , we have that $c^M \in X$. □

Exercise 6.3.17. Let N be an $\mathcal{L}$ -structure and let M be its substructure. Prove that $[M]$ is closed in N . □

Exercise 6.3.18. Let M and N be $\mathcal{L}$ -structures and let $\phi : M \rightarrow N$ be a morphism. Prove that $\text{ran}(\phi)$ is closed in N . □

Exercise 6.3.19. Let N be an $\mathcal{L}$ -structure and let X be a closed subset of its universe. Prove that there is a unique full substructure M of N such that $[M] = X$. □

Exercise 6.3.20. Prove every monoid is isomorphic to a submonoid of $\mathbb{M}_E$, for some set E . □

Exercise 6.3.21. Prove every group is isomorphic to a subgroup of $\mathbb{S}_E$, for some set E . □

Exercise 6.3.22. Prove that every group is isomorphic to a subgroup of its automorphism group. □

6.4 Products

Terminology 6.4.1. Some reminders follow.

- A family indexed by a set I is just a function whose domain is the set I .
- If x is a family indexed by I and $i \in I$, then we write x_i to mean the same as $x(i)$.
- To make it clear that an object x is a family, we write $\vec{x}$. This arrow has no particular meaning – its role is to improve readability.
- If $\vec{x}$ is a family indexed by I , then we use the notation

$$\vec{x} = (x_i : i \in I).$$

- If $(X_i : i \in I)$ is a family of sets, then we define its product, denoted

$$\prod_{i \in I} X_i,$$

to be the set of all families $(x_i : i \in I)$ such that

$$\forall i \in I, x_i \in X_i.$$

- If $\vec{X}$ is a family of sets indexed by I , then for $i \in I$, we denote by $\pi_i^{\vec{X}}$ the canonical projection

$$\pi_i^{\vec{X}} : \prod_{j \in I} X_j \rightarrow X_i : (x_j : j \in I) \mapsto x_i.$$

If $\vec{X}$ is clear from the context, then we omit it from the superscript.

- The product of the family $(X : i \in I)$ is equal to $\mathcal{F}(I, X)$, and it is also denoted by X^I . □

Exercise 6.4.2. Let $(X_i : i \in I)$ be a family of non-empty sets. Prove that

$$\prod_{i \in I} X_i \neq \emptyset.$$

Hint. You will have to use the Axiom of Choice. □

Definition 6.4.3. Let $(M_i : i \in I)$ be a family of $\mathcal{L}$ -structures. Then we define $\prod_{i \in I} M_i$ to be the structure N given by:

- $[N] = \prod_{i \in I} [M_i]$;
- for every operation symbol f , of arity n , and all $a^0, a^1, \dots, a^{n-1} \in [N]$, it holds that

$$f^N(a^0, a^1, \dots, a^{n-1}) = (f^{M_i}(a_i^0, \dots, a_i^{n-1}))_{i \in I};$$

- for every constant symbol c , it holds that

$$c^N = (c^{M_i})_{i \in I};$$

- for every operation symbol R , of arity n , and all $a^0, a^1, \dots, a^{n-1} \in [N]$,

$$R^N(a^0, a^1, \dots, a^{n-1})$$

holds if and only if

$$(\forall i \in I) R^{M_i}(a_i^0, \dots, a_i^{n-1}).$$

□

Exercise 6.4.4. Let $(M_i : i \in I)$ be a family of monoids. Prove that its product is a monoid. □

Exercise 6.4.5. Let $(M_i : i \in I)$ be a family of groups. Prove that its product is a group. □

Exercise 6.4.6. Let $(M_i : i \in I)$ be a family of $\mathcal{L}$ -structures. Prove that for all $i_0 \in I$, the map

$$\pi_{i_0} : \prod_{i \in I} M_i \rightarrow M_{i_0}$$

is a morphism. □

Exercise 6.4.7. Suppose:

1. $(M_i : i \in I)$ is a family of $\mathcal{L}$ -structures;
2. N is an $\mathcal{L}$ -structure;
3. $(\phi_i : i \in I)$ is a family of morphisms

$$\phi_i : N \rightarrow M_i$$

(all $i \in I$).

Prove that there is a unique morphism

$$\psi : N \rightarrow \prod_{i \in I} M_i$$

such that for all $i \in I$, the diagram

$$\begin{array}{ccc} N & \xrightarrow{\psi} & \prod_{j \in I} M_j \\ & \searrow \phi_i & \downarrow \pi_i \\ & & M_i \end{array}$$

commutes. □

Definition 6.4.8. In the previous exercise, the morphism ψ is called the **tupling** of the family $\vec{\phi}$, and is denoted by

$$\langle \phi_i \rangle_{i \in I}.$$

□

Exercise 6.4.9. Suppose:

1. $(M_i : i \in I)$ is a family of $\mathcal{L}$ -structures;
2. N is an $\mathcal{L}$ -structure;
3. $(\phi_i : i \in I)$ is a family of morphisms

$$\phi_i : N \rightarrow M_i$$

(all $i \in I$).

Prove that for all $z \in N$, we have

$$\langle \phi_i \rangle_{i \in I}(z) = (\phi_i(z))_{i \in I}.$$

□

Notation 6.4.10. The product of a family $(M_i : i \in I)$ of $\mathcal{L}$ -structures is denoted by M^I . □

6.5 Algebraic Structures and Theories

Definition 6.5.1. We use terms “**operational language**” and “**algebraic language**” interchangeably and to mean “the language with no relation symbols”. □

Remark 6.5.2. In other words, algebraic language is one where $\mathcal{R} = \emptyset$. Notice that we always have the equality symbol. □

Terminology 6.5.3. An **algebraic structure** is a structure for an algebraic language. □

Definition 6.5.4. A **relation language** is one that has neither function nor constant symbols. □

Example 6.5.5. The language of monoids and the language of groups are algebraic languages, while the languages of orders and graphs are relational. The only language which is both algebraic and relational is the language with no symbols, also called the **empty language**. In the empty language, every atomic formula is of the form $\perp$ or $v_i = v_j$. $\square$

Example 6.5.6. Monoids and groups are both examples of algebraic structures. $\square$

Definition 6.5.7. Let $\mathcal{L}$ be a language and let ϕ be an $\mathcal{L}$ -formula. Its **universal closure** is defined to be the sentence

$$(\forall v_{i_0})(\forall v_{i_1}) \dots (\forall v_{i_{n-1}})\phi,$$

where $i_0 < i_1 < \dots < i_{n-1}$ are the natural numbers such that

$$\text{fvar}(\phi) = \{i_k : k < n\}.$$

$\square$

Definition 6.5.8. An **algebraic identity** (of a given language) is any formula of the form $s = t$, where s and t are terms. $\square$

Remark 6.5.9. Algebraic identities are **not** necessarily sentences. When we need to work with sentences, we replace them by their universal closure. $\square$

Definition 6.5.10. Let T be a theory and let ϕ be an algebraic identity. Then $T \models \phi$ holds iff $T \models \phi'$ holds, where ϕ' is the universal closure of ϕ . $\square$

Exercise 6.5.11. Let T be a theory and let $s = t$ be an algebraic identity. Prove that $T \models (s = t)$ iff for every model M of T and every M -valuation μ , we have $s^M[\mu] = t^M[\mu]$. $\square$

Definition 6.5.12. An **algebraic theory** is a theory in an algebraic language consisting only of universal closures of algebraic identities. $\square$

Example 6.5.13. The theory of monoids and the theory of groups are examples of algebraic theories. $\square$

Exercise 6.5.14. Let $\mathcal{L}$ be an algebraic language and let T be an $\mathcal{L}$ -theory. Prove:

- a. if N is an $\mathcal{L}$ -structure that satisfies T and if M is a substructure of N , then N satisfies T ;
- b. the product of an arbitrary family of models of T is a model of T . $\square$

Remark 6.5.15. From the previous exercise we immediately get that a substructure of a monoid is a monoid and that a substructure of a group is a group, but also much more. $\square$

Exercise 6.5.16. Let $\mathcal{L}$ be an algebraic language and let T be an algebraic $\mathcal{L}$ -theory. Suppose M is an arbitrary $\mathcal{L}$ -structure, N is a model of T , and $\phi : M \rightarrow N$ is a morphism. Prove that M is also a model of T . $\square$

6.6 Congruences

Definition 6.6.1. Let $\mathcal{L}$ be an algebraic language. A **congruence** on an $\mathcal{L}$ -structure M is an equivalence relation $\equiv$ on its universe such that for every function symbol f , of arity n , we have

$$(\forall \vec{x} \in [M]^n)(\forall \vec{y} \in [M]^n)(x_0 \equiv y_0 \wedge \cdots \wedge x_{n-1} \equiv y_{n-1} \implies f^M(\vec{x}) \equiv f^M(\vec{y})).$$

□

Exercise 6.6.2. The **language of rings** the algebraic language given by:

- $\mathcal{F}_{\mathcal{L}} = \{+, \cdot\};$
- $\text{ar}_{\mathcal{L}}(+) = \text{ar}_{\mathcal{L}}(\cdot) = 2;$
- $\mathcal{C}_{\mathcal{L}} = \{0, 1\}.$

The **ring of integers** Z is the structure in the language of rings given by:

- $[Z] = \mathbb{Z};$
- $+^Z$ is the usual addition of integers;
- $\cdot^Z$ is the usual multiplication of integers;
- $0^Z = 0$ and $1^Z = 1.$

Let $n \in \mathbb{N} - \{0\}$ be given. Prove that $\equiv_n$ is a congruence on Z .

□

Exercise 6.6.3. Let G be a group and let H be its subgroup. Recall that H is a **normal subgroup** iff

$$(\forall g \in G)(gH = Hg).$$

Now, define the binary relation $\equiv_H$ on G as follows:

$$(\forall x, y \in G)(x \equiv_H y \iff (\exists h \in H)(x = hy)).$$

- a. Prove that $\equiv_H$ is an equivalence relation on G .
- b. Prove that $\equiv_H$ is a congruence on G if and only if H is a **normal** subgroup of G .

□

Remark 6.6.4. Let $\mathcal{L}$ be an algebraic language. Notice:

- every morphism of $\mathcal{L}$ -structures is full;
- every substructure of an $\mathcal{L}$ -structure is a full substructure.

□

Exercise 6.6.5. Let $\mathcal{L}$ be an algebraic language, let M be an $\mathcal{L}$ -structure, and let E be an equivalence relation on $[M]$.

- a. Prove that there exists an $\mathcal{L}$ -structure N such that $[N] = [M]/E$ and $\pi_E : M \rightarrow N$ is a morphism if and only if E is a congruence on M .

- b. Suppose now that E is a congruence on M . Prove that there exists a **unique** $\mathcal{L}$ -structure N such that $[N] = [M]/E$ and $\pi_E : M \rightarrow N$ is a morphism. $\square$

Definition 6.6.6. Let $\mathcal{L}$ be an algebraic language, let M be an $\mathcal{L}$ -structure, and let E be a congruence on it. The **quotient** of M by E , denoted also M/E , is the structure is the unique $\mathcal{L}$ -structure N such that $[N] = [M]/E$ and $\pi_E : M \rightarrow N$ is a morphism. $\square$

Exercise 6.6.7. Let $\mathcal{L}$ be an algebraic language and let T be an algebraic theory. Let M be a model of T and let E be a congruence relation on M . Prove that $M/E \models T$. $\square$

Remark 6.6.8. Thus, a quotient of a group by congruence is a group. $\square$

Exercise 6.6.9. Let $\mathcal{L}$ be an algebraic language, let M and N be algebraic structures, and let ϕ be a morphism between them.

- a. Prove that $\equiv_\phi$ is a congruence on M .
b. Prove that there is a unique morphism

$$\phi_* : M / \equiv_\phi \rightarrow N$$

such that the diagram

$$\begin{array}{ccc} M & \xrightarrow{\phi} & N \\ \downarrow \pi_{\equiv_\phi} & \nearrow \phi_* & \\ M / \equiv_\phi & & \end{array}$$

commutes.

- c. Prove that ϕ_* is injective. $\square$

Exercise 6.6.10. Let $\mathcal{L}$ be an algebraic language, let M and N be algebraic structures, let ϕ be a morphism between them, and let E be a congruence on M .

- a. Prove that there is a morphism

$$\phi_* : M/E \rightarrow N$$

such that the diagram

$$\begin{array}{ccc} M & \xrightarrow{\phi} & N \\ \downarrow \pi_E & \nearrow \phi_* & \\ M/E & & \end{array}$$

commutes if and only if $E \subseteq \equiv_\phi$.

b. Suppose $E \subseteq \equiv_\phi$. Prove that there is a **unique** morphism

$$\phi_* : M/E \rightarrow N$$

such that the diagram

$$\begin{array}{ccc} M & \xrightarrow{\phi} & N \\ \downarrow \pi_E & \nearrow \phi_* & \\ M/E & & \end{array}$$

commutes.

c. Prove that ϕ_* is injective if and only if $E = \equiv_\phi$.

d. Prove that $\text{ran}(\phi_*) = \text{ran}(\phi)$. □

Exercise 6.6.11. Let $\mathcal{L}$ be an algebraic language, let M and N be $\mathcal{L}$ -structures, and let ϕ be a morphism between them.

a. Prove that $\text{ran}(\phi)$ is closed N .

We denote by $\text{Im}(\phi)$ the unique substructure of N whose domain is $\text{ran}(\phi)$.

b. Prove that

$$\phi_* : M / \equiv_\phi \xrightarrow{\sim} \text{Im}(\phi)$$

(cf. Exercise 6.6.10). □

7 Boolean Algebras

7.1 Boolean Algebras

Definition 7.1.1. The **language of boolean algebras** is the language $(\mathcal{F}, \mathcal{C}, \mathcal{R}, \text{ar})$ given by:

- $\mathcal{F} = \{\wedge, \vee, \neg\}$;
- $\mathcal{C} = \{0, 1\}$;
- $\mathcal{R} = \emptyset$;
- $\text{ar}(\wedge) = \text{ar}(\vee) = 2$ and $\text{ar}(\neg) = 1$. □

Definition 7.1.2. The **theory of boolean algebras** is the theory in the language of boolean algebras consisting of the universal closures of the following algebraic identities:

- $v_0 \wedge v_1 = v_1 \wedge v_0$ (commutativity of the meet);
- $v_0 \vee v_1 = v_1 \vee v_0$ (commutativity of the join);

- $(v_0 \wedge v_1) \wedge v_2 = v_0 \wedge (v_1 \wedge v_2)$ (associativity of the meet);
- $(v_0 \vee v_1) \vee v_2 = v_0 \vee (v_1 \vee v_2)$ (associativity of the join);
- $v_0 \wedge (v_1 \vee v_2) = (v_0 \wedge v_1) \vee (v_0 \wedge v_2)$ (distributivity of the meet towards the join);
- $v_0 \vee (v_1 \wedge v_2) = (v_0 \vee v_1) \wedge (v_0 \vee v_2)$ (distributivity of the join towards the meet);
- $v_0 \wedge (v_0 \vee v_1) = v_0$ (the absorption law for the meet);
- $v_0 \vee (v_0 \wedge v_1) = v_0$ (the absorption law for the join);
- $v_0 \wedge \neg v_0 = 0$ (the complementation law for the meet);
- $v_0 \vee \neg v_0 = 1$ (the complementation law for the join). □

Terminology 7.1.3. In a boolean algebra B , the operation $\wedge^B$ is called the **meet**, the operation $\vee^B$ is called the **join**, and the operation $\neg^B$ is called the **complementation**. □

Definition 7.1.4. A **boolean algebra** is a model of the theory of boolean algebras. □

Fact 7.1.5.

- The language of boolean algebras is an algebraic language.*
- The theory of boolean algebras is an algebraic theory.* □

Corollary 7.1.6.

- A substructure of a boolean algebra is a boolean algebra.*
- The product of a family of boolean algebras is a boolean algebra.*
- The quotient of a boolean algebra by a congruence is a boolean algebra.* □

Exercise 7.1.7. Let E be a set. The structure M for the language of boolean algebras is given as follows:

- $[M] = \mathcal{P}(E)$;
- for all $X, Y \in [M]$, $X \wedge^M Y = X \cap Y$;
- for all $X, Y \in [M]$, $X \vee^M Y = X \cup Y$;
- for all $X \in [M]$, $\neg^M X = X^c$;
- $0^M = \emptyset$ and $1^M = E$.

Prove that M is a boolean algebra. When there is no danger of confusion, we denote this boolean algebra simply by $\mathcal{P}(E)$. □

Exercise 7.1.8. The structure **2** in the language of boolean algebras is given by:

- $[2] = \{0, 1\}$;
- for all $x, y \in \{0, 1\}$, $x \wedge^2 y = \min\{x, y\}$;
- for all $x, y \in \{0, 1\}$, $x \vee^2 y = \max\{x, y\}$;
- for all $x \in \{0, 1\}$, $\neg^2 x = 1 - x$;
- $0^2 = 0$ and $1^2 = 1$.

Prove that 2 is a boolean algebra. □

Exercise 7.1.9. Let E be an **infinite** set and let $\sim$ the binary relation on $\mathcal{P}(E)$ given by:

$$(\forall X, Y \in \mathcal{P}(E))(X \sim Y \iff (X - Y) \cup (Y - X) \text{ is finite}).$$

Prove that $\sim$ is a congruence on the boolean algebra $\mathcal{P}(E)$. The quotient algebra $\mathcal{P}(E)/\text{Fin}$ for the quotient boolean algebra. □

Definition 7.1.10. Let B be a boolean algebra. The structure $\check{B}$ in the language of boolean algebras is defined as follows:

- $[\check{B}] = [B]$;
 - $\wedge^{\check{B}} = \vee^B$;
 - $\vee^{\check{B}} = \wedge^B$;
 - $\neg^{\check{B}} = \neg^B$;
 - $0^{\check{B}} = 1^B$ and $1^{\check{B}} = 0^B$.
-

Exercise 7.1.11. Let B be a boolean algebra. Prove that $\check{B}$ is also a boolean algebra. □

Terminology 7.1.12. Let B be a boolean algebra. The structure $\check{B}$ is called the **dual boolean algebra** of B . □

Exercise 7.1.13. Let B be a boolean algebra. Prove that $\check{\check{B}} = B$. □

Definition 7.1.14. Let $\mathcal{L}$ be the language of boolean algebras. The function $t \mapsto \check{t}$ from the set of $\mathcal{L}$ -terms to itself is defined by recursion on complexity of a term t as follows:

- if t is a variable, then $\check{t} = t$;
- if $t = 0$, then $\check{t} = 1$;
- if $t = 1$, then $\check{t} = 0$;
- if t is of the form $t_0 \wedge t_1$, then $\check{t}$ is the term $\check{t}_0 \vee \check{t}_1$;
- if t is of the form $t_0 \vee t_1$, then $\check{t}$ is the term $\check{t}_0 \wedge \check{t}_1$;

- if t is of the form $\neg s$, then $\check{t}$ is the term $\neg \check{s}$. □

Exercise 7.1.15. Let t be a term in the language of boolean algebras. Prove that the term $\check{\check{t}}$ is equal to the term t . □

Exercise 7.1.16. Let B be a boolean algebra and let $\mu : \mathcal{V} \rightarrow B$. Prove that for every term t , we have $t^B[\mu] = \check{t}^{\check{B}}[\mu]$. □

Fact 7.1.17. Let T be the theory of boolean algebras and let $s = t$ be an algebraic identity. If $T \models (s = t)$, then $T \models (\check{s} = \check{t})$.

Proof. Let B be an arbitrary boolean algebra and let $\mu : \mathcal{V} \rightarrow B$. We want to see that $\check{s}^B[\mu] = \check{t}^B[\mu]$ (see Exercise 6.5.11). We compute:

$$\check{s}^B[\mu] = \check{\check{s}}^{\check{B}}[\mu] = s^{\check{B}}[\mu] = t^{\check{B}}[\mu] = \check{t}^{\check{\check{B}}}[\mu] = \check{t}^B[\mu];$$

here are the justifications of the equalities (in order):

1. Exercise 7.1.16;
2. Exercise 7.1.15;
3. because $\check{B} \models T$ and $T \models (s = t)$;
4. Exercise 7.1.16;
5. Exercise 7.1.13.

□

Fact 7.1.18 (Identity laws). Let B be a boolean algebra and let $x \in B$. Then $x \wedge 1 = x$ and $x \vee 0 = x$.

Proof. Let T be the theory of boolean algebras. We are actually claiming that $T \models (v_0 \wedge 1 = v_0)$ and $T \models (v_0 \vee 0 = v_0)$. Let us prove the first relation.

Let B be a boolean algebra and let $x \in B$. We want to prove that $x \wedge 1 = x$. We use the complementation law for the meet, followed by the absorption law for the meet:

$$x \wedge 1 = x \wedge (x \vee \neg x) = x.$$

Thus, $T \models (v_0 \wedge 1 = v_0)$. By Fact 7.1.17, we immediately get that $T \models (v_0 \vee 0 = v_0)$. □

Remark 7.1.19. We will normally shorten the argument of the above kind as follows: we prove that $x \wedge 1 = x$, and then we say: “The second part follows by duality.” □

Fact 7.1.20 (Uniqueness of complements). Let B be a boolean algebra and let $x, y \in B$. If $x \vee y = 1$ and $x \wedge y = 0$, then $x = \neg y$.

Proof. We use the identity, complement, distributivity laws, then the assumption, then the commutativity law:

$$x = x \wedge 1 = x \wedge (y \vee \neg y) = (x \wedge y) \vee (x \wedge \neg y) = 0 \vee (x \wedge \neg y) = (x \wedge \neg y) \vee 0 =$$

then we use the complement, commutativity, distributivity laws, then the assumption, then the identity law:

$$= (x \wedge \neg y) \vee (y \wedge \neg y) = (\neg y \wedge x) \vee (\neg y \wedge y) = \neg y \wedge (x \vee y) = \neg y \wedge 1 = \neg y.$$

□

Corollary 7.1.21 (Double-complement law). *Let B be a boolean algebra and let $x \in B$. Then $\neg\neg x = x$.*

Proof. By the complement laws:

$$x \vee \neg x = 1 \text{ and } x \wedge \neg x = 0.$$

By the uniqueness of complement;

$$x = \neg\neg x.$$

□

Corollary 7.1.22. *Let B be a boolean algebra. Then $\neg 0 = 1$ and $\neg 1 = 0$.*

Proof. We show the first equality, whereas the second one follows by duality. We have that $1 \vee 0 = 1$ by the identity law for $\vee$ and that $1 \wedge 0 = 0$ by the identity law for $\wedge$. It follows by uniqueness of complement that $1 = \neg 0$. □

Fact 7.1.23 (Domination laws). *Let B be a boolean algebra and let $x \in B$. Then $x \vee 1 = 1$ and $x \wedge 0 = 0$.*

Proof. We prove the first equality – the second follows by duality. We compute:

$$1 = 1 \wedge (1 \vee x) = (1 \vee x) \wedge 1 = 1 \vee x = x \vee 1;$$

by the absorption, commutativity, identity, commutativity laws. □

Fact 7.1.24 (De Morgan laws). *Let B be a boolean algebra and let $x, y \in B$. Then*

$$\neg(x \wedge y) = \neg x \vee \neg y,$$

$$\neg(x \vee y) = \neg x \wedge \neg y.$$

Proof. We prove the first equality – the second follows by duality. By the uniqueness of complements, it suffices to prove:

$$(\neg x \vee \neg y) \vee (x \wedge y) = 1;$$

$$(\neg x \vee \neg y) \wedge (x \wedge y) = 0.$$

We prove the first equality – the second is an **exercise**. We compute:

$$\begin{aligned} (\neg x \vee \neg y) \vee (x \wedge y) &= \neg x \vee (\neg y \vee (x \wedge y)) = \neg x \vee ((\neg y \vee x) \wedge (\neg y \vee y)) = \\ &= \neg x \vee ((\neg y \vee x) \wedge 1) = \neg x \vee (\neg y \vee x) = \neg x \vee (x \vee \neg y) = (\neg x \vee x) \vee \neg y = 1 \vee \neg y = 1. \end{aligned}$$

□

Exercise 7.1.25. Let B be a boolean algebra. Prove that the map

$$B \rightarrow \check{B} : x \mapsto \neg x$$

is an isomorphism of boolean algebras. □

Exercise 7.1.26 (Idempotence laws). Let B be a boolean algebra and $x \in B$. Prove that $x \vee x = x$ and $x \wedge x = x$. □

Exercise 7.1.27. Let B be a boolean algebra such that $0^B = 1^B$. Prove that B has exactly one element.

Comment. Another way to express this is to say

$$T, 0 = 1 \models (\forall v_0)(v_0 = 0).$$

□

Definition 7.1.28. A boolean algebra B is **proper** iff $0^B \neq 1^B$. □

Remark 7.1.29. We care mostly about proper boolean algebras. The reason why we did not include the sentence “ $\neg(0 = 1)$ ” in the theory of boolean algebras is because this sentence is not an algebraic identity (indeed, it is the negation of one). □

Exercise 7.1.30. Let B be a boolean algebra.

a. Prove that B is proper if and only if there is an injective morphism $f : \mathbf{2} \rightarrow B$.

b. Prove that if B is proper, then there is a **unique** injective morphism $f : \mathbf{2} \rightarrow B$. □

Exercise 7.1.31. Let E be an arbitrary set. Prove that the boolean algebra 2^E is isomorphic to the boolean algebra $\mathcal{P}(E)$. □

7.2 Lindenbaum Algebras

Definition 7.2.1. On the set $\text{Prop}_{\mathbb{P}}$, we define binary operations $\dot{\wedge}$ and $\dot{\vee}$, a unary operation $\dot{\neg}$, and constants $\dot{0}$ and $\dot{1}$ as follows:

- $\forall \phi, \psi \in \text{Prop}_{\mathbb{P}}, \phi \dot{\wedge} \psi = (\phi \wedge \psi);$
- $\forall \phi, \psi \in \text{Prop}_{\mathbb{P}}, \phi \dot{\vee} \psi = (\phi \vee \psi);$
- $\forall \phi \in \text{Prop}_{\mathbb{P}}, \dot{\neg} \phi = (\neg \phi);$
- $\dot{0} = \perp;$
- $\dot{1} = (\neg \perp).$ □

Notation 7.2.2. When there is no risk of confusion, we will denote by $\text{Prop}_{\mathbb{P}}$ the structure M for the language of boolean algebras given by:

- $[M] = \text{Prop}_{\mathbb{P}};$

- $\wedge^M = \dot{\wedge}$;
- $\vee^M = \dot{\vee}$;
- $\neg^M = \dot{\neg}$;
- $0^M = \dot{0}$ and $1^M = \dot{1}$.

□

Exercise 7.2.3. Prove that the structure $\text{Prop}_{\mathbb{P}}$ is **not** a boolean algebra.

□

Remark 7.2.4. Notice that a propositional letter p can be understood as either the character ‘ p ’ or the string “ p ”. Even though these objects are formally distinct, we will, nevertheless, identify them. In this sense, $\mathbb{P}$ is both a subset of $\text{Prop}_{\mathbb{P}}$ and the domain of every valuation.

□

Notation 7.2.5. Let $\mu : \mathbb{P} \rightarrow 2$ be a valuation. We denote by $\hat{\mu}$ the function

$$\text{Prop}_{\mathbb{P}} \rightarrow 2 : \phi \mapsto \|\phi\|_{\mu}.$$

□

Exercise 7.2.6. Let $\mu : \mathbb{P} \rightarrow 2$ be a valuation. Prove $\hat{\mu} : \text{Prop}_{\mathbb{P}} \rightarrow 2$ is a morphism.

□

Notation 7.2.7. Let T be a propositional theory. We denote by $\text{Mod}(T)$ the set of all models of T , i.e.

$$\text{Mod}(T) = \{\mu \in \mathcal{F}(\mathbb{P}, 2) : \mu \models T\}.$$

If $T = \emptyset$, then we write Mod for $\text{Mod}(T)$.

□

Exercise 7.2.8. Let T be a propositional theory. Consider the morphism:

$$\langle \hat{\mu} \rangle_{\mu \in \text{Mod}(T)} : \text{Prop}_{\mathbb{P}} \rightarrow 2^{\text{Mod}(T)} : \phi \mapsto (\|\phi\|_{\mu} : \mu \in \text{Mod}(T)).$$

Prove that

$$\equiv_{\langle \hat{\mu} \rangle_{\mu \in \text{Mod}(T)}} = \equiv_T.$$

Conclude that the structure $\text{Prop}_{\mathbb{P}} / \equiv_T$ is a boolean algebra, isomorphic to a subalgebra of $\mathcal{P}(\text{Mod}(T))$.

□

Definition 7.2.9. Let T be a propositional theory. We define the **Lindenbaum algebra** of T , denoted $\mathbb{L}_T$, to be the boolean algebra $\text{Prop}_{\mathbb{P}} / \equiv_T$. If $T = \emptyset$, then we write $\mathbb{L}$ for $\mathbb{L}_T$.

□

Exercise 7.2.10. Let T and S be propositional theories.

- a. Prove if $T \subseteq S$, then there is a unique morphism of boolean algebras $f : \mathbb{L}_T \rightarrow \mathbb{L}_S$ such that the diagram

$$\begin{array}{ccc} \text{Prop}_{\mathbb{P}} & \xrightarrow{\pi_{\equiv_S}} & \mathbb{L}_S \\ \downarrow \pi_{\equiv_T} & \nearrow f & \\ \mathbb{L}_T & & \end{array}$$

commutes.

b. Show by example that the converse may fail in general. □

Exercise 7.2.11. We define the *closure for semantic consequences* to be the function

$$\text{scl} : \mathcal{P}(\text{Prop}_{\mathcal{P}}) \rightarrow \mathcal{P}(\text{Prop}_{\mathcal{P}}) : T \mapsto \{\phi \in \text{Prop}_{\mathcal{P}} : T \models \phi\}.$$

Prove that for all theories T and S :

- a. $T \subseteq \text{scl}(T)$;
- b. if $T \subseteq S$, then $\text{scl}(T) \subseteq \text{scl}(S)$;
- c. $\text{scl}(\text{cl}(T)) = \text{scl}(T)$;
- d. T and S are semantically equivalent if and only if $\text{scl}(T) = \text{scl}(S)$;
- e. the following are equivalent:
 - i. $T \subseteq \text{cl}(S)$;
 - ii. there is a unique morphism of boolean algebras $f : \mathbb{L}_T \rightarrow \mathbb{L}_S$ such that the diagram

$$\begin{array}{ccc} \text{Prop}_{\mathcal{P}} & \xrightarrow{\pi_{\equiv_S}} & \mathbb{L}_S \\ \downarrow \pi_{\equiv_T} & \nearrow f & \\ \mathbb{L}_T & & \end{array}$$

commutes.

□

Remark 7.2.12. Notice that propositional formulas ϕ and ψ are semantically equivalent if and only if their equivalence classes are equal in $\mathbb{L}$ (i.e. $[\phi]_{\equiv} = [\psi]_{\equiv}$). So:

- to show $\phi \equiv \psi$, it suffices to exhibit a computation in $\mathbb{L}$ showing $[\phi] = [\psi]$;
- to show that ϕ is a tautology, it suffices to exhibit a computation in $\mathbb{L}$ showing $[\phi] = 1$.

The following exercise is helpful to translate away $\implies$ and $\iff$. □

Exercise 7.2.13. Prove that in $\mathbb{L}$:

$$[\phi \implies \psi] = \neg[\phi] \vee [\psi];$$

$$[\phi \iff \psi] = ([\phi] \wedge [\psi]) \vee (\neg[\phi] \wedge \neg[\psi]).$$

□

Exercise 7.2.14. Through a computation in $\mathbb{L}$, prove:

$$\neg(\phi \implies \neg\psi) \equiv \phi \wedge \psi.$$

□

Exercise 7.2.15.

a. Prove that in every boolean algebra B , one has, for all $a, b \in B$,

$$\neg(a \vee (\neg a \wedge b)) = \neg a \wedge \neg b.$$

b. Conclude that for all $\phi, \psi \in \text{Prop}_{\mathbf{P}}$,

$$\neg(\phi \vee (\neg\phi \wedge \psi)) \equiv \neg\phi \wedge \neg\psi.$$

□

Exercise 7.2.16.

a. Prove that in every boolean algebra B , one has, for all $a, b \in B$,

$$a \wedge b \leq a \vee b.$$

b. Conclude that for all $\phi, \psi \in \text{Prop}_{\mathbf{P}}$, the formula

$$\phi \wedge \psi \implies \phi \vee \psi$$

is a tautology.

□

7.3 Boolean Algebras as Orders

Definition 7.3.1. Let B be a boolean algebra. Then its **canonical order relation** is the binary relation $\leq_B$ given by:

$$\forall x, y \in B, (x \leq_B y \iff x \wedge y = x).$$

□

Exercise 7.3.2. Let B be a boolean algebra. Prove:

a. $\leq_B$ is an order relation;

b. for all $x, y \in B$,

$$x \leq_B y \iff x \vee y = y,$$

$$x \leq_B y \iff x \wedge \neg y = 0,$$

$$x \leq_B y \iff \neg x \vee y = 1;$$

c. for all $x, y, z \in B$,

$$x \leq_B y \implies x \vee z \leq_B y \vee z,$$

$$x \leq_B y \implies x \wedge z \leq_B y \wedge z,$$

$$x \leq_B y \implies \neg y \leq_B \neg x;$$

d. for all $x \in B$, $0 \leq_B x \leq_B 1$;

e. for all $x \in B$,

$$0 <_B x <_B 1 \iff x \leqslant_B \neg x \text{ and } \neg x \leqslant_B x.$$

f. $(B, \leqslant_B)$ is total if and only if $B = 2$. □

Exercise 7.3.3. Prove:

a. the canonical order relation of $\mathcal{P}(E)$ is equal to $\subseteq$;

b. the canonical order relation of 2 is equal to the order relation inherited from the real numbers. □

Exercise 7.3.4. Let E be an infinite set. Prove that the canonical order relation $\leqslant$ on $\mathcal{P}(E)/\text{Fin}$ satisfies that for all $X, Y \in \mathcal{P}(E)$, the relation $[X] \leqslant [Y]$ holds if and only if the set $X - Y$ is finite. □

Exercise 7.3.5. Prove that in $\mathbb{L}_T$, the canonical order relation satisfies that for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$

$$[\phi]_T \leqslant [\psi]_T \iff T, \phi \models \psi.$$

□

Exercise 7.3.6. Prove that the order $(\mathbb{L}, \leqslant)$ is not total and not well-founded. Furthermore, prove that there is an infinite subset $A \subseteq \mathbb{L}$ such that every two elements of A are incomparable. □

Exercise 7.3.7. Let $\mathbb{B}$ be a boolean algebra and let $\leqslant$ be the canonical order relation on it. Prove that the canonical order relation on the dual algebra of $\mathbb{B}$ is exactly the order relation $\geqslant$. □

Remark 7.3.8. We started from boolean algebras as algebraic structures and derived a canonical order relation on them. We could have also proceeded in reverse, i.e. start from an order with right properties and derive from it the algebraic structure of a boolean algebra. The orders with the “right properties” are so-called distributive-complementing lattices. We elaborate on this below. □

Definition 7.3.9. Suppose:

1. $(E, \leqslant)$ is an order;
2. A is a subset of E ;
3. η is an element of E .

We say that:

- η is an **upper bound** for A , written $A \leqslant \eta$, iff $\forall a \in A, a \leqslant \eta$;
- η is an **lower bound** for A , written $\eta \leqslant A$, iff $\forall a \in A, \eta \leqslant a$. □

Definition 7.3.10. Suppose:

1. $(E, \leqslant)$ is an order;
2. A is a subset of E ;

3. ξ is an element of E .

We say that:

- ξ is a **supremum** of A iff it is the minimum of the set of all upper bounds of A ;
- ξ is an **infimum** of A iff it is the maximum of the set of all lower bounds of A . □

Exercise 7.3.11. Prove that a set can have at most one infimum and at most one supremum. Give an example of an order and its subset which has neither. □

Notation 7.3.12. If A has a supremum, then this supremum is unique, and we denote it by **sup** A . If we need to make explicit the order $(E, \leq)$ in which we are computing the supremum, we write

$$\sup_{(E, \leq)} A.$$

Similarly, in the case of an infimum, we use the notation **inf** A . □

Exercise 7.3.13. Let $A = \{q \in \mathbb{Q}_{\geq 0} : q^2 < 2\}$.

- a. Prove that A has no supremum in $(\mathbb{Q}, \leq)$.
- b. Prove that $\sup A = \sqrt{2}$ in $(\mathbb{R}, \leq)$. □

Exercise 7.3.14. Suppose:

1. $(E, \leq)$ is an order;
2. A is a subset of E ;
3. ξ is an element of E .

Prove that $\xi = \sup A$ if and only if

$$\forall \eta \in E, (A \leq \eta \iff \xi \leq \eta).$$

□

Exercise 7.3.15. Suppose:

1. $(E, \leq)$ is an order;
2. A is a subset of E which has a maximum.

Prove that A has a supremum and $\sup A = \max A$. □

Definition 7.3.16. An order $(E, \leq)$ is a **lattice** iff for all $a, b \in E$, the set $\{a, b\}$ has a supremum and an infimum. □

Exercise 7.3.17. Let B be a boolean algebra. Prove that $(B, \leq_B)$ is a lattice. Furthermore, prove that for all $x, y \in B$, one has

$$x \wedge y = \inf\{x, y\},$$

$$x \vee y = \sup\{x, y\}.$$

□

Exercise 7.3.18 (Xiang Li). Let $(E, \leq)$ be an order.

- a. Prove that $(E, \leq)$ is a lattice if and only if there are binary operations $\wedge$ and $\vee$ on E which satisfy the commutativity, associativity, and absorption laws, and such that

$$(\forall a, b \in E)(a \leq b \iff a \wedge b = a).$$

- b. Prove that if $(E, \leq)$ is a lattice, then there are **unique** binary operations $\wedge$ and $\vee$ on E which satisfy the commutativity, associativity, and absorption laws, and such that

$$(\forall a, b \in E)(a \leq b \iff a \wedge b = a).$$

□

Notation 7.3.19. In an arbitrary lattice $(E, \leq)$, we write $a \wedge b$ for $\inf\{a, b\}$ and $a \vee b$ for $\sup\{a, b\}$. □

Exercise 7.3.20. Prove that every total order is a lattice. □

Exercise 7.3.21. We denote by P the set of prime natural numbers. For $n \in \mathbb{N} - \{0\}$ and $p \in P$, we define $v_p(n)$ to be the greatest $k \in \mathbb{N}$ such that $p^k | n$.

- a. Prove that for all $n \in \mathbb{N} - \{0\}$, the set $\{p \in P : v_p(n) > 0\}$ is finite.

- b. Prove that for all finite sets $F \subseteq P$ such that

$$\{p \in P : v_p(n) > 0\} \subseteq F,$$

the product $\prod_{p \in F} p^{v_p(n)}$ has the same value.

This means that for $n \in \mathbb{N} - \{0\}$, we can make sense of the product

$$\prod_{p \in P} p^{v_p(n)}$$

by defining it to be the common value of all these finite products.

- c. Prove that for all $n \in \mathbb{N} - \{0\}$,

$$n = \prod_{p \in P} p^{v_p(n)}.$$

- d. Let $\mathcal{F}^*(P, \mathbb{N})$ consist of all $f \in \mathcal{F}(P, \mathbb{N})$ with the property that the set $\{p \in P : f(p) > 0\}$ is finite. Prove that the function

$$\mathbb{N} - \{0\} \rightarrow \mathcal{F}^*(P, \mathbb{N}) : n \mapsto (v_p(n) : p \in P)$$

is a bijection.

e. Prove that $(\mathbb{N}, |)$ is a lattice. Prove also that for all $a, b \in \mathbb{N} - \{0\}$ and all $p \in P$, we have

$$v_p(a \wedge b) = \min\{v_p(a), v_p(b)\},$$

$$v_p(a \vee b) = \max\{v_p(a), v_p(b)\}.$$

□

Exercise 7.3.22. Let G be a group and let $\mathcal{G}_G$ be the set of all of its subgroups. Prove that $(\mathcal{G}, \subseteq)$ is a lattice. □

Exercise 7.3.23. Prove that every finite subset of a lattice has a supremum and an infimum. □

Notation 7.3.24. Let $(E, \leq)$ be a lattice, let X be a subset of E , and let $(x_i : i \in I)$ be a family of elements of E .

- a. If X has a supremum, then we may denote it by $\bigvee X$.
- b. Similarly, its infimum, when it exists, may be denoted by $\bigwedge X$.
- c. We write $\bigwedge_{i \in I} x_i$ and $\bigvee_{i \in I} x_i$ to mean

$$\bigwedge \{x_i : i \in I\} \text{ and } \bigvee \{x_i : i \in I\},$$

respectively.

- d. If $I = [n]$, for some $n \in \mathbb{N}$, then we also write $\bigwedge_{i < n} x_i$ and $\bigvee_{i < n} x_i$ to mean $\bigwedge_{i \in I} x_i$ and $\bigvee_{i \in I} x_i$, respectively. □

Exercise 7.3.25. Let $\mathbb{B}$ be a boolean algebra and let $(a_i : i < n) \in \mathbb{B}^{<\mathbb{N}}$. Prove:

$$\bigwedge_{i < n} a_i = a_0 \wedge a_1 \wedge \cdots \wedge a_{n-1},$$

$$\bigvee_{i < n} a_i = a_0 \vee a_1 \vee \cdots \vee a_{n-1}.$$

□

Definition 7.3.26. A lattice is called **bounded** iff it has a maximum and a minimum. □

Exercise 7.3.27.

- a. Prove that every boolean algebra is a bounded lattice.
- b. Prove that every complete lattice is bounded.
- c. Give an example of a lattice which is not bounded and not a totally ordered. □

Exercise 7.3.28. Let $(E, \leq)$ be a lattice and let $a, b, c \in E$. Prove that

$$(a \wedge b) \vee (a \wedge c) \leq a \wedge (b \vee c).$$

Show by example that the reverse inequality may fail in general.

Hint. You may want to use Exercise 3.2.50. □

Definition 7.3.29. A lattice $(E, \leq)$ is said to be **distributive** iff for all $a, b, c \in E$, one has

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c),$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c).$$

□

Exercise 7.3.30. Let $\mathbb{B}$ be a boolean algebra. Prove that $(\mathbb{B}, \leq_{\mathbb{B}})$ is a distributive lattice. □

Exercise 7.3.31. Give an example of a lattice $(E, \leq)$ which has a minimum and a maximum, but in which the following condition **fails**:

$$\forall x \in E, \exists y \in E, (x \wedge y = \min E \text{ and } x \vee y = \max E).$$

□

Definition 7.3.32. A **bounded** lattice $(E, \leq)$ is said to be **complementing** iff for every $x \in E$, there is $y \in E$ such that $x \wedge y = \min E$ and $x \vee y = \max E$. □

Exercise 7.3.33. Let $\mathbb{B}$ be a boolean algebra. Prove that $(\mathbb{B}, \leq_{\mathbb{B}})$ is a complementing lattice. □

Exercise 7.3.34. Let $(E, \leq)$ be a lattice which is bounded, distributive, and complementing.

a. Prove that for every $x \in E$, there is a **unique** $y \in E$ such that $x \wedge y = \min E$ and $x \vee y = \max E$.

For $x \in E$, we denote by $\neg x$ the unique $y \in E$ such that $x \wedge y = \min E$ and $x \vee y = \max E$.

b. Prove that $(E, \wedge, \vee, \neg, 0, 1)$ is a boolean algebra.

c. Prove that the canonical order relation associated to this boolean algebra equal to the starting order relation $\leq$. □

Exercise 7.3.35 (Xiang Li). A lattice $(H, \leq)$ is a **Heyting algebra** iff it is bounded and there is a binary operation $\rightarrow$ on it such that

$$(\forall a, b, c \in H)(a \wedge b \leq c \iff a \leq (b \rightarrow c)).$$

a. Prove that if $(H, \leq)$ is a Heyting algebra, then there is a **unique** binary operation $\rightarrow$ on it such that

$$(\forall a, b, c \in H)(a \wedge b \leq c \iff a \leq (b \rightarrow c)).$$

b. Prove that every boolean algebra is a Heyting algebra.

c. Prove that every finite distributive algebra is a Heyting algebra. □

7.4 Filters

Definition 7.4.1. Let $\mathbb{B}$ be a boolean algebra and let $F \subseteq \mathbb{B}$. We say that F is a **filter** iff

- a. $1 \in F$;
- b. for all $a, b \in F$, it holds that $a \wedge b \in F$;
- c. for all $a \in F$ and all $b \geq_{\mathbb{B}} a$, it holds that $b \in F$. □

Remark 7.4.2. In every boolean algebra, its universe is a filter. We will see that this filter corresponds to the improper boolean algebras, so very often we will want to exclude it from the consideration. □

Terminology 7.4.3. A filter F in a boolean algebra B is **proper** iff $F \neq B$. □

Exercise 7.4.4. Prove that a filter F is proper if and only if $0 \notin F$. □

Exercise 7.4.5. In an arbitrary boolean algebra, the set $\{1\}$ is a filter. □

Exercise 7.4.6. Let E be an infinite set and let F be the collection of all **co-finite** subsets of E , i.e. all $A \subseteq E$ with the property that $E - A$ is finite. Prove that F is a filter in $\mathcal{P}(E)$. □

Exercise 7.4.7. Let T be a propositional theory.

- a. Prove that the set

$$\mathbf{F}_T := \{[\phi] \in \mathbb{L} : T \models \phi\}$$

is a filter in $\mathbb{L}$.

- b. Prove that $\mathbf{F}_T$ is proper if and only if T is satisfiable. □

Exercise 7.4.8. Let $\mathbb{B}$ be a boolean algebra.

- a. Let E be a congruence on $\mathbb{B}$. Prove that $[1_{\mathbb{B}}]_E$ is a filter in $\mathbb{B}$.
- b. Let F be a filter in $\mathbb{B}$. The binary relation $\equiv_F$ is defined as follows:

$$\forall a, b \in \mathbb{B}, (a \equiv_F b \iff (\exists c \in F)(a \wedge c = b \wedge c)).$$

Prove that $\equiv_F$ is a congruence on $\mathbb{B}$.

- c. Prove that if E is a congruence on $\mathbb{B}$, then

$$\equiv_{[1_{\mathbb{B}}]_E} = E.$$

- d. Prove that if F is a filter in $\mathbb{B}$, then

$$[1_{\mathbb{B}}]_{\equiv_F} = F.$$

□

Exercise 7.4.9. Let E be an infinite set and let F be the filter of all co-finite subsets of E . Prove that for all $X, Y \in \mathcal{P}(E)$,

$$X \equiv_F Y \iff (X - Y) \cup (Y - X) \text{ is finite.}$$

(Thus, $\mathcal{P}(E)/F = \mathcal{P}(E)/\text{Fin.}$) □

Exercise 7.4.10. Let $\phi : \mathbb{B} \rightarrow \mathbb{C}$ be a morphism of boolean algebras. Prove that the set

$$\phi^{-1}(1_{\mathbb{C}}) := \{b \in \mathbb{B} : \phi(b) = 1_{\mathbb{C}}\}$$

is a filter in $\mathbb{B}$. □

Notation 7.4.11. Let F be a filter in a boolean algebra $\mathbb{B}$. We denote by $\mathbb{B}/F$ and π_F the objects $\mathbb{B}/\equiv_F$ and $\pi_{\equiv_F}$ (respectively). □

Remark 7.4.12. So, $\pi_F : \mathbb{B} \rightarrow \mathbb{B}/F$ is a morphism. □

Exercise 7.4.13. Let $\mathbb{B}$ be a boolean algebra and let F be a filter in it. Prove that $\pi_F^{-1}(1_{\mathbb{B}/F}) = F$. □

Exercise 7.4.14. Let $\mathbb{B}$ be a boolean algebra, let F be a filter in it, and let $\phi : \mathbb{B} \rightarrow \mathbb{C}$ be a morphism of boolean algebras.

a. Prove that there is a morphism $\phi_* : \mathbb{B} \rightarrow \mathbb{C}$ such that the diagram

$$\begin{array}{ccc} \mathbb{B} & \xrightarrow{\phi} & \mathbb{C} \\ \downarrow \pi_F & \nearrow \phi_* & \\ \mathbb{B}/F & & \end{array}$$

commutes if and only if $F \subseteq \phi^{-1}(1_{\mathbb{C}})$.

b. Prove that if $F \subseteq \phi^{-1}(1_{\mathbb{C}})$, then there is a **unique** morphism ϕ_* such that the diagram

$$\begin{array}{ccc} \mathbb{B} & \xrightarrow{\phi} & \mathbb{C} \\ \downarrow \pi_F & \nearrow \phi_* & \\ \mathbb{B}/F & & \end{array}$$

commutes.

c. Prove that $\text{Im}(\phi_*) = \text{Im}(\phi)$.

d. Prove that ϕ_* is injective if and only if $F = \phi^{-1}(1_{\mathbb{C}})$.

e. Conclude that $\phi_* : \mathbb{B}/\phi^{-1}(1_{\mathbb{C}}) \xrightarrow{\sim} \text{Im}(\phi)$. □

Exercise 7.4.15. Let T be a propositional theory and let

$$\mathbf{F}_T := \{[\phi] \in \mathbb{L} : T \models \phi\}.$$

Prove that $\mathbb{L}/\mathbf{F}_T \simeq \mathbb{L}_T$. □

Exercise 7.4.16. Let $\mathbb{B}$ be a boolean algebra and let F and G be two filters in it.

a. There is a morphism ϕ such that the diagram

$$\begin{array}{ccc} \mathbb{B} & \xrightarrow{\pi_G} & \mathbb{B}/G \\ \downarrow \phi_F & \nearrow \phi & \\ \mathbb{B}/F & & \end{array}$$

commutes if and only if $F \subseteq G$.

b. If $F \subseteq G$, then:

i. there is a unique morphism ϕ such that the diagram

$$\begin{array}{ccc} \mathbb{B} & \xrightarrow{\pi_G} & \mathbb{B}/G \\ \downarrow \phi_F & \nearrow \phi & \\ \mathbb{B}/F & & \end{array}$$

commutes;

ii. ϕ is surjective;

iii. ϕ is injective if and only if $F = G$.

iv. Conclude that $\phi : \mathbb{B}/F \xrightarrow{\sim} \mathbb{B}/G$ if and only if $F = G$. □

Definition 7.4.17. Let $\mathbb{B}$ be a boolean algebra and let $X \subseteq \mathbb{B}$. We say that X has the **finite meet property** iff for all finite $Y \subseteq X$, it holds that $\bigwedge Y > 0$. □

Exercise 7.4.18. Let $\mathbb{B}$ be a boolean algebra and let $X \subseteq \mathbb{B}$ be finite. Prove that X has the finite meet property if and only if $\bigwedge X > 0$. □

Exercise 7.4.19. Prove that X has the finite meet property if and only if every finite subset of X has the finite meet property. □

Exercise 7.4.20. Let $\mathbb{B}$ be a boolean algebra.

a. Let F be a filter in $\mathbb{B}$. Then F is proper if and only if F has the finite meet property.

b. If X is a subset of $\mathbb{B}$, then the set

$$\mathbb{F}_X := \left\{ y \in \mathbb{B} : (\exists (x_i)_{i < n} \in X^{<\mathbb{N}}) \left(\bigwedge_{i < n} x_i \leq y \right) \right\}$$

is a filter in $\mathbb{B}$.

c. If X is a subset of $\mathbb{B}$ with finite meet property, then the filter $\mathbb{F}_X$ is proper.

d. F_X is the least element of the order $(\mathcal{F}_X, \subseteq)$, where $\mathcal{F}$ is the collection of all filters of $\mathbb{B}$ that extend X .

e. Prove that if X is a filter, then $X = F_X$. □

Terminology 7.4.21. The set F_X is called the **filter generated by X** . □

Exercise 7.4.22. Let $B = \{\mathbb{N} - [n] : n \in \mathbb{N}\} \subseteq \mathcal{P}(\mathbb{N})$. Prove that B has the finite meet property in $\mathcal{P}(\mathbb{N})$. What is the (proper) filter generated by it? □

7.5 Ultrafilters

Definition 7.5.1. Let $\mathbb{B}$ be a boolean algebra and let us denote by $\mathcal{F}$ the set of all **proper filters** in $\mathbb{B}$. A subset of $\mathbb{B}$ is called an **ultrafilter** iff it is a maximal element of $(\mathcal{F}, \subseteq)$. □

Exercise 7.5.2. Let E be an arbitrary set. For $a \in E$, we define

$$\mathcal{U}_a := \{A \subseteq E : a \in A\}.$$

a. Prove that for all $a \in E$, the set $\mathcal{U}_a$ is an ultrafilter in $\mathcal{P}(E)$.

These ultrafilters are called **principal**.

b. Prove that if E is finite, then every ultrafilter in $\mathcal{P}(E)$ is principal.

c. Suppose that E is infinite. Prove that an ultrafilter $\mathcal{U}$ in $\mathcal{P}(E)$ is non-principal if and only if it extends the filter of co-finite sets. □

Fact 7.5.3. Let $\mathbb{B}$ be a boolean algebra and let F be a filter in it. Then F is an ultrafilter if and only if there is a morphism $h : \mathbb{B} \rightarrow 2$ such that $F = h^{-1}(1)$.

Proof.

Direct implication. We want to see that $\mathbb{B}/F = \{[0_{\mathbb{B}}], [1_{\mathbb{B}}]\}$. Assume towards contradiction that there is $x \in \mathbb{B}$ such that

$$[0_{\mathbb{B}}] < [x] < [1_{\mathbb{B}}].$$

We claim that $F \cup \{x\}$ has the finite meet property.

Assume otherwise. Then there is a finite set $Y \subseteq F \cup \{x\}$ such that $\bigwedge Y = 0_{\mathbb{B}}$. Since F does have the finite meet property, it must be the case that $x \in Y$. Since $Y - \{x\}$ is a finite subset of the filter F , it holds that

$$\bigwedge (Y - \{x\}) \in F.$$

We now have:

$$x \wedge \bigwedge (Y - \{x\}) = \bigwedge Y = 0_{\mathbb{B}} = 0_{\mathbb{B}} \wedge \bigwedge (Y - \{x\}),$$

whence $x \equiv_F 0_{\mathbb{B}}$. In other words $[x] = [0_{\mathbb{B}}]$, which is a contradiction.

Thus, the set $F \cup \{x\}$ has the finite meet property. It follows that it generates a filter, say F' . Since F is a maximal filter (=ultrafilter), we get $F = F'$, i.e. $x \in F$. But this now means:

$$x \wedge x = x = 1_{\mathbb{B}} \wedge x,$$

so $x \equiv_F 1_{\mathbb{B}}$. In other words, $[x] = [1_{\mathbb{B}}]$ – contradiction.

Thus, $\mathbb{B}/F = \{[0_{\mathbb{B}}], [1_{\mathbb{B}}]\}$. It now easily follows that the function $f : \mathbb{B}/F \rightarrow 2$ given by $f([0_{\mathbb{B}}]) = 0$ and $f([1_{\mathbb{B}}]) = 1$ is an isomorphism of boolean algebras. This means that the map

$$h := f \circ \pi_F : \mathbb{B} \rightarrow 2$$

is a morphism of boolean algebras with the property that

$$h^{-1}(1) = \pi_F^{-1}([1_{\mathbb{B}}]) = F.$$

Reverse implication. Since h is a morphism of boolean algebras, the set $F = h^{-1}(1)$ is a filter.

Assume towards contradiction that there is a filter G in $\mathbb{B}$ with the property $G \supset F$. Pick $x \in G - F$. Then $h(x) = 0$ (because $x \notin h^{-1}(1)$ and $\text{ran}(h) = \{0, 1\}$). It follows that

$$1 = \neg 0 = \neg h(x) = h(\neg x),$$

whence $\neg x \in h^{-1}(1) = F \subseteq G$. We now have that both x and $\neg x$ belong to G , which contradicts the assumption that G is a filter.

Therefore, F is a maximal filter, c-à-d. an ultrafilter. □

Exercise 7.5.4. Let $\mathbb{B}$ be a boolean algebra and let F be a filter in it. Then the following are equivalent:

- a. F is an ultrafilter;
- b. F is a maximal element of $(M, \subseteq)$, where M is the set of all subsets of $\mathbb{B}$ with the finite meet property;
- c. for all $a \in \mathbb{B}$, either $a \in F$ or $\neg a \in F$ (and not both);
- d. F is proper and for all $a, b \in \mathbb{B}$, if $a \vee b \in F$, then $a \in F$ or $b \in F$. □

Exercise 7.5.5. Let X be an arbitrary set and let $*$ $\in X$. Prove that there is a well-order R on X such that

$$\min_R X = *.$$

□

Exercise 7.5.6. Suppose:

- 1. $\mathbb{B}$ is a boolean algebra;
- 2. $(I, \leq_I)$ is a total order;
- 3. $(X_i : i \in I)$ is a family of subsets of $\mathbb{B}$;
- 4. for all $i \in I$, the set X_i has the finite meet property;
- 5. for all $i, j \in I$ such that $i \leq_I j$, we have $X_i \subseteq X_j$.

Prove that

$$\bigcup_{i \in I} X_i$$

has the finite meet property. □

Exercise 7.5.7. Let $\mathbb{B}$ be a boolean algebra, let X be its subset that has the finite meet property, and let $b \in \mathbb{B}$. Suppose there is a finite $Z \subseteq X$ such that $\bigwedge Z \leq b$. Prove that the set $X \cup \{b\}$ also has the finite meet property. □

Theorem 7.5.8 (Ultrafilter Theorem). Let $\mathbb{B}$ be a boolean algebra and let $X \subseteq \mathbb{B}$. Then X has the finite meet property if and only if it is contained in an ultrafilter.

Proof. The reverse implication follows by Exercise 7.4.20, so let us prove the direct one. Let $*$ be an object that does **not** belong to $\mathbb{B}$. By Exercise 7.5.5, there is a well-order $\leq$ on $\mathbb{B} \cup \{*\}$ whose least element is $*$. We define by recursion for the well-order $(\mathbb{B} \cup \{*\}, \leq)$ (see Exercise 3.2.46) a sequence

$$(Y_b : b \in \mathbb{B} \cup \{*\})$$

of subsets of $\mathbb{B}$.

Suppose $b \in \mathbb{B} \cup \{*\}$ and the sequence $(Y_a : a < b)$ has been defined. We want to define Y_b . If $b = *$, then we set $Y_* := X$. Suppose now that $b > *$. Let

$$Y_{<b} := \bigcup_{a < b} Y_a.$$

If the set $Y_{<b} \cup \{b\}$ has the finite meet property, then we define

$$Y_b := Y_{<b} \cup \{b\};$$

otherwise, we define

$$Y_b := Y_{<b} \cup \{\neg b\}.$$

This finishes the recursion.

Claim 7.5.9. For all $a, b \in \mathbb{B} \cup \{*\}$ such that $a \leq b$, we have

$$Y_a \subseteq Y_b.$$

Proof. If $a = b$, then the conclusion is obvious; so, let us assume that $a < b$. In particular, $b > *$, so $Y_{<b}$ is defined and contained in Y_b . We have: $Y_a \subseteq Y_{<b} \subseteq Y_b$; as required. □

Claim 7.5.10. For all $b \in \mathbb{B} \cup \{*\}$, the set Y_b has the finite meet property.

Proof. The proof is by induction for the well-order $(\mathbb{B} \cup \{*\}, \leq)$ (see Exercise 3.2.45).

Suppose $b \in \mathbb{B} \cup \{*\}$ and for all $a < b$, the set Y_a has the finite meet property. If $b = *$, then $Y_b = X$, which has the finite meet property by assumption. So, let us assume that $b > *$.

If the set $Y_{<b} \cup \{b\}$ has the finite meet property, then Y_b is equal to this set and consequently, has the finite meet property. So, let us assume that $Y_{<b} \cup \{b\}$ does **not** have the finite meet property. Then $Y_b = Y_{<b} \cup \{\neg b\}$.

Assume towards contradiction that Y_b does not have the finite meet property. Then there is a finite subset Z of it such that $\bigcap Z = 0$. By Exercise 7.5.6 and Claim 7.5.9, the set $Y_{\prec b}$ has the finite meet property, whence it **cannot** be that $Z \subseteq Y_{\prec b}$; in other words, $\neg b \in Z$. This means

$$\bigwedge (Z - \{\neg b\}) \wedge \neg b = 0,$$

whence

$$\bigwedge (Z - \{\neg b\}) \leq b.$$

By Exercise 7.5.7, the set $Y_{\prec b} \cup \{b\}$ has the finite meet property – contradiction.

Therefore, Y_b has the finite meet property.

This concludes the induction. □

By Exercise 7.5.6 and Claim 7.5.9, the set

$$Y := \bigcup_{b \in B \cup \{*\}} Y_b$$

has the finite meet property. Let us prove it is maximal with respect to this property.

Assume towards contradiction that there is $W \supset Y$ which has the finite meet property. Pick $b \in W - Y$. Then $Y_{\prec b} \cup \{b\} \subseteq W$ has the finite meet property, whence $Y_b = Y_{\prec b} \cup \{b\}$. Hence, $b \in Y_b \subseteq Y$ – contradiction.

Therefore, Y is maximal among sets with the finite meet property, so it is an ultrafilter by Exercise 7.5.4. Since $X = Y_* \subseteq Y$, we are done. □

Corollary 7.5.11. *Every filter of every boolean algebra can be extended to an ultrafilter of that algebra.* □

Exercise 7.5.12. *Let E be an infinite set and let A be an infinite subset of it. Prove that there is a **non-principal** ultrafilter in $\mathcal{P}(E)$ that contains A .* □

7.6 Stone's Representation Theorem

Exercise 7.6.1. *Let ϕ be a function from a boolean algebra B to a boolean algebra C . Then ϕ is a morphism of boolean algebras B and C if and only if for all $x, y \in B$, it holds that $\phi(x \wedge y) = \phi(x) \wedge \phi(y)$ and $\phi(\neg x) = \neg \phi(x)$.* □

Exercise 7.6.2. *Let B be a boolean algebra, a, b its elements, and F its filter. Prove that $a \wedge b \in F$ if and only if $a \in F$ and $b \in F$.* □

Exercise 7.6.3. *Let $\phi : B \rightarrow C$ be a morphism of boolean algebras. Prove that ϕ is injective if and only if $\phi^{-1}(1) = \{1\}$.* □

Exercise 7.6.4. *Let $\mathcal{U}$ be the set of all ultrafilters of a boolean algebra B . Prove that $\bigcap \mathcal{U} = \{1_B\}$.* □

Exercise 7.6.5. *Let $\phi : B \rightarrow C$ be a morphism of boolean algebras. Prove that ϕ is injective if and only if $\phi^{-1}(1^C) = \{1^B\}$.* □

Corollary 7.6.6 (Stone's Representation Theorem). *Let B be an arbitrary boolean algebra and let $\mathcal{U}$ be the set of all ultrafilters of B . Then B is isomorphic to a subalgebra of $\mathcal{P}(\mathcal{U})$.*

Proof. We define

$$\phi : \mathbb{B} \rightarrow \mathcal{P}(\mathcal{U}) : b \mapsto \{F \in \mathcal{U} : b \in F\}.$$

We want to see that ϕ is a morphism of boolean algebras. By Exercise 7.6.1, it suffices to prove that for all $a, b \in \mathbb{B}$, we have

$$\begin{aligned}\phi(a \wedge b) &= \phi(a) \cap \phi(b), \\ \phi(\neg a) &= \phi(a)^c.\end{aligned}$$

By Exercise 7.6.2, we have

$$\begin{aligned}\phi(a \wedge b) &= \{F \in \mathcal{U} : a \wedge b \in F\} = \{F \in \mathcal{U} : a \in F \text{ and } b \in F\} = \\ &= \{F \in \mathcal{U} : a \in F\} \cap \{F \in \mathcal{U} : b \in F\} = \phi(a) \cap \phi(b).\end{aligned}$$

On the other hand, for every ultrafilter F , we have that either $a \in F$ or $\neg a \in F$ (see Exercise 7.5.4), whence:

$$a \notin F \iff \neg a \in F.$$

We get:

$$\phi(\neg a) = \{F \in \mathcal{U} : \neg a \in F\} = \{F \in \mathcal{U} : a \notin F\} = \{F \in \mathcal{U} : a \in F\}^c = \phi(a)^c.$$

Therefore, ϕ is a morphism. Let us compute $\phi^{-1}(1_{\mathcal{P}(\mathcal{U})})$.

Let $b \in \mathbb{B}$ be such that $\phi(b) = 1_{\mathcal{P}(\mathcal{U})}$. This means:

$$\{F \in \mathcal{U} : b \in F\} = \mathcal{U};$$

i.e. b belongs to every ultrafilter. By Exercise 7.6.4, $b = 1^{\mathbb{B}}$.

Thus, $\phi^{-1}(1_{\mathcal{P}(\mathcal{U})}) = \{1^{\mathbb{B}}\}$, which, by Exercise 7.6.5, gives that ϕ is injective. It now follows that $\phi : \mathbb{B} \rightarrow \text{Im}(\phi)$ is an isomorphism and that $\text{Im}(\phi)$ is a subalgebra of $\mathcal{P}(\mathcal{U})$, as required. $\square$

Remark 7.6.7. Exercises 6.3.20 and 6.3.21 are results which are, in their form, analogous to Stone's Representation Theorem. $\square$

Exercise 7.6.8. Let $\mathbb{B}$ be a boolean algebra and let $a, b \in \mathbb{B}$. Prove that $a \neq b$ if and only if there is an ultrafilter $\mathcal{U}$ such that $|\{a, b\} \cap \mathcal{U}| = 1$. $\square$

Notation 7.6.9. Let $\mathcal{L}$ be a first-order language, let $s = t$ be an algebraic identity, and let M be an $\mathcal{L}$ -structure. We write $M \models (s = t)$ to mean $M \models \phi$, where ϕ is the universal closure of $s = t$. $\square$

Exercise 7.6.10. Let $\mathbb{B}$ be a proper boolean algebra and let $s = t$ be an algebraic identity. Prove that

$$\mathbb{B} \models (s = t) \iff 2 \models (s = t).$$

$\square$

Hint. The direct implication follows from the fact that 2 can be seen as a subalgebra of any proper boolean algebra. For the reverse inclusion, assume otherwise. Then there is a valuation $\mu : \mathcal{V} \rightarrow \mathbb{B}$ such that $s^{\mathbb{B}}[\mu] \neq t^{\mathbb{B}}[\mu]$. Use Exercise 7.6.8 to get a morphism $h : \mathbb{B} \rightarrow 2$ such that

$$h(s^{\mathbb{B}}[\mu]) \neq h(t^{\mathbb{B}}[\mu]).$$

$\square$

7.7 Compactness of Propositional Logic

Theorem 7.7.1 (Compactness of Propositional Logic). *A propositional theory T is satisfiable if and only if every finite subset of T is satisfiable.*

Proof. The direct implication is trivial because any model of T is also a model of every subset of T . Thus, we focus on the reverse implication.

We are assuming that every finite subset of T has a model. This means that the set

$$\{[\psi] \in \mathbb{L} : \psi \in T\}$$

has the finite meet property in $\mathbb{L}$ (**explain**). Therefore, it is contained in an ultrafilter $\mathcal{U}$.

For all $p \in \mathbb{P}$, we have that either $[p] \in \mathcal{U}$ or $\neg[p] \in \mathcal{U}$ (and not both), so we can define:

$$\mu(p) := \begin{cases} 1, & [p] \in \mathcal{U} \\ 0, & \neg[p] \in \mathcal{U}. \end{cases}$$

In this way, we obtain a valuation $\mu : \mathbb{P} \rightarrow \{0, 1\}$. We want to show that $\mu \models T$.

Recall that $\mathbb{L}/\mathcal{U}$ is a two-element boolean algebra. Thus, there is a unique isomorphism $h : \mathbb{L}/\mathcal{U} \rightarrow \mathbf{2}$. So, we have, for all $\psi \in \text{Prop}_{\mathbb{P}}$, that

$$h([\psi]_{\equiv}) = \begin{cases} 1, & [\psi]_{\equiv} \in \mathcal{U} \\ 0, & [\psi]_{\equiv} \notin \mathcal{U}. \end{cases}$$

Claim 7.7.2. *For all $\psi \in \text{Prop}_{\mathbb{P}}$,*

$$\|\psi\|_{\mu} = h([\psi]_{\equiv}).$$

Proof. This is proved by induction on complexity of a formula. If $\psi = p$, for some $p \in \mathbb{P}$, then

$$\|\psi\|_{\mu} = \mu(p) = h([p]) = h([\psi])$$

(by definition of μ). If $\psi = \perp$, then

$$\|\psi\|_{\mu} = 0 = h(0_{\mathbb{L}}) = h([\perp]).$$

If $\psi = \neg\theta$, for some $\theta \in \text{Prop}_{\mathbb{P}}$, then

$$\|\psi\|_{\mu} = \neg\|\theta\|_{\mu} = \neg h([\theta]) = h(\neg[\theta]) = h([\neg\theta]),$$

where in the second equality we are applying the inductive hypothesis. If $\psi = (\theta \implies \chi)$, for some $\theta, \chi \in \text{Prop}_{\mathbb{P}}$, then

$$\|\psi\|_{\mu} = \neg\|\theta\|_{\mu} \vee \|\chi\|_{\mu} = \neg h([\theta]) \vee h([\chi]) = h(\neg[\theta] \vee [\chi]) = h([\theta \implies \chi]) = h([\psi]).$$

The remaining cases are left as an **exercise**. □

Now, for every $\psi \in T$, we have $[\psi] \in \mathcal{U}$, whence $\|\psi\|_{\mu} = h([\psi]) = 1$. Thus, $\mu \models T$, so T is satisfiable. □

Exercise 7.7.3 (Xiang Li).

- Give an example of a theory which is **not** satisfiable, but all of its members are.
- Give an example of a theory which is **not** satisfiable, but all of its subsets with at most two members are.
- Give an example of a theory which is **not** satisfiable, but all of its subsets with at most three members are. □

Exercise 7.7.4.

- Let F be a filter in $\mathbb{L}$. Prove that

$$\mathbf{T}_F := \{\phi \in \text{Prop}_{\mathbb{P}} : [\phi] \in F\}$$

is a satisfiable theory.

- Let T be a satisfiable propositional theory and let

$$\mathbf{F}_T := \{[\phi] : T \models \phi\}.$$

You already proved that $\mathbf{F}_T$ is a filter in $\mathbb{L}$. Now prove that

$$\mathbf{T}_{\mathbf{F}_T} = \text{scl}(T).$$

- Let T be a satisfiable propositional theory and let $\mathcal{U}$ be an ultrafilter of $\mathbb{L}$ that contains $\mathbf{F}_T$. Prove that $\mathbf{T}_{\mathcal{U}}$ is a maximal satisfiable theory that contains T .
- Prove that every satisfiable theory is contained in a maximal satisfiable theory. □

Exercise 7.7.5. Let T be a propositional theory and let ϕ be a propositional formula. If $T \models \phi$, then there is a **finite** subset T_0 of T such that $T_0 \models \phi$. □

Exercise 7.7.6. Prove that a theory T is finitely axiomatizable if and only if there is a finite theory $T_0 \subseteq T$ such that T_0 and T are semantically equivalent. □

Remark 7.7.7. For the remainder of this section, we do not admit the Axiom of Choice. □

Definition 7.7.8. The **Ultrafilter Principle** is the statement: “In every boolean algebra, every subset with the finite meet property is contained in an ultrafilter.” □

Definition 7.7.9. The **Compactness of Propositional Logic** is the statement: “Over an arbitrary set of propositional letters, if every finite subset of a given theory has a model, then the theory has a model.” □

Remark 7.7.10. Assuming the Axiom of Choice, we proved both the Ultrafilter Principle and the Compactness of Propositional Logic. However, if we do not admit the Axiom of Choice, neither statement is provable. □

Exercise 7.7.11 (Xiang Li). *Prove that the Ultrafilter Principle is equivalent to the Compactness of Propositional logic.*

Hint. Verify that the direct implication follows by the proof of Theorem 7.7.1. For the reverse inclusion, you may reason as follows.

Let $\mathbb{B}$ be a boolean algebra and let X be its subset with the finite meet property. Consider the set $\mathbb{P} = \{p_a : a \in \mathbb{B}\}$ of propositional letters and the theory T over $\mathbb{P}$ consisting of the following formulas:

- $p_{1_{\mathbb{B}}} \wedge \neg p_{0_{\mathbb{B}}}$;
- for all $a, b \in \mathbb{B}$ such that $a \leq_{\mathbb{B}} b$, the formula $p_a \implies p_b$;
- for all $a, b \in \mathbb{B}$, the formula $p_a \wedge p_b \implies p_{a \wedge b}$;
- for all $a \in \mathbb{B}$, the formula $p_a \vee p_{\neg a}$;
- for all $x \in X$, the formula p_x .

Prove that every finite subset of T has a model. Use the Compactness principle to get a model μ of T . Prove that

$$\{a \in \mathbb{B} : \mu \models p_a\}$$

is an ultrafilter of $\mathbb{B}$ that contains X . □

Definition 7.7.12. Let X be a set. A **tree on X** is a subset T of $X^{<\mathbb{N}}$ that satisfies

$$(\forall s, t \in X^{<\mathbb{N}})(t \in T \wedge s \subseteq t \implies s \in T).$$

□

Remark 7.7.13. Recall that if $s, t \in X^{<\mathbb{N}}$, then there are $m, n \in \mathbb{N}$ such that $\text{dom}(s) = [m]$ and $\text{dom}(t) = [n]$. Then $s \subseteq t$ if and only if $m \leq n$ and $t \upharpoonright [m] = s$. □

Exercise 7.7.14. Let X be a set and let T be a tree on X . Prove that $(T, \subseteq)$ is a well-founded order. Prove that $\emptyset$ is the minimum of this order. □

Definition 7.7.15. Let T be a tree on X and let $n \in \mathbb{N}$. The **n^{th} level** of T is the set

$$\{s \in T : \text{dom}(s) = [n]\}.$$

□

Exercise 7.7.16 (Xiang Li). *Assume the Compactness of Propositional Logic. Let T be a tree on X . Suppose that every level of T is non-empty and finite. Prove that there is $f : \mathbb{N} \rightarrow X$ such that*

$$(\forall n \in \mathbb{N})(f \upharpoonright [n] \in T).$$

Hint. The function f is called a *branch* through T .

For $s, t \in X^{<\mathbb{N}}$, we say that t is an *immediate successor* of s if $s \subseteq t$ and for some $n \in \mathbb{N}$, it holds that $\text{dom}(s) = [n]$ and $\text{dom}(t) = [n + 1]$.

Let $\mathbb{P}$ be the set $\{p_s : s \in T\}$ of propositional letters and let T be the theory over $\mathbb{P}$ consisting of the following formulas:

- $p_\emptyset$;
- for all $s, t \in T$ such that $s \subseteq t$, the formula $p_t \implies p_s$;
- for every $s \in T$, every $k \in \mathbb{N}$, and every bijection f from the set $[k]$ to the set of all immediate successors of s , the formula

$$p_s \implies \bigvee_{i < k} \left(p_{f(i)} \wedge \bigwedge_{\substack{j < k \\ j \neq i}} \neg p_{f(j)} \right).$$

Prove that every finite subset of T has a model. Apply the Compactness of Propositional Logic to get a model μ of T . Put together elements of

$$\{s \in T : \mu \models p_s\}$$

to get a branch through T . □

7.8 Boolean-valued Valuations

Remark 7.8.1. We could have defined valuations to take values in arbitrary boolean algebra instead of the boolean algebra 2 . We explore this possibility in the current section. Exercise 7.8.13 suggests that not much would be gained in this way. □

Definition 7.8.2. Let $\mathbb{B}$ be a boolean algebra. A **B-valuation** is any function

$$\mu : \mathbb{P} \rightarrow \mathbb{B}.$$

□

Remark 7.8.3. A 2-valuation is just a valuation in the old sense. □

Definition 7.8.4. Let $\mathbb{B}$ be a boolean algebra and let μ be a $\mathbb{B}$ -valuation. We define the value $\|\phi\|_\mu$, for all $\phi \in \text{Prop}_{\mathbb{P}}$, by recursion on complexity:

- for all $p \in \mathbb{P}$, it holds $\|p\|_\mu = \mu(p)$;
- $\|\perp\|_\mu = 0$;
- for all $\phi \in \text{Prop}_{\mathbb{P}}$, it holds $\|\neg\phi\|_\mu = \neg\|\phi\|_\mu$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$, it holds $\|\phi \wedge \psi\|_\mu = \|\phi\|_\mu \wedge \|\psi\|_\mu$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$, it holds $\|\phi \vee \psi\|_\mu = \|\phi\|_\mu \vee \|\psi\|_\mu$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$, it holds $\|\phi \implies \psi\|_\mu = \neg\|\phi\|_\mu \vee \|\psi\|_\mu$;
- for all $\phi, \psi \in \text{Prop}_{\mathbb{P}}$, it holds $\|\phi \iff \psi\|_\mu = (\|\phi\|_\mu \wedge \|\psi\|_\mu) \vee (\neg\|\phi\|_\mu \wedge \neg\|\psi\|_\mu)$. □

Exercise 7.8.5. If μ is 2-valuation, prove that this new meaning of $\|\phi\|_\mu$ coheres with the old one. $\square$

Exercise 7.8.6. Let T be a propositional theory and let μ be the $\mathbb{L}_T$ -valuation given by:

$$(\forall p \in \mathbb{P})(\mu(p) = [p]_T).$$

Prove that for all $\phi \in \text{Prop}_{\mathbb{P}}$, it holds that $\|\phi\|_\mu = [\phi]_T$. $\square$

Exercise 7.8.7. Let μ be a $\mathbb{B}$ -valuation. Prove:

a. $\|\phi\|_\mu \implies \|\psi\|_\mu = 1$ if and only if $\|\phi\|_\mu \leq \|\psi\|_\mu$;

b. $\|\phi\|_\mu \iff \|\psi\|_\mu = 1$ if and only if $\|\phi\|_\mu = \|\psi\|_\mu$. $\square$

Exercise 7.8.8. Let $\mathbb{B}$ be a boolean algebra, let μ be a $\mathbb{B}$ -valuation, and let

$$\hat{\mu} : \text{Prop}_{\mathbb{P}} \rightarrow \mathbb{B} : \phi \mapsto \|\phi\|_\mu.$$

a. Prove that $\hat{\mu} : \text{Prop}_{\mathbb{P}} \rightarrow \mathbb{B}$ is a morphism.

b. Prove that $\equiv \subseteq \equiv_{\hat{\mu}}$.

c. Conclude that there is a unique morphism

$$\hat{\mu}_* : \mathbb{L} \rightarrow \mathbb{B}$$

such that the diagram

$$\begin{array}{ccc} \text{Prop}_{\mathbb{P}} & \xrightarrow{\hat{\mu}} & \mathbb{B} \\ \downarrow \pi_{\equiv} & \nearrow \hat{\mu}_* & \\ \mathbb{L} & & \end{array}$$

commutes. $\square$

Definition 7.8.9. Let $\mathbb{B}$ and $\mathbb{C}$ be boolean algebras. We denote by $\text{Hom}(\mathbb{B}, \mathbb{C})$ the set of all morphisms of $\mathbb{B}$ and $\mathbb{C}$. $\square$

Exercise 7.8.10. Prove that the function

$$\text{Hom}(\mathbb{L}, \mathbb{B}) \rightarrow \mathcal{F}(\mathbb{P}, \mathbb{B}) : f \mapsto (f \circ \pi_{\equiv}) \upharpoonright \mathbb{P}$$

is a bijection.

Comment. In other words, the set of all $\mathbb{B}$ -valuations can be identified with the set $\text{Hom}(\mathbb{L}, \mathbb{B})$. $\square$

Exercise 7.8.11. Let μ be a $\mathbb{B}$ -valuation, let

$$\text{Th}(\mu) := \{\phi \in \text{Prop}_{\mathbb{P}} : \|\phi\|_\mu = 1\},$$

and let

$$\mathbb{C} := \{\|\phi\|_\mu : \phi \in \text{Prop}_{\mathbb{P}}\}.$$

a. Prove that $\mathbb{C}$ is a sub-algebra of $\mathbb{B}$.

b. Prove that $\mathbb{L}_{\text{Th}(\mu)} \simeq \mathbb{C}$.

c. Prove that the theory $\text{Th}(\mu)$ is maximal satisfiable if and only if $\mathbb{C} = \{0, 1\}$. $\square$

Exercise 7.8.12. Let $\mathbb{B}$ and $\mathbb{C}$ be boolean algebras and let $h : \mathbb{B} \rightarrow \mathbb{C}$ be a morphism. We denote

$$h_* : \mathcal{F}(\mathbb{P}, \mathbb{B}) \rightarrow \mathcal{F}(\mathbb{P}, \mathbb{C}) : \mu \mapsto h \circ \mu.$$

Prove that for all $\phi \in \text{Prop}_{\mathbb{P}}$,

$$\|\phi\|_{h_*\mu} = h(\|\phi\|_{\mu}).$$

$\square$

Exercise 7.8.13. Let $\mathbb{B}$ be a boolean algebra and let ϕ and ψ be propositional formulas. Then

$$\phi \equiv \psi \iff (\forall \mu \in \mathcal{F}(\mathbb{P}, \mathbb{B}))(\|\phi\|_{\mu} = \|\psi\|_{\mu}).$$

Hint. The direct implication follows from Exercise 7.8.8. In the reverse implication, reason by contradiction. You would have a $\mathbb{B}$ -valuation μ such that $\|\phi\|_{\mu} \neq \|\psi\|_{\mu}$. Use Exercises 7.6.8 and 7.8.12 to obtain a 2-valuation ν such that $\|\phi\|_{\nu} \neq \|\psi\|_{\nu}$. $\square$

Remark 7.8.14. Notice that the equivalence of the previous exercise is true by definition of $\equiv$ in the case $\mathbb{B} = 2$. The exercise can be understood as saying that if we tried to get a different notion of semantic equivalence by looking at $\mathbb{B}$ -valuations, for a $\mathbb{B}$ more complicated than 2, we would still end up with the same notion of semantic equivalence as in the case 2. $\square$

7.9 Complete Lattices

Definition 7.9.1. A **complete lattice** is an order whose every subset has a supremum and an infimum. $\square$

Remark 7.9.2. In other words, a complete lattice is an order where $\bigwedge X$ and $\bigvee X$ exist for every subset X . Notice that $\bigwedge E$ is the minimum and $\bigvee X$ is the maximum of a complete lattice E . $\square$

Example 7.9.3.

- a. $\mathbb{Q}$ is not a complete lattice because the set $\{q \in \mathbb{Q}_{\geq 0} : q^2 < 2\}$ does not have a supremum.
- b. $\mathbb{R}$ is not a complete lattice because the subset $\mathbb{R} \subseteq \mathbb{R}$ does not have a supremum. The sets that have suprema are exactly the sets which are non-empty and which have upper bounds.
- c. $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ is a complete lattice. Notice that

$$\inf \emptyset = +\infty, \quad \sup \emptyset = -\infty.$$

- d. $[0, 1]$ is also a complete lattice. $\square$

Exercise 7.9.4. Let $(E, \leq)$ be an order with at least two elements. Suppose $A \subseteq E$ has both an infimum and a supremum. Prove that $\inf A \leq \sup A$ if and only if $A \neq \emptyset$. $\square$

Exercise 7.9.5. Let $(E, \leq)$ be an order. Assume only that every subset of E has a supremum. Verify that this suffices for the conclusion that $(E, \leq)$ is a complete lattice. $\square$

Exercise 7.9.6. Prove that $(\mathbb{N}, |)$ is a complete lattice. $\square$

Exercise 7.9.7. Let X be an arbitrary set. Prove that $(\mathcal{P}(X), \subseteq)$ is a complete lattice. Prove moreover that for all $\mathcal{U} \subseteq \mathcal{P}(X)$, we have

$$\sup \mathcal{U} = \bigcup \mathcal{U}, \quad \inf \mathcal{U} = \bigcap \mathcal{U}.$$

$\square$

Exercise 7.9.8. Let G be a group and let $\mathcal{G}_G$ be the set of all of its subgroups. Prove that $(\mathcal{G}_G, \subseteq)$ is a complete lattice. $\square$

Exercise 7.9.9 (Lattice Isomorphism Theorem for groups). For a group G , we denote by $\mathcal{G}_G$ the set of all its subgroups. Now, fix a group G and a normal subgroup N . Prove that

$$\{H \in \mathcal{G}_G : N \subseteq H\} \rightarrow \mathcal{G}_{G/N} : H \mapsto H/N$$

is an order isomorphism, where the order relation on both sets is given by $\subseteq$. $\square$

Exercise 7.9.10 (Tarski's Fixed-point Theorem). Let $(E, \leq)$ be a complete lattice and let $f : E \rightarrow E$ be increasing. Prove that there is $a \in E$ such that $f(a) = a$.

Hint. Consider

$$a = \sup\{x \in E : x \leq f(x)\}.$$

$\square$

Corollary 7.9.11. Every increasing function $f : [0, 1] \rightarrow [0, 1]$ has a fixed point, i.e. there is $x \in [0, 1]$ such that $f(x) = x$. $\square$

Exercise 7.9.12. Let X and Y be arbitrary sets.

- We say that X has *lesser cardinality* than Y , and write $X \preceq Y$, iff there is an injection from X to Y .
- We say that X has the *same cardinality* as Y , and write $X \simeq Y$, iff there is a bijection from X to Y .

Prove that for all sets X, Y, Z :

- $X \simeq X$;
- if $X \simeq Y$, then $Y \simeq X$;
- if $X \simeq Y$ and $Y \simeq Z$, then $X \simeq Z$;
- $X \preceq X$;
- $X \preceq Y$ and $Y \preceq X$ if and only if $X \simeq Y$;

f. if $X \preceq Y$ and $Y \preceq Z$, then $X \preceq Z$.

Hint. The only hard part is the direct implication of [e](#), which is known as Cantor-Bernstein theorem. It is proved as follows. Fix injections

$$f : X \rightarrow Y, \quad g : Y \rightarrow X.$$

We have to define a bijection $h : X \rightarrow Y$. We do so by putting f and g together. Consider the function

$$\mathcal{P}(X) \rightarrow \mathcal{P}(X) : A \mapsto X - g[Y - f[A]].$$

Prove that this function is increasing, and apply Tarski's fixed point theorem. Let Z be a fixed point. Define h to be equal to f on Z and to the inverse of g on $X - Z$. Fill in the details. $\square$

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